

environment and prioritize based on the more critical (priority) goals. In addition, [Chapter 5](#) covered the design options that help to overcome RR suboptimal routing and low convergence time issues that can be incorporated with BGP RR design.

Note

In IP (non-MPLS) environments, it is recommended to have the RR in the forwarding path (no peering over nonclients for example) to avoid possibilities of suboptimal routing or routing information loops.

Partitioned MPLS-VPN RR

With this design model, network designers can optimize the performance efficiency of the BGP RR for MPLS-VPN by distributing a range of VPNv4/v6 to each RR or pair of RRs (usually by carving up a range of RTs per RR cluster). This model can offer large-scale MPLS-VPN operators control to distribute the peering of the VPN based on certain criteria; for example, global service providers can allocate a pair of RRs to serve each local VPN customer per region. Although this design option offers more structured RR design than a typical flat MPLS-VPN RR design, it requires more iBGP peering sessions per PE, which means that RRs will send all the VPNv4v6 routes to the PEs and that the PEs will usually drop the routes that have no importing VRF (extra unnecessary processing), as shown in [Figure 6-68](#).

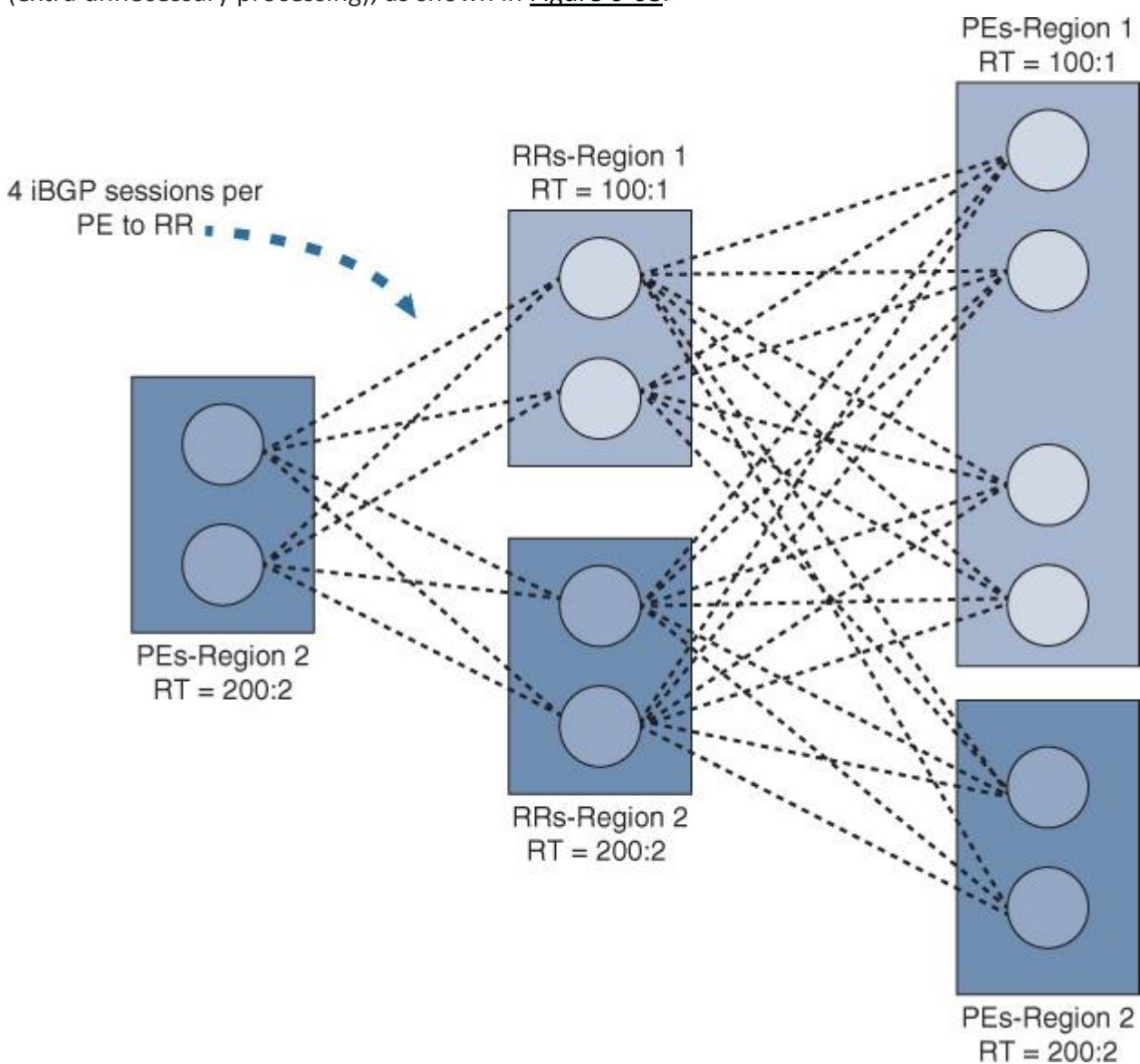


Figure 6-68 Partitioned BGP RR (RT Based)

Furthermore, this design can be enhanced with the RT constraint feature (RTC). With the RTC, each PE will receive VPNv4/v6 prefixes that have important matching VRF RT values per VRF per PE, which helps to avoid unnecessarily sending of VPNv4/v6 prefixes to the PE routers by the RRs (RFC 4684).

Table 6-9 compares the different BGP RR design options covered in this chapter.

	P as BGP RR	BGP RR for MPLS VPN and BGP RR for IP	BGP RR per Service	MPLS VPN Partitioned RR	Hierarchical BGP RR
Scalability	Limited	Moderate	High	High	Very high
Targeted environment	Native IP and MPLS VPN	Native IP and MPLS VPN	Native IP and MPLS VPN	MPLS VPN	Native IP and MPLS VPN
Operational complexity	Moderate	Moderate	High	High	High
Flexibility	Limited	Moderate	High	Moderate	High
Physical and logical topology should be kept congruent	For native IP only	For native IP only	For native IP only	No	For native IP only
Potential suboptimal routing*	Yes, mainly for native IP	Yes, mainly for native IP	Yes, mainly for native IP	Yes, low	Yes, mainly for native IP
Optimize RR routing table size	No	Yes, moderate	Yes, high	Yes, moderate	Yes, high

* Typically in MPLS L3VPN environments there is very low possibility of suboptimal routing because each CE/prefix of a given VPN is advertised by the directly connected PE only, unless there is a dual-homed CE of the same VPN customer connected to two different PEs and advertised with the same RD value.

Table 6-9 BGP RR Design Options

Hierarchical LSP (Unified MPLS)

This concept, also known as the *seamless MPLS* (or *unified MPLS*), is primarily based on a divide-and-conquer strategy where the core, aggregation, and access networks are partitioned in different MPLS LDP/IGP domains. Typically, by partitioning a single large routed domain into independent and isolated IGP/LDP domains, it helps optimize the size of the routing and forwarding tables of the routers in each domain, which in turn will significantly enhance overall network stability and result in faster convergence time. As a result, each domain can host a larger number of nodes without impacting network stability or performance.

In this design model, the reachability is achieved among the network nodes within each routing domain via intradomain LDP LSPs, while the reachability across the routed domains is achieved using RFC 3107. In RFC 3107, BGP allocates labels to routes between BGP peers to build hierarchical LSPs across the routed domains. In other words, for end-to-end reachability between two PEs on different access networks, there will be a transport LSP with label stacking that carries the inner label driven by BGP and the outer label driven by LDP per routed domain, as shown in [Figure 6-69](#).

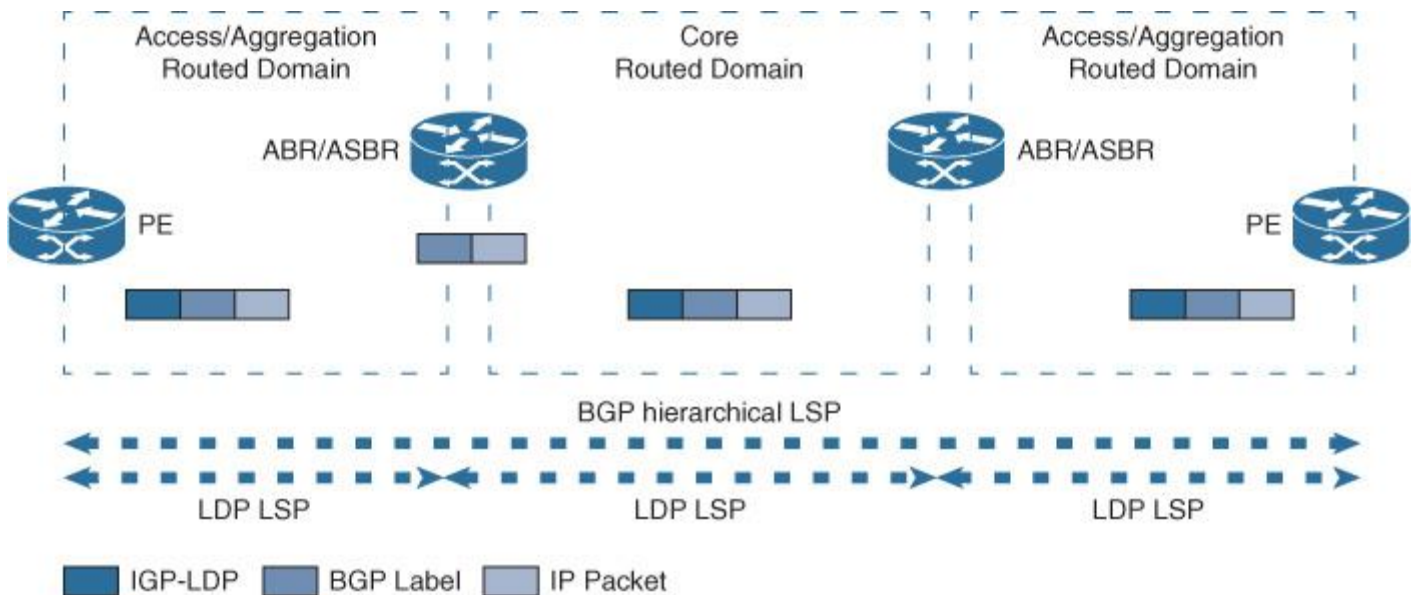


Figure 6-69 Hierarchical LSP

This design offers operators (like mobile operators with next-generation mobile backhaul for data transport 3G/LTE) that have a large number of external nodes and prefixes to be carried across the network. It usually maintains the routing and forwarding tables of the small IGP in each domain (such as link-state database). This is because only PE loopbacks are required to be leaked between the routed domains; all external reachability information is carried via BGP, which is designed to scale to the order of millions of routes. In addition, this architecture supports an easy provisioning of services such as L3VPN and L2VPN without the reliance on complex mechanisms such as PW stitching, as shown in [Figure 6-70](#). From a business point of view, in addition to the previously mentioned benefits, it offers simplified operation of such huge networks with the flexibility to support various access networks.

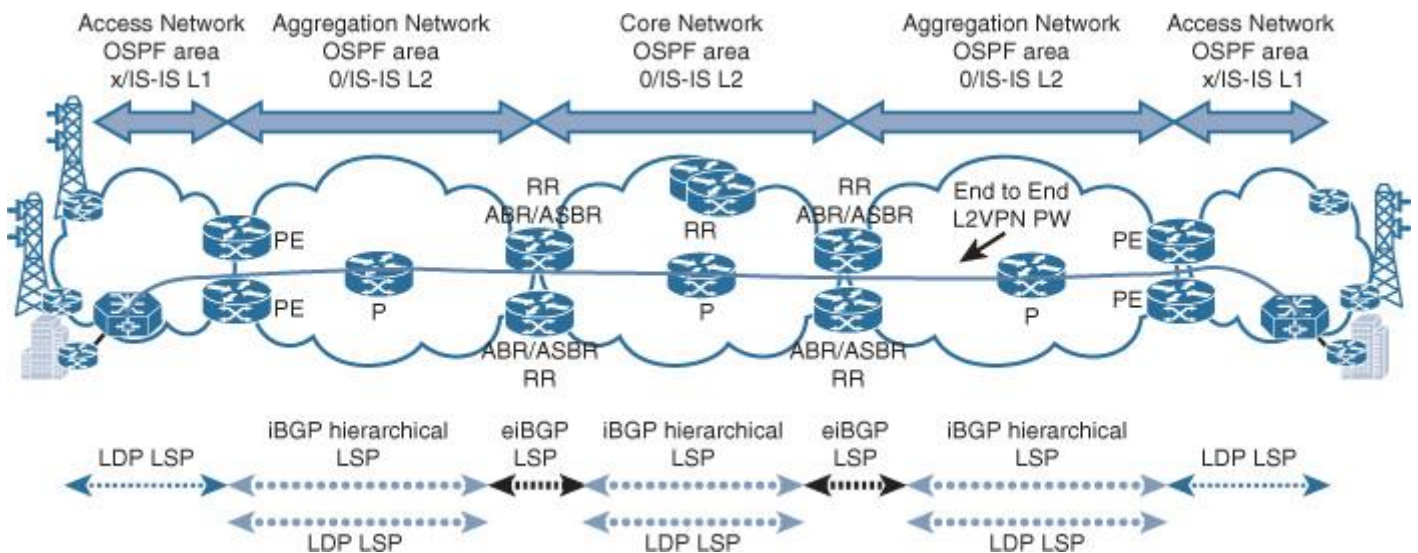


Figure 6-70 End-to-End Seamless MPLS Architecture

Note

The design shown in [Figure 6-70](#) supports both single BGP AS and multi-BGP AS design models between the routed domains.

Note

For the node in the access domains, if IGP is not supported (such as on legacy or low-end devices), you can use static routing instead, considering IP information requirements for the LDP setup covered earlier in the “[IGP Scalability](#)” section in this chapter.

Summary

This chapter covered how MPLS VPN services have become primary business enablers for service providers by meeting enterprise customer connectivity requirements, whether it is a Layer 3 or Layer 2 type of connectivity. In addition, the different design models and considerations of each have been discussed from the service provider point of view. These design models can directly or indirectly impact the service provider design efficiency and the ability to meet certain customer requirements, such as the load-balancing requirement for customers with multihomed sites to the MPLS L3VPN cloud.

In addition to the design models and considerations, this chapter covered the different design approaches that offer a scalable design to support large-scale service provider networks with a very large number of nodes and prefixes. In multiple sections in this chapter, comparison tables were included to simplify the design decision for network designers from various design angles, where each fits different design and business requirements. Last but not least, the design decision of selecting a certain design approach or protocol has to be based on the holistic approach, to avoid designing in isolation of other parts of the network, regardless of whether it is for a physical, virtual, underlay, or overlay entity.

Further Reading

RFC 4364, *BGP/MPLS IP Virtual Private Networks (VPNs)*, <http://www.ietf.org>

RFC 4762, *Virtual Private LAN Service (VPLS) Using Label Distribution Protocol (LDP) Signaling*, <http://www.ietf.org>

RFC 7432, *BGP MPLS-Based Ethernet VPN*, <http://www.ietf.org>

RFC 4761, *Virtual Private LAN Service (VPLS) Using BGP for Auto-Discovery and Signaling*, <http://www.ietf.org>

GP MPLS Based Ethernet VPN IETF draft, <https://tools.ietf.org/html/draft-ietf-l2vpn-evpn-11>

RFC 6368, *Internal BGP as the Provider/Customer Edge Protocol for BGP/MPLS IP Virtual Private Networks (VPNs)*, <http://www.ietf.org>

RFC 4577, *OSPF as the Provider/Customer Edge Protocol for BGP/MPLS IP Virtual Private Networks (VPNs)*, IETF <http://www.ietf.org>

ISI-S as the PE/CE Protocol in BGP/MPLS VPNs, IETF draft, <https://tools.ietf.org/html/draft-sheng-isis-bgp-mpls-vpn-00>

RFC 5654, *Requirements of an MPLS Transport Profile*, <http://www.ietf.org>

Chapter 7. Multi-AS Service Provider Network Design

The previous chapters of this book focused on Multiprotocol Label Switching virtual private network (MPLS VPN) services within one domain where all VPN customers are connected to one autonomous system (AS). This chapter, in contrast, covers the design options and considerations of service providers that need to interconnect different autonomous systems and offer end-to-end services for multiple VPN and non-VPN customers (including native IP, L2VPN, and L3VPN). Typically, in multi-AS scenarios, two or more carriers need to work jointly to facilitate end-to-end-connectivity for their VPN customers who need connectivity across geographically dispersed networks. However, the varied connectivity needs can have different design models. For example, for clients who have VPN sites that need to interconnect over different MPLS VPN

backbones belonging to different service providers, an inter-AS solution is required to maintain end-to-end connectivity. The same concept is applicable in scenarios where the same provider network owns multiple autonomous systems, similarly when two provider networks merge while each retains its Border Gateway Protocol (BGP) AS and possibly its administrative control. However, sometimes the service provider itself needs to interconnect two of its network islands located in different regions or countries over a transitional (backbone) provider, also known as *carrier supporting carrier* (CSC), to maintain its end-to-end connectivity, as illustrated in Figure 7-1.

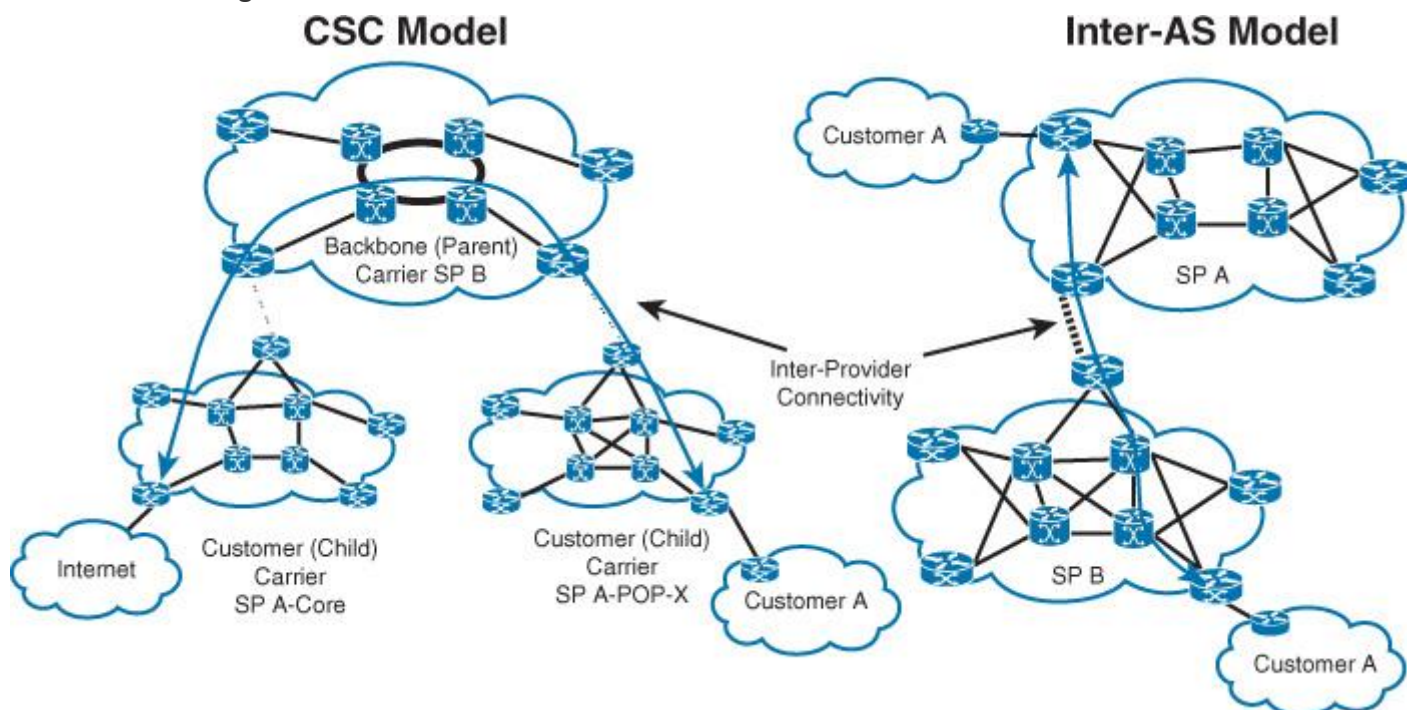


Figure 7-1 Inter-AS Versus CSC

This chapter covers the different design and connectivity options of both models (inter-AS and CSC).

Inter-AS Design Options and Considerations

According to RFC 4364, there are three design options for the inter-AS scenarios. These design options are commonly referred to by their respective subsection number in RFC 4364 as Option A, B, and C.

Inter-AS Option A: Back-to-Back VRF (VRF-to-VRF)

In this inter-AS design option, the provider border routers located at the boundary of the different autonomous systems (Autonomous System Boundary Routers [ASBRs]) interconnect directly over either a single physical link with logical interfaces (subinterfaces) or multiple physical links. Each ASBR associates one VPN/virtual routing and forwarding (VRF) to one of the logical or physical interfaces (1:1 mapping) for those VPNs that require inter-AS communication, as illustrated in Figure 7-2. This inter-AS option is considered the simplest and most secure among the other options to exchange VPN routes across different autonomous system boundaries where per-VPN filtering, accounting, and policing can be applied in a more controlled manner.

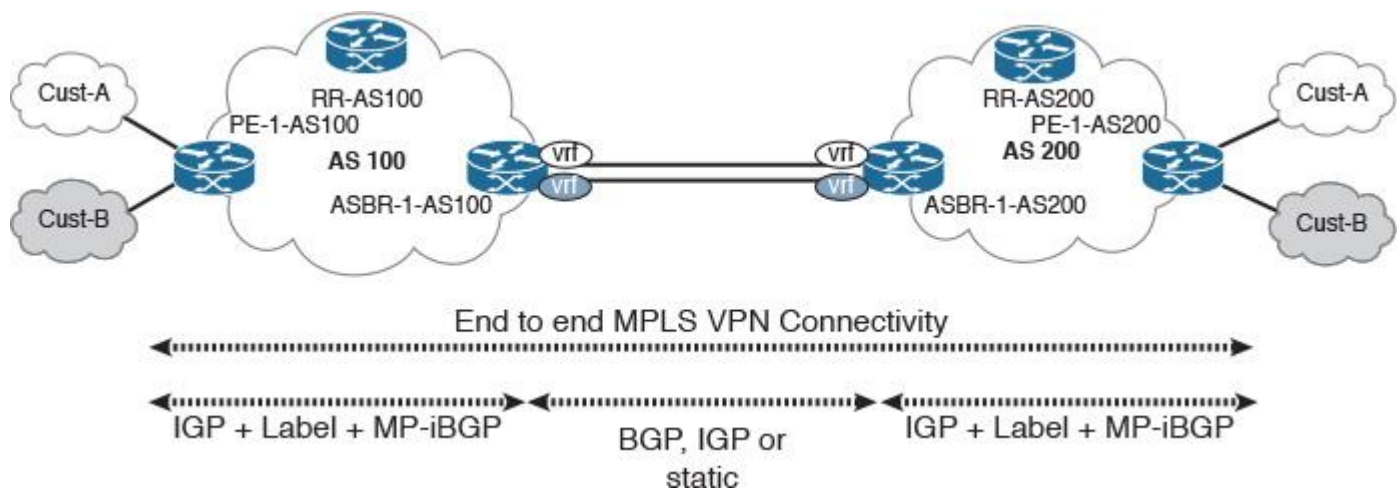


Figure 7-2 *Inter-AS Option A*

Each VRF can run external Border Gateway Protocol (eBGP), an interior gateway protocol (IGP), or static routing to exchange the respective VPN routes with its adjacent peer. Simply, each ASBR treats the facing ASBR link/VRF as a customer edge (CE) peer, where they exchange IP routes per VRF (exactly the same concept as the PE (provider edge)-CE routing concept); typically, native IP forwarding between the two autonomous systems is used (without MPLS) over this link. However, eBGP is commonly used in scenarios like this (and it is recommended) because eBGP can provide optimized scalability and policy control mechanisms. One of the major issues with this design option is the limited scalability when there is a large number of VPNs to be exchanged between the peering providers, which typically leads to an increased operational complexity.

Inter-AS Option B: ASBR to ASBR with MP-eBGP Approach

This inter-AS design option is based on the concept of having a single direct multiprotocol eBGP (MP-eBGP) session between the ASBRs of the two adjacent autonomous systems that need to be interconnected. Therefore, inter-AS Option B eliminates the need to have per-VPN configuration on the ASBRs. Moreover, in L3VPN deployments, only VPN customer prefixes need to be installed and not the local VRF Routing Information Base (RIB) at the ASBR, as compared to inter-AS Option A, which typically will allow MPLS VPN prefixes to be exchanged across multiple providers.

To allow the transport of VPN prefixes, the interconnect link between the ASBRs of the two autonomous systems should normally support Label Distribution Protocol (LDP). However, there are three primary approaches to deploy this interconnect link, where it is not always the case that the LDP must be enabled on the interprovider link. Therefore, network designers must be aware of the design variations of the inter-AS Option B to be able to consider the desired approach that meets the design requirements and constraints. For instance, in some situations, the peering provider might not allow MPLS/LDP to be enabled on the physical interconnecting link, which will compel the designer to consider an approach that can accommodate this design constraint. The following sections highlight the different approaches that you can use to achieve inter-AS Option B.

Option B-1: Next-Hop-Self Approach

With Option B-1, each AS ASBR is seen as the next-hop for the MP-eBGP routes received from the adjacent AS/ASBR. Normally with this approach, the next hop and the VPN labels are rewritten when propagated over the Inter-AS MP-eBGP session. Typically, almost always there is no transport LDP distribution over the inter-AS Option B link. Therefore, the receiving ASBR will receive only the VPN label over the MP-eBGP session, and because this label was not assigned by this ASBR, it will either not be able to forward the packet or might conflict with a label already assigned by the ASBR for something else. The same concept is applicable when the receiving PE of the local AS receives a VPN label assigned by the remote ASBR (of the peering AS); it will not be able to forward the packet properly to the respective VPN/CE destination. That is why, with MPLS

and MP-BGP when the next-hop is modified a new label has to be generated for the BGP prefixes. Accordingly, when you use the *next-hop-self approach* with inter-AS Option B, the MP-BGP next hop will technically be changed twice in this case: when the route is advertised from the local AS (ASBR) to the adjacent AS (remote ASBR) over the MP-eBGP, and when the route is propagated by the local AS ASBR (with next-hop-self) to the internal BGP peers such as PEs or BGP route reflectors (RRs). Consequently, a new VPN label will be generated by each border ASBR when the route is propagated to the local AS MP-iBGP (internal BGP) peers (PEs or RRs) and to the adjacent ASBR of the neighbor AS, as illustrated in Figure 7-3. (For simplicity, in this figure the topmost IGP transport label is not shown.)

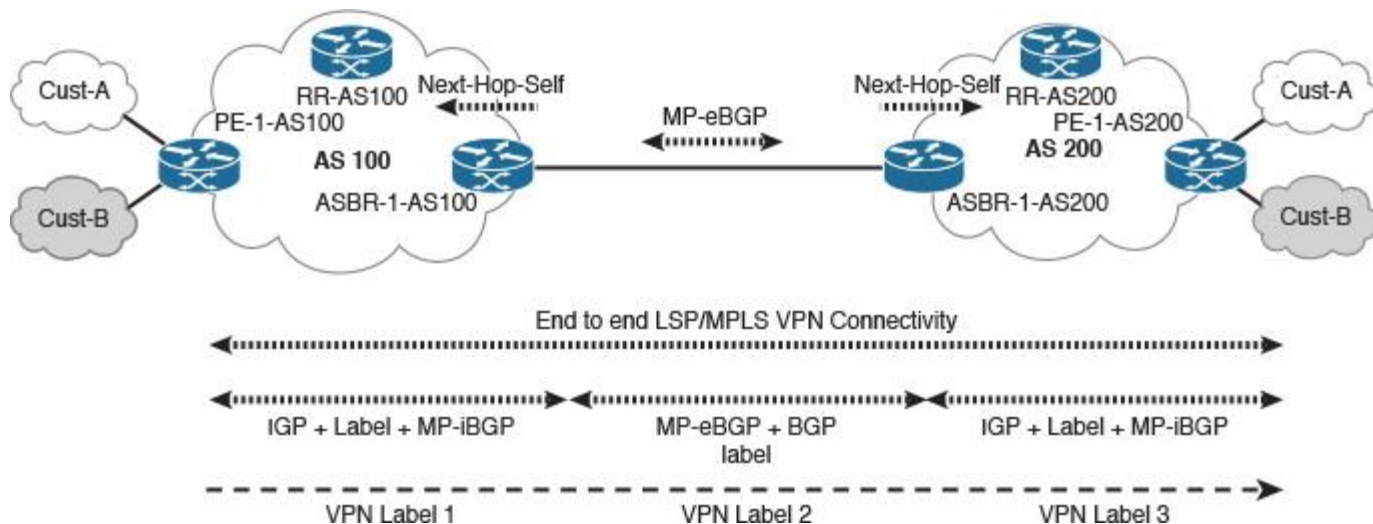


Figure 7-3 Inter-AS Option-B: Next-Hop-Self Approach

Note

eBGP label exchange is used in this approach. Therefore, there is no need to enable LDP on the physical link.

Option B-2: Redistribute Connected Approach

In Option B-2, the receiving ASBR accepts the route without modifying the next-hop attribute (the peering ASBR IP as MP-eBGP next hop will be advertised “redistribute connected” into the local AS), and the VPN label information will stay intact. With this approach, the MPLS VPN label is changed only when the ASBR advertises the VPN routes to the neighbor ASBR of the other AS. This is where the next hop is changed (once only along the path). In other words, the VPN label will be changed once with this approach, as illustrated in Figure 7-4.

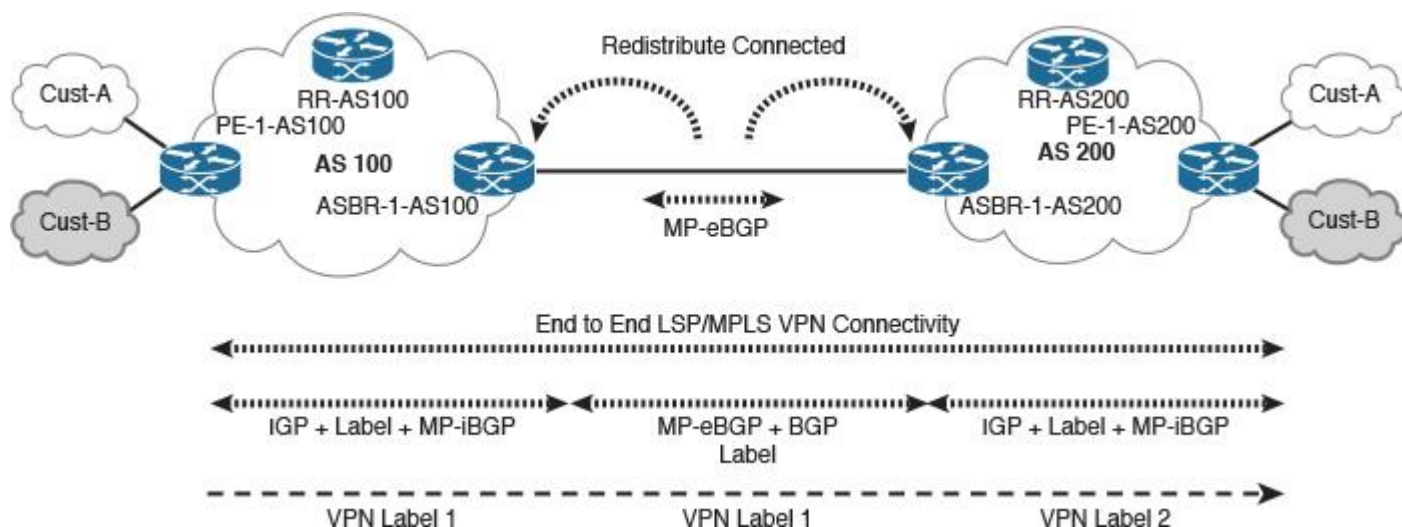


Figure 7-4 Inter-AS Option-B: Redistribute Connected Approach

However, this design option may break some load-balancing requirements. For instance, if ASBR-1-100 is interconnected to two ASBRs of AS 200, and AS 100 needs to load balance traffic from its PEs toward AS 200 over these two links, this approach can break this design requirement because technically it will announce the peering ASBRs' next-hop IPs to the PEs of AS 100. Therefore, traffic will be tied directly to one of these remote ASBRs' IP as a MP-BGP next hop from AS 100 PE's point of view. As illustrated in Figure 7-5, in the next-hop-self approach, ASBR-1-AS100 will be the next hop to reach all AS 200 prefixes. In this case, ASBR-1-AS100 can perform the load balancing/sharing (assuming BGP attributes are equal and BGP multipathing is enabled at this ASBR).

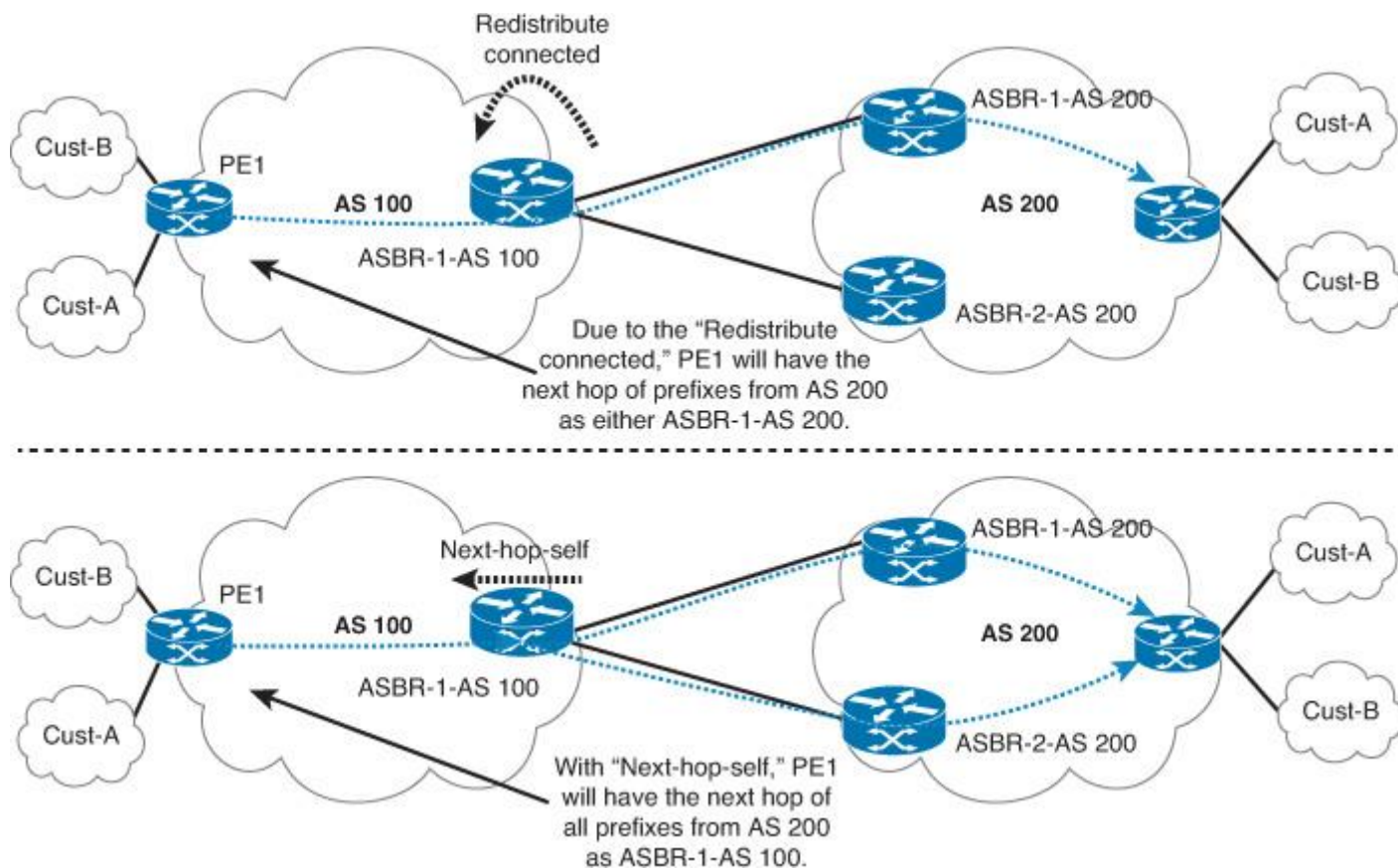


Figure 7-5 Inter-AS Option B: Load-Balancing Consideration

Option B-3: Multihop MP-eBGP Approach

In Option B-3, the MP-eBGP session is established using the ASBRs' loopback address of the two adjacent autonomous systems. One of the main incentives to using this approach is when there are multiple links between the two ASBRs and load balancing across these multiple links is required to increase bandwidth capacity and optimize the links' return on investment (ROI). In this scenario, to maintain an end-to-end LSP path, MPLS is enabled between the ASBRs, as illustrated in Figure 7-6.

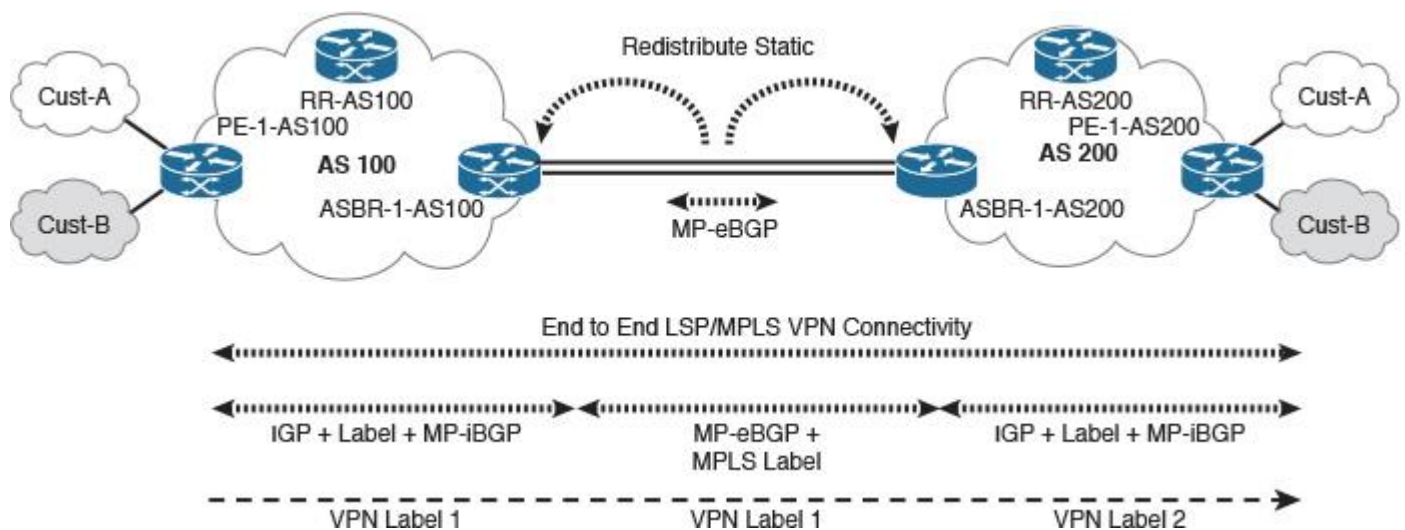


Figure 7-6 Inter-AS Option B: Multihop MP-eBGP Approach

Note

For security purposes, sometimes the service provider (SP) does not enable LDP on the inter-AS link unless both autonomous systems belong to the same company. In this case, you might consider using a combination of static label binding, static route, and eBGP label exchange.

In addition, this approach is considered as a variation of options B-1 and B-2 previously discussed; therefore, we can use both approaches to propagate the next-hop IP (next-hop-self or redistribute static).

Note

The default behavior of a PE router or ASBR is to keep only the routes that carry a route target (RT) for which the receiving router has a VRF defined with the corresponding **import** statement configured within that VRF. This functionality is called *automatic route filtering* (ARF). Therefore, ARF must be disabled with Inter-AS Option B to avoid defining all the VRFs on the ASBR manually to install the VPN's prefixes.

Note

All three approaches of achieving inter-AS Option B are viable solutions. The difference between them is minimal from a design point of view, because all of them are driven from the same inter-AS design model, which is Option B. However, each approach is achieved by a different technical method. Therefore, network designers must take these technical differences into consideration, particularly their impact on the design requirements and on operational complexity, along with any technical or design constraint.

Inter-AS Option C: Multihop MP-eBGP Between RR

Option B, discussed in previous section, still requires that all VPN routes be maintained and advertised by the ASBR, which makes the solution not very scalable in situations where there is a significantly large number of VPN prefixes. Inter-AS design Option C, however, overcomes this issue. The principle of this approach is based on establishing the inter-AS MP-eBGP session directly between the RR of each AS. In other words, the ASBRs in this option only exchange BGP next-hop address using IPv4 labels, and the RRs form an MP-eBGP session to transport VPNv4 information. As a result, the ASBRs do not need to carry any of the VPN routes, which makes this option the most scalable compared to the other options discussed earlier in this section, as illustrated in Figure 7-7.

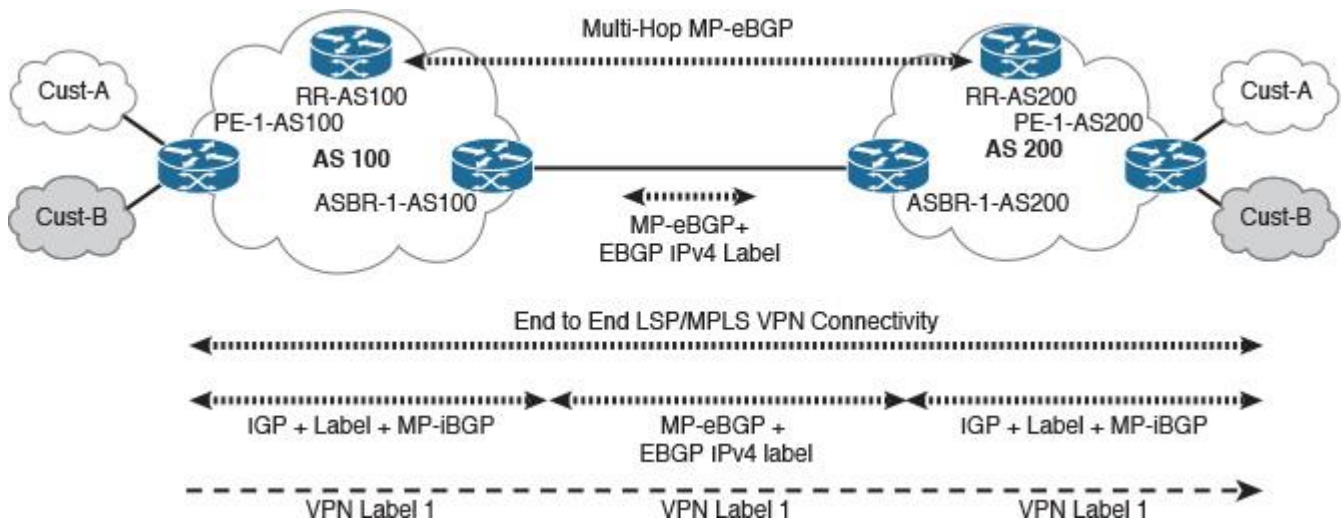


Figure 7-7 Inter-AS Option C

Note

This design option requires that the BGP next hop of the VPN-IPv4 routes exchanged across the multihop MP-eBGP session not be changed.

Note

Because Option C requires sharing internal IPs (at least loopback IPs of the PEs) of the both interconnected autonomous systems, it is most commonly used by providers who own multiple autonomous systems in different regions (for example, a global SP) or in the same region and operate under the same administrative authority. Because each AS will have visibility of the other autonomous systems to a great extent, this design option is typically not a common practice between different SPs / autonomous systems that are under different administrative authorities and owned by different stakeholders. This is due to several reasons, the foremost of which is network security.

Inter-AS Option D

The MPLS VPN inter-AS Option D, also known as AB, combines the best functionalities of both inter-AS Option A and Option B (defined in RFC 4364) to allow MPLS VPN providers to facilitate the interconnection of different autonomous systems with end-to-end VPN services.

From the control plane perspective, inter-AS Option (AB) addresses the scalability limitations of inter-AS Option A by using a single MP-BGP session between the peering ASBRs. VRF routes are exchanged between the two adjacent ASBRs using a unique MP-BGP session referred to as an *Option AB peer*. Similarly, the VRFs are defined in a special way in each ASBR, called *Option AB VRFs*.

From the forwarding plane perspective, each VRF will have a next-hop IP configured under the VRF configurations specifying the next-hop IP (the VRF interface/subinterface) between the ASBRs. In other words, the route will be received per VRF/VPN over one MP-BGP session, while the forwarding will use the respective VRF interface/subinterface, as illustrated in Figure 7-8. This design option requires two types of interfaces between the adjacent ASBRs: one type allocated per VRF basis (main or subinterfaces), and the other interface has to be in global routing to be used for the MP-eBGP peering session.

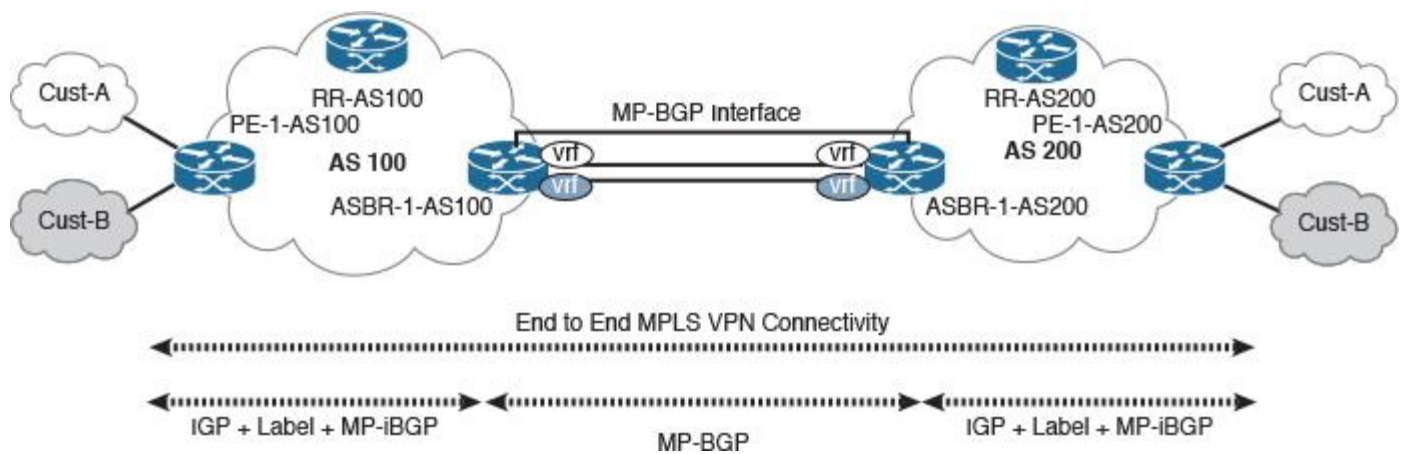


Figure 7-8 Inter-AS option-D

Isolating the forwarding plane traffic on a per-VRF basis offers an optimized security control and reasonable quality of service (QoS) granularity between ASBRs to maintain customer service level agreements (SLAs).

Inter-AS IPv6 VPN

The design options discussed earlier in this section support transporting IPv6 routes between two MPLS VPN providers in the same manner for MP-BGP VPNv6 routes (6VPE), where IPv4 peering can be preserved for PE to ASBR or RR, and for ASBR to ASBR. For example, in Figure 7-9, the inter-AS Option B is formed between AS 100 and AS 200. In this particular scenario, the IPv6 VPN prefix will be seen by each ASBR associated with the neighbor ASBR IPv4 as the BGP next hop (NH).

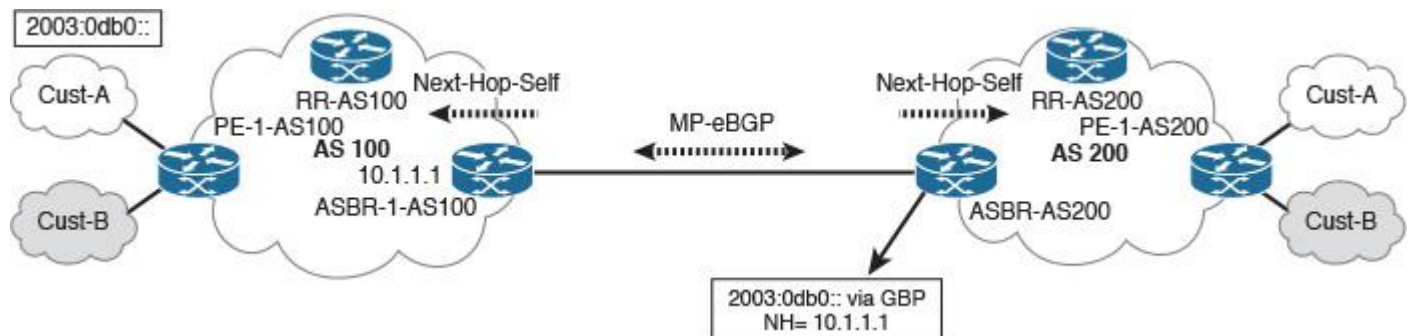


Figure 7-9 Inter-AS IPv6 VPN

Similarly, if inter-AS Option C is used, the BGP NH will be the actual remote PE's IPv4/32 loopback addresses. This offer operators the flexibility to retain their current MPLS and IPv4 infrastructure and enables IPv6 transport for their VPN customers.

Inter-AS MPLS-TE

Large and global service providers do not always have the infrastructure to cover the geographic distances across countries or sometimes within the same country. Therefore, they partner with other service providers, either local or international providers, to extend their presence. In this type of scenario, there are usually separate routing islands per AS or provider. Normally, for security purposes, not many providers are willing to expose their internal network topology or internal IP addressing. Similarly, when two providers merge, each with different routing domains (AS), a mechanism is required here to facilitate the setup of the end-to-end Multiprotocol Label Switching Traffic Engineering (MPLS-TE) label switched-paths (LSPs) and to provide the capability to enforce policy control at the provide edge (ASBR), such as bandwidth policing, message types, and authentication.

As discussed Chapter 5, "Service Provider Network Architecture Design," the interdomain MPLS-TE (whether its IGP interarea or BGP inter-AS) both share the same attributes in terms of the limited information in the

TE database to set up end-to-end MPLS-TE LSPs by relying only on the headend node. Therefore, a different approach has to be adopted in scenarios like this (commonly by either considering Explicit Route Object [ERO] loose object expansion or the Path Computation Element [PCE]). As illustrated in Figure 7-10, the MPLS-TE headend R1 defines ASBR3 and ASBR4 as a loose explicit route object (ERO) of the MPLS-TE path messages along with the tail-end node (R10). Based on this setup, the Constrained Shortest Path First (CSPF) will be completed along with the creation of TE LSP setup.

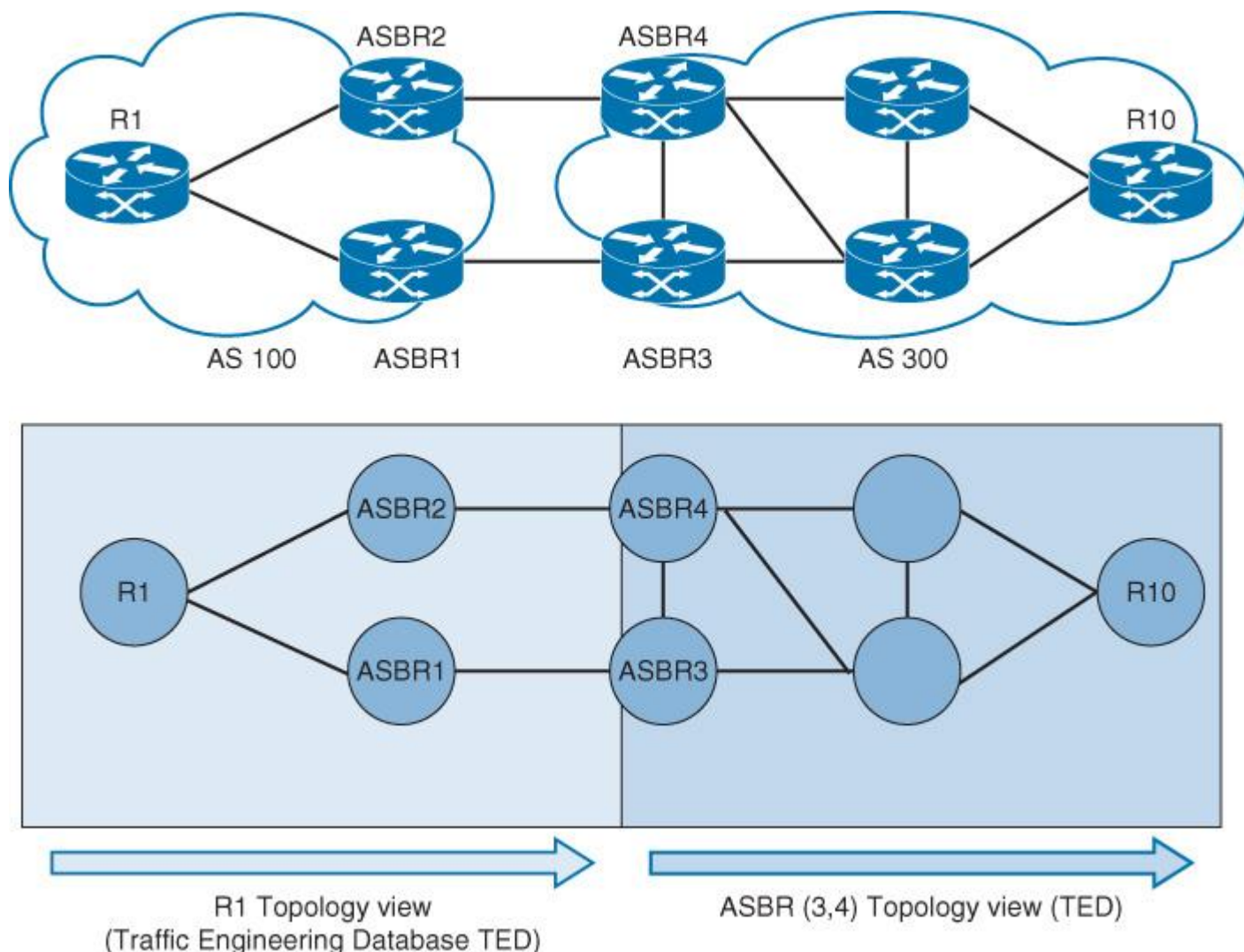


Figure 7-10 *Inter-AS MPLS-TE*

Note

As highlighted in Chapter 5, PCE and BGP Link State (BGP-LS) are both valid mechanisms to be considered in MPLS-TE inter-AS scenarios, because both offer the ability to share TE information between different routing domains; however, these options when used between different service provider businesses must cater for constraining the type and amount of the shared information between the different provider networks.

Inter-AS L2VPN

Provisioning L2VPN over inter-AS links is one of the common and primary connectivity requirements in today's modern service provider networks. This is applicable to any of the following scenarios:

- Inter-AS or provider integration scenario (between two separate service providers)
- Inter-AS scenario across different autonomous systems that belong to a single administrative authority
- Service provider merger and acquisition scenarios

Technically, the main inter-AS connectivity options (A, B and C) discussed earlier in this chapter are applicable and commonly used to provide L2VPN inter-AS connectivity for both point-to-point pseudo wires (PWs) (E-

Line) and multipoint VPLS (E-LAN). Like L3VPN, each inter-AS option has different attributes when used to provision inter-AS L2VPN connectivity, as summarized here:

■ **Inter-AS Option A:** In this model, each of the L2VPN PWs terminates at the local AS ASBR, and the inter-AS connectivity is based on the used native attachment circuit technology. (At the time of this writing, it has to be Ethernet with Virtual Private LAN Services [VPLS].) The ASBR at the remote AS (peering AS) must create a separate (new) PW to link it to the relevant attachment circuit to maintain end-to-end connectivity, as illustrated in Figure 7-11. In other words, there will be two separate PWs segments, across each AS, while over the inter-AS link the transport will be native based on the technology used, such as Ethernet, time-division multiplexing (TDM), or Frame Relay. Although this option offers a more controlled and secure connectivity model, the high level of operational complexity, along with scalability limitations, makes it least preferred in large environments where a large number of PWs and customers need to be extended over the inter-AS link. As with the IP/VPN, where a large number of prefixes add more load on the ASBRs, with the L2VPN the number of VLANs needs to be maintained, and the number of MAC addresses in the case of VPLS can introduce hardware resource constraints in this option.

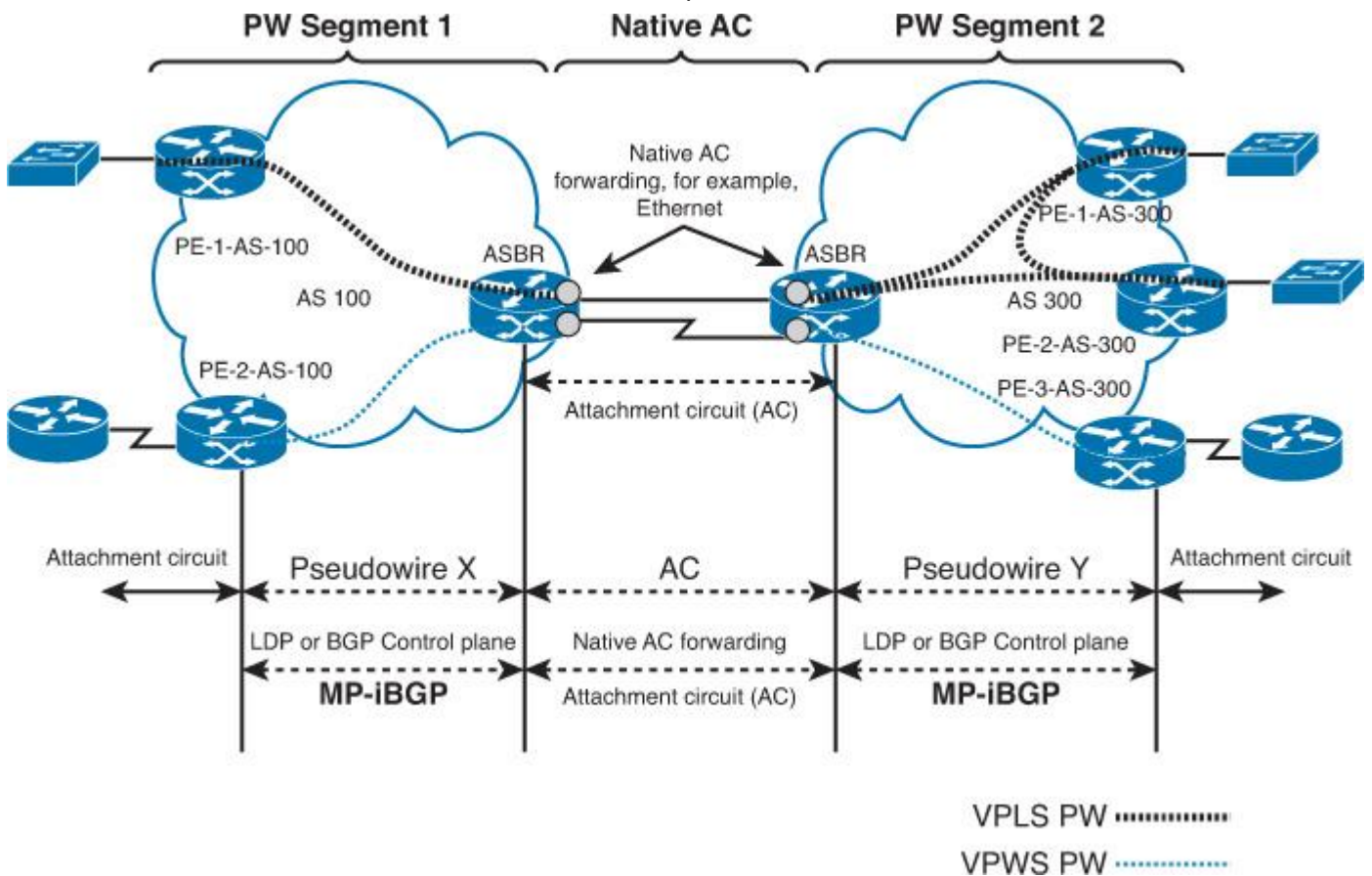


Figure 7-11 L2VPN Inter-AS Option A

■ **Inter-AS Option B:** In this model, each of the L2VPN PWs terminates at the local ASBR. In turn, this ASBR will create another PW (performing what is known as *PW stitching* or *PW switching*) that terminates at the remote AS ASBR over the inter-AS link (inter-AS link does not have to be Ethernet). The remote AS ASBR again will create a PW that terminates at the egress PE node where the CE attachment circuit is connected, as illustrated in Figure 7-12. It is obvious that this option with the multisegment PW (as specified in RFC 5659) provides the ability to pass the PWs over single inter-AS link with targeted LDP (T-LDP), which offers more granular and strict security policies for control plane traffic going across the autonomous systems. However, this multisegment PW approach makes it a more complicated option to manage, and its scalability is limited in large network with a large number of PWs because the PWs' control plane will be constrained to the hardware/software limits of the ASBR.

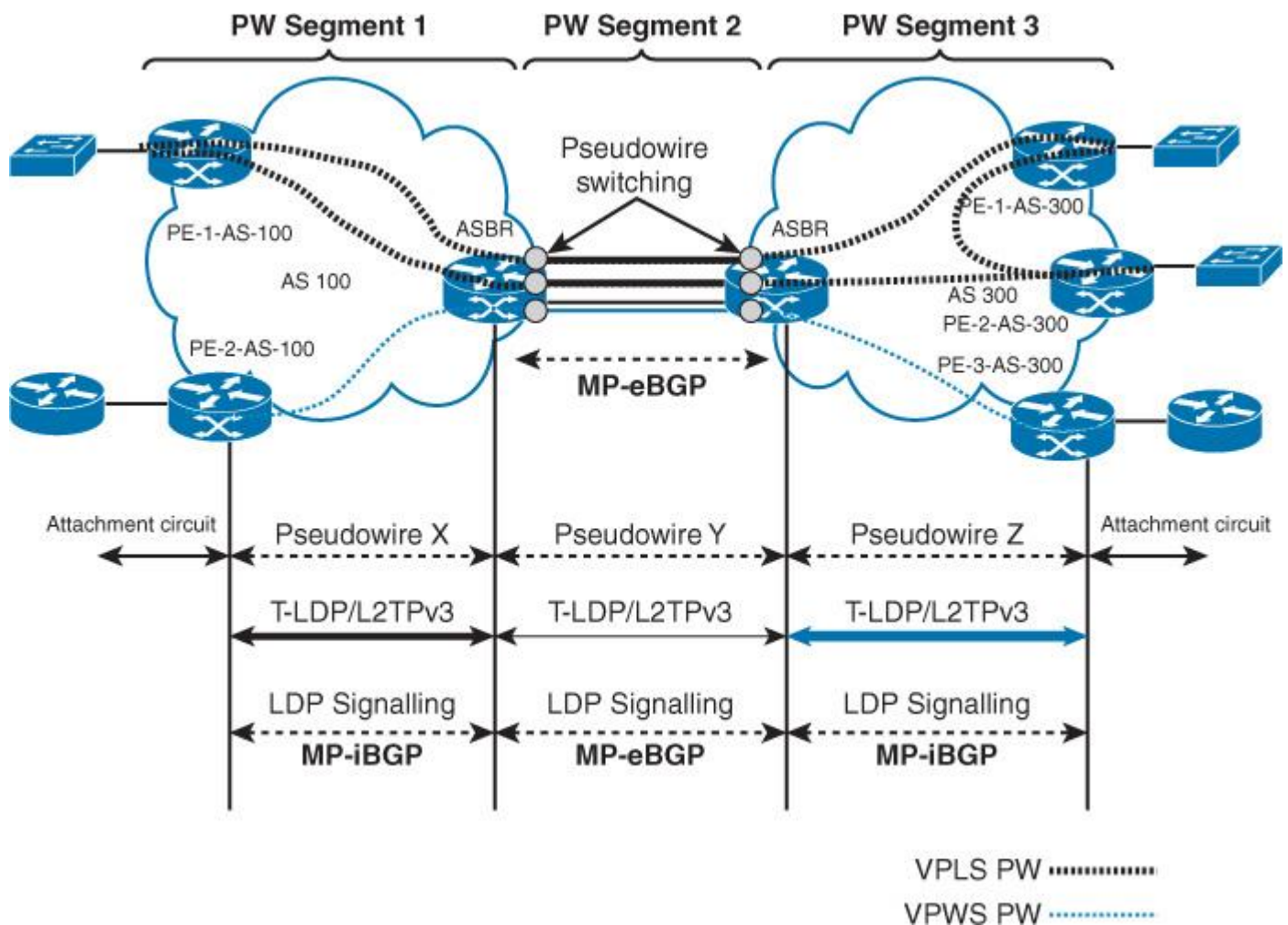


Figure 7-12 Multisegment PW L2VPN: Inter-AS Option B + LDP Signaling + LDP/BGP Autodiscovery

Note

The introduction of VPLS BGP signaling for L2VPN inter-AS Option B, that supports RFC 4761, offers a more simplified PE nodes discovery (autodiscovery) as well as signaling of all participating PE devices per VPLS instance by using BGP for both functions. However, before the introduction of the combined BGP autodiscovery and signaling over inter-AS Option B, some L2VPN inter-AS deployments used LDP for the signaling and BGP for autodiscovery, as specified in RFC 6074. Technically, RFC 6074 and RFC 4761 use a different network layer reachability information (NLRI) encoding format for the autodiscovery protocol. Therefore, network designers must avoid mixing the implementation of these RFCs together, because they are incompatible with each other (at the time of this writing). In addition, it can be a design concern when interconnecting two autonomous systems and each uses one of these RFCs for BGP autodiscovery. In this case, inter-AS Option A can be considered as one solution if it's not possible to migrate any of the autonomous systems to be compatible with the other one with regard to the deployed BGP autodiscovery RFC.

Note

For interprovider VPLS Option B, there are two common possible deployment flavors. The first one (described above) allows direct PWs (LDP signaled) among all PEs in both autonomous systems per VPLS instance, and the ASBRs in this case hold all the control plane for each VPLS instance (performing PW switching). The second flavor is based on BGP signaling over the inter-AS link, in which the number of PWs can be reduced to a single ASBR to ASBR BGP signaled PW session, which helps to significantly optimize the scalability and reduce the overhead of the control plane of this inter-

AS solution in general, as illustrated in Figure 7-13. With this model, the ASBR might participate in the data plane (MAC learning) when interconnecting different domains with different signaling protocols (LDP and BGP).

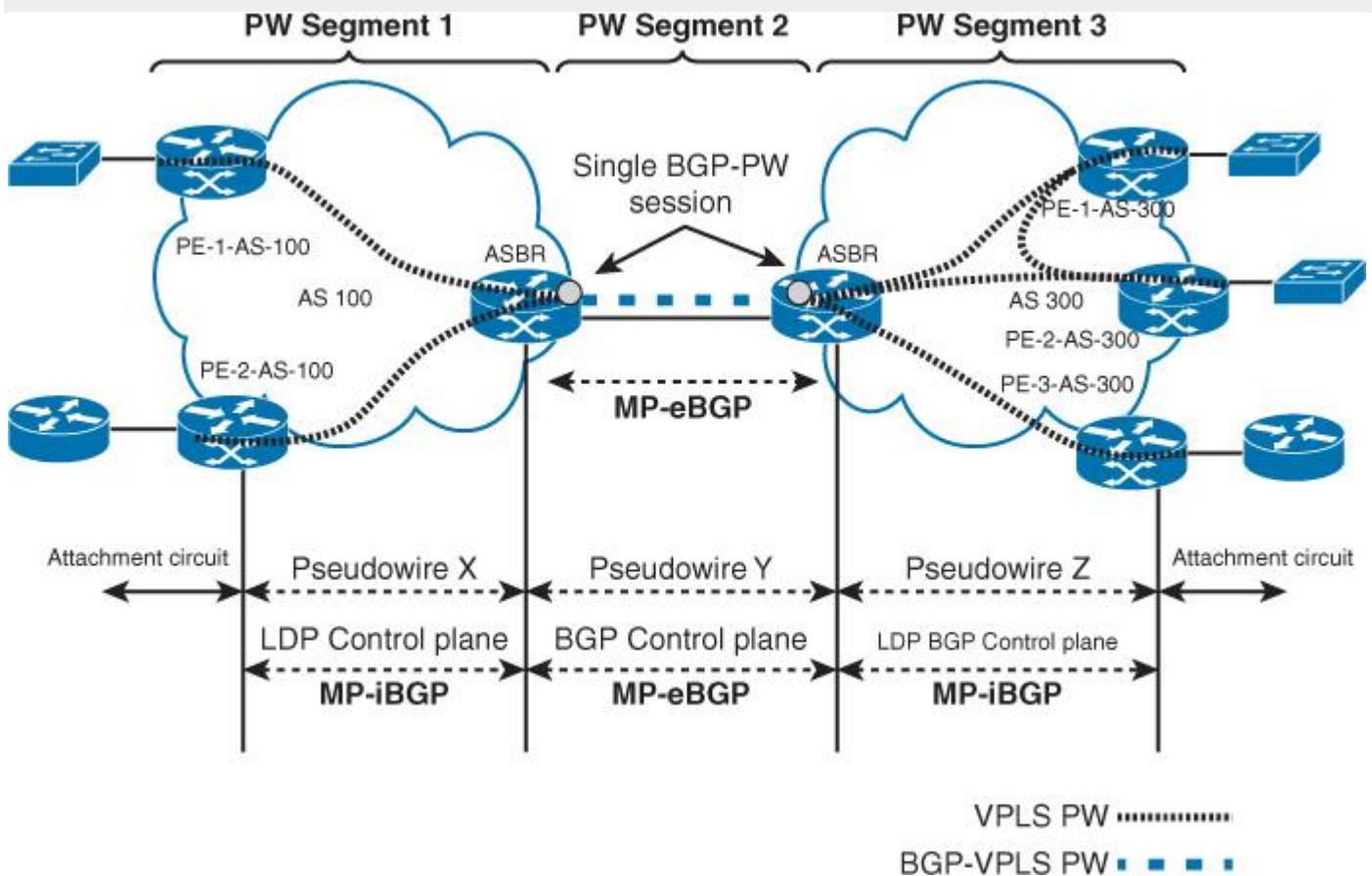


Figure 7-13 Multisegment PW L2VPN: Inter-AS Option B with BGP for Signaling and Autodiscovery

■ **Inter-AS Option C:** This connectivity model is the most scalable option compared to the other inter-AS options with regard to L2VPN, because the ingress and egress PEs can establish the L2VPN PWs directly over a targeted LDP (T-LDP) session or a direct PE-to-PE L2TPv3 PW, without any termination point along the path, as illustrated in Figure 7-14. This option is most commonly used in environments where both autonomous systems belong to the same administrative authority, such as a global server provider with multiple autonomous systems. Because this inter-AS option technically requires some of the IP addressing of the internal IP infrastructure (typically loopback IPs) to be shared (reachable directly) between the interconnected autonomous systems, it is not common and may not even be applicable to the L2VPN “interprovider” scenario, where each provider belongs to completely a separate business/stakeholders. In addition, there might be many LDP or L2TPv3 PW sessions traversing the inter-AS link. Technically, these PWs may “connect to many devices within an AS. In the case where the ASes belong to different providers, one might imagine that providers would like to have fewer signaling sessions crossing the AS boundary and that the entities that terminate the sessions could be restricted to a smaller set of devices.”¹ A merger and acquisition scenario, in contrast, is another example where the L2VPN inter-AS Option C is commonly used.

1. IETF RFC 6074, <http://www.ietf.org>

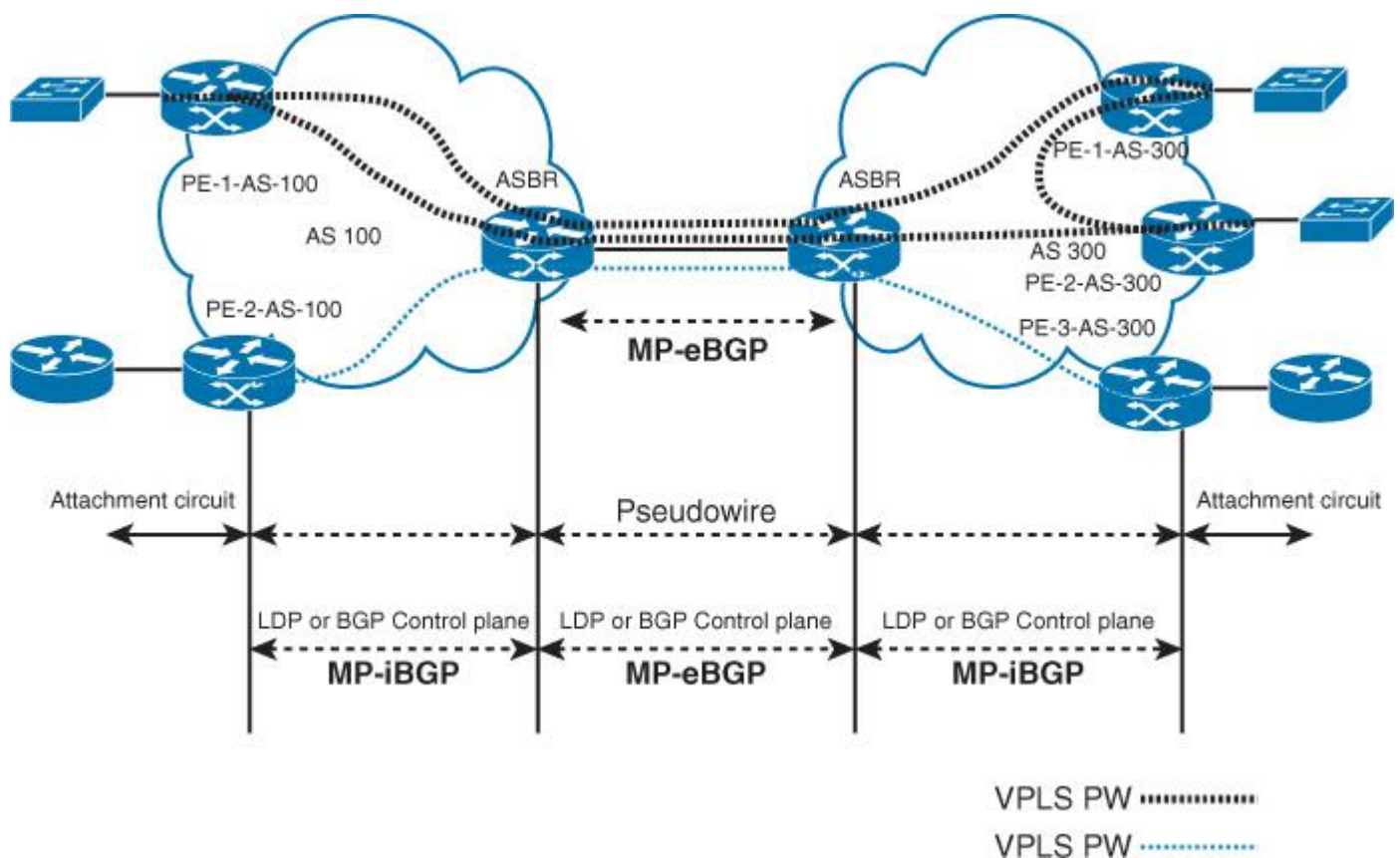


Figure 7-14 L2VPN Inter-AS Option C

Furthermore, with VPLS another layer of constraints must be taken into account: MAC learning. In other words, in addition to the number and setup of the PWs considerations, network designers must take into account the scalability and manageability with regard to MAC address learning and flooding per VPLS virtual switching interface (VSI). For instance, VPLS along with provider backbone bridging (PBB) and with inter-AS Option B can offer an optimized and scalable solution. As with PBB, MAC learning and forwarding within the core will be based on the backbone MACs only (B-MAC) using the PBB MAC-in-MAC encapsulation concept. Similarly, the VPLS with PBB over inter-AS Option C will offer the most scalable and seamless connectivity model. However, as highlighted earlier, this model is the least secure among the other options when used between different service provider businesses, due to the fact that each provider (AS) will have visibility on the other AS network (leaking IP information between the two provider networks to create end-to-end LSPs for transport, T-LDP, and MP-BGP for PW, and so on).

Ethernet VPN (EVPN), in contrast, as highlighted earlier in Chapter 6, is based on the BGP control plane using a separate NLRI for MAC address learning and forwarding. This means that all the design considerations discussed in this chapter with regard to Layer 3 inter-AS options apply to EVPN. The only difference here is the addresses learned and shared by this BGP NLRI are MACs and not IPs.

Inter-AS QoS

Quality of service (QoS) is one of the primary influencing factors with regard to selecting the interprovider (inter-AS) connectivity model because it directly impacts the end-to-end differentiated services offered as part of the SLA between each provider and its customers. Table 7-1 summarizes the characteristics of the inter-AS connectivity models discussed in this chapter that pertains to end-to-end QoS provisioning and consistency.

	Option A	Option B	Option C	Option D
Traffic control and QoS policy granularity per VPN	High	Low	Low	High
QoS policy scalability with regard to the number of VPN/VRFs	Limited*	High	High	Limited*
Operational complexity to set up and maintain QoS policies	High	Low	Low	High
Supports mapping between different providers' class of service (CoS) models	Yes (re-marking over the interprovider link based on the original IP differentiated services code point [DSCP] per VPN customer) (Original customer's DSCP marking must be left intact.)	Yes (re-marking over the interprovider link based on MPLS EXP) (Mapping normally happens at EXP tag, while the original customer's DSCP marking is left intact.)	Yes (re-marking over the interprovider link based on MPLS EXP) (Mapping normally happens at EXP tag, while the original customer's DSCP marking is left intact.)	Yes (re-marking over the interprovider link based on the original IP DSCP marking per VPN customer) (Original customer's DSCP marking must be left intact.)

* This is hardware/software dependent.

Table 7-1 Comparison of QoS Considerations per Inter-AS Option

Note

The selection of the MPLS DiffServ tunneling mode, such as uniform or pipe mode, can influence to a large extent the preservation and consistency of the original customer's IP DSCP marking across a single or different autonomous systems. DiffServ tunneling modes are covered in more detail in Chapter 10, "Design of Other Network Technologies and Services."

Comparison of Inter-AS Connectivity Options

Technically, all the inter-AS options discussed in this chapter are all valid and proven design options; however, each has its own characteristic that fit certain design requirements.

Table 7-2 briefly compares the four MPLS L3VPN inter-AS design options discussed in this section from different design aspects.

	Option A	Option B	Option C	Option D
Scalability	Low	Moderate	High	Moderate
L2/L3VPN Security/Control	High	Moderate (e.g. Label spoofing concerns)	Low (For example, the PE's IPs are visible to the remote AS.)	High
Requires leaking some internal IP information between the interconnected autonomous systems	No	No	Yes	No
Flexibility (IPv6, L2VPN, Multicast)	Moderate	Moderate	High	Moderate
State/prefixes at ASBR	High (per VRF state)	Moderate- VPN routes states maintained	Minimal (Only remote AS PE's IP are maintained.)	High (per VRF state)
VRF configuration at the ASBR	Yes	No	No	Yes
Label at the AS boundary (interconnect)	No	Yes (LDP/BGP)	Yes	Not required for data plane

ASBR control plane peering sessions	Multiple (per VRF state)	Single MP-eBGP session per peering ASBR	Single multihop MP-eBGP session per RR	Single MP-eBGP session per peering ASBR
QoS granularity per VPN at the ASBR	High	Low	Low	High
End-to-end seamless QoS marking	Yes (Complex-mapping between DSCP and MPLS EXP may be required at each ASBR.)	Yes (simple; end-to-end MPLS EXP based)	Yes (simple; end-to-end MPLS EXP based)	Yes (complex; mapping between DSCP and MPLS EXP may be required at each ASBR.)
End-to-end L2VPN simplicity and scalability	Low	Moderate	High	Low (not common)
MPLS-TE end-to-end simplicity (seamless)	Low	Moderate	High	Low
ASBR operational complexity	High	Moderate	Low	Moderate

Table 7-2 Inter-AS Connectivity Options Comparison

Carrier Supporting Carrier

As described earlier in this chapter, the CSC model aims to facilitate the connectivity between two or more dispersed points of presence (POPs) or backbones (customer carriers) over a middle carrier (backbone carrier). For the backbone carrier to facilitate end-to-end route propagation for the customers' carriers, an LSP path is required (label stacking). Hence, the link between the customer carrier CE and the backbone carrier PE must be MPLS enabled. There are two primary ways to achieve this:

■ **Label exchange over IGP:** This is used when the PE-CE routing is IGP and MPLS needs to be enabled on the interface level.

■ **Label exchange over BGP:** This method relies on using BGP for label exchange (the same concept used with inter-AS).

In some scenarios, however, this backbone network runs native IP, in which case other mechanisms are required here, such an overlay to facilitate the CSC connectivity model. This section covers the following CSC connectivity scenarios:

- Non-MPLS customer over MPLS VPN carrier
- MPLS customer over MPLS VPN carrier
- MPLS VPN customer over MPLS VPN carrier
- MPLS VPN customer over MPLS carrier
- MPLS VPN customer over IP-only carrier

Note

The customer provider in this section can be an enterprise that needs to interconnect multiple large sites that has either a large number of routes or MPLS-VPN enabled, application service providers, MPLS VPN service providers, or ISPs. In addition, some backbone service providers do not offer CSC as a connectivity option to enterprise customers and this connectivity model is only offered to service provide businesses (customers). Therefore, the network designer should make it clear whether this is a supported connectivity model by asking the SP before producing the design for the customer.

Non-MPLS Customer over MPLS VPN Carrier

In this design model, the customer carrier runs native IP (not MPLS enabled) across its internal network. For example, this can be an ISP that needs to interconnect different IP backbone islands of the same provider, or a global enterprise with a very large number of prefixes to be shared across the different locations. To maintain the end-to-end LSP connectivity of the CSC backbone carrier, you can enable IGP with MPLS on the interface level over the PE-CE link, as illustrated in Figure 7-15. From the carrier backbone point of view, each CSC customer will be seen as a typical MPLS VPN customer, where no route overlapping can occur, nor does the backbone carrier need to participate and process the routes, except at the PE edge (VRF level). In addition, this minimizes the number of prefixes that need to be installed and processed by the backbone carrier (the parent carrier in the middle) because the only prefixes to be processed in this case are the internal routes, such as BGP next-hop IPs of the different customer BGP islands. Other external prefixes will be MPLS encapsulated (for instance, if an ISP needs to propagate full Internet routes to a remote POP over the CSC backbone). In this particular scenario, the CSC backbone carrier does not need to install all the Internet routes for this ISP customer. Instead, it will provide only next-hop reachability for the customer BGP speakers (ASBRs), along with end-to-end LSP over the backbone carrier network.

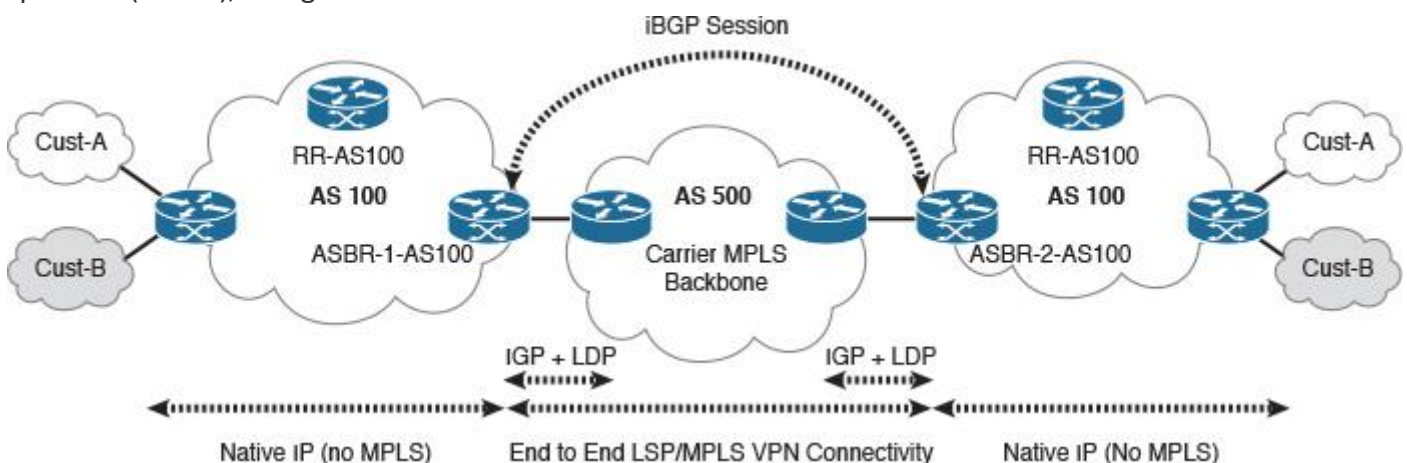


Figure 7-15 Non-MPLS over MPLS VPN Backbone

Note

If BGP is used as the PE-CE routing protocol, it can also be used for label exchange without the need to manually enable MPLS/LDP on the interface level.

MPLS Customer over MPLS VPN Carrier

In this design model, the customer carrier is running MPLS but not MPLS VPN. This design model is similar to the one described earlier. The main difference is that MPLS is enabled across the customer carrier backbone/POPs, and hence forwarding is based on label switching (LSP) end to end. At the edge of the customer carrier, the label must be exchanged with the carrier backbone to maintain end-to-end LSP inside the transit AS. As described earlier, label exchange can be done using either BGP label or IGP with LDP manually enabled on the link level. As illustrated in Figure 7-16, the iBGP peering can be formed directly between the ASBRs of AS 100 over the backbone carrier. Also, this iBGP session can be established directly between the BGP RRs in each network island of AS 100. However, terminating the iBGP session at the ASBR offers more “control plane” security control at the network boundary.

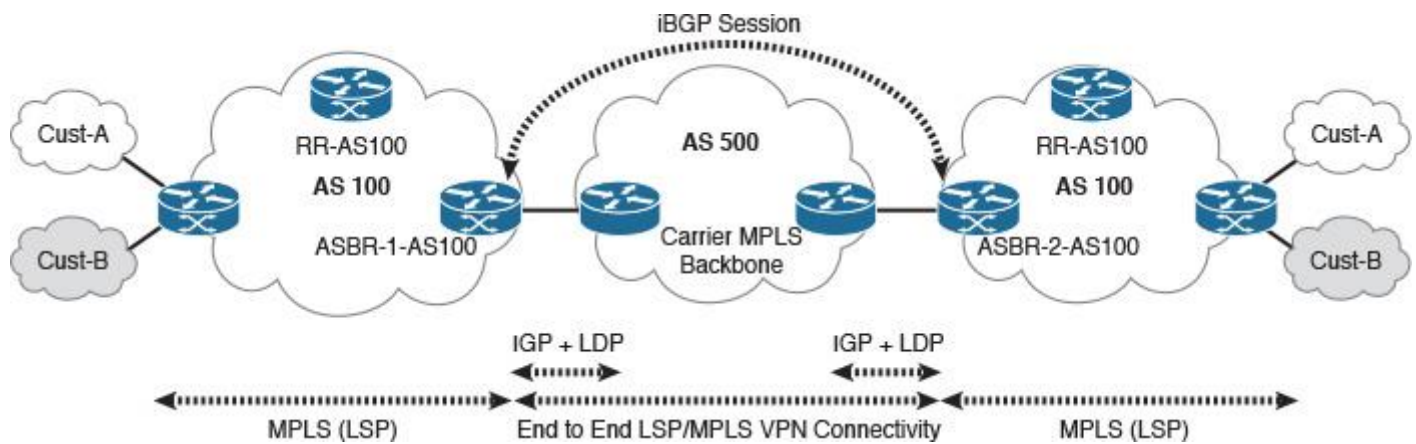


Figure 7-16 MPLS Customer over MPLS VPN Carrier

MPLS VPN Customer over MPLS VPN Carrier

In this model, the customer carrier performs MP-BGP VPN peering between the dispersed sites (for example, between the BGP RRs in each location) to exchange MPLS VPN routes over the carrier backbone, where only the loopback IPs required for the MP-BGP session are propagated across the backbone carrier. From the backbone carrier point of view, each customer carrier is an MPLS VPN customer. In other words, the MPLS VPN router will be the carrier over the MPLS VPN of the CSC backbone carrier. Accordingly, this design model is also known as *hierarchical VPN* and is illustrated in Figure 7-17.

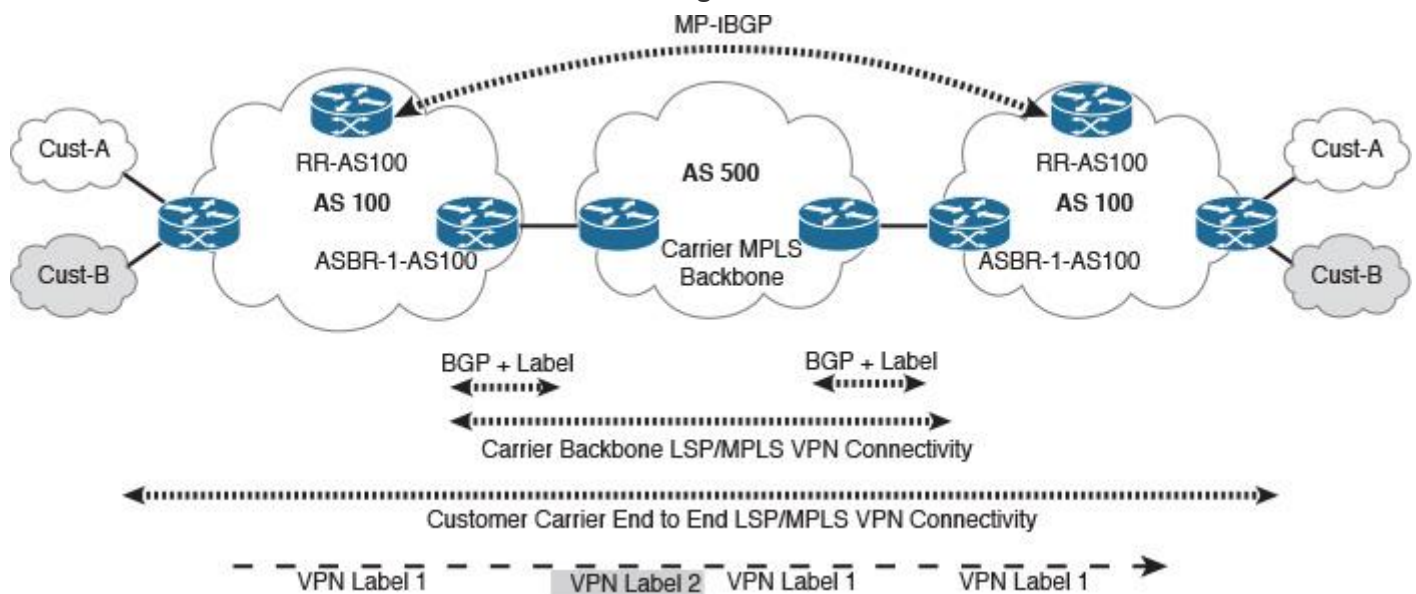


Figure 7-17 MPLS VPN over MPLS VPN Backbone

MPLS VPN Customer over MPLS Carrier

Like the previous CSC design model, in this model the customer carrier performs MP-BGP VPN peering between the dispersed sites (for example, RRs each location) to exchange MPLS VPN routes over the carrier backbone, where only the loopback IPs required for the MP-BGP session are propagated across the backbone carrier. The backbone carrier, however, performs only label switching (Non-MPLS VPN); it is usually LDP or hierarchical MPLS (such as the seamless MPLS model), where both LDP and BGP labels are used across the backbone carrier network to carry loopback IP addresses of the customer carriers or at least the loopback IPs of the MPLS VPN RRs at the dispersed domains, as illustrated in Figure 7-18.

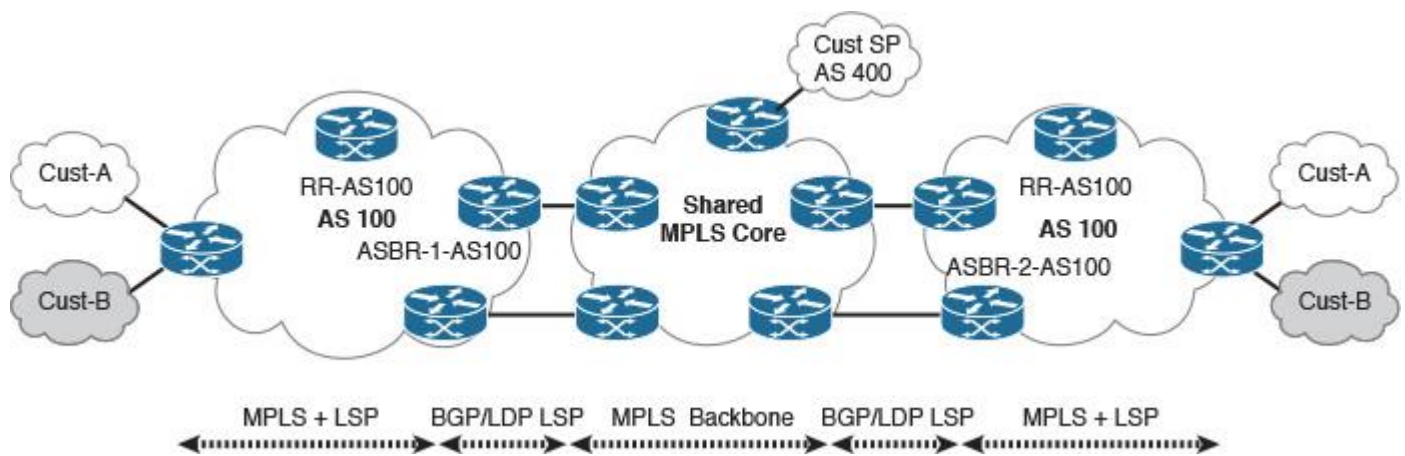


Figure 7-18 *MPLS VPN over MPLS Non-VPN Backbone*

One of the primary concerns about this model is the “none separated” control plane per customer across the backbone carrier network, because each customer PE may have visibility or reachability to other customer’s IPs. Therefore, this design model is more feasible for environments where the same network operator owns both the backbone MPLS network and the MPLS VPN islands.

MPLS VPN Customer over IP-Only Carrier

In some scenarios, dispersed MPLS-enabled networks have to interconnect over a native IP network. The following are common scenarios of interconnecting MPLS islands over native IP non-MPLS infrastructure:

- Enabling MPLS applications among multiple remote MPLS networks interconnected over a native IP core carrier infrastructure (where MPLS is enabled at the edge networks only)
- During migration and acquisition scenarios

Therefore, MPLS must be tunneled across the native IP core to extend and connect these MPLS islands. A common and simple method is to use MPLS over a manual tunneling/encapsulation mechanism. As shown in Figure 7-19, the edge PEs/ASBRs can set up a manual MPLS over Layer 2 Tunneling Protocol (L2TP), generic routing encapsulation (GRE) tunnel, or MPLS over IP. In all cases, the source and destination addresses are set to the addresses of the encapsulating and decapsulating label switching routers (LSRs) over the native IP core. The main difference between these options is that with GRE and L2TPv3, there will be additional encapsulation overhead for the packet header. In contrast, MPLS over IP is considered the easiest to spoof among other options; therefore, IPsec can be added as an optional protection mechanism if required (RFC 4023).

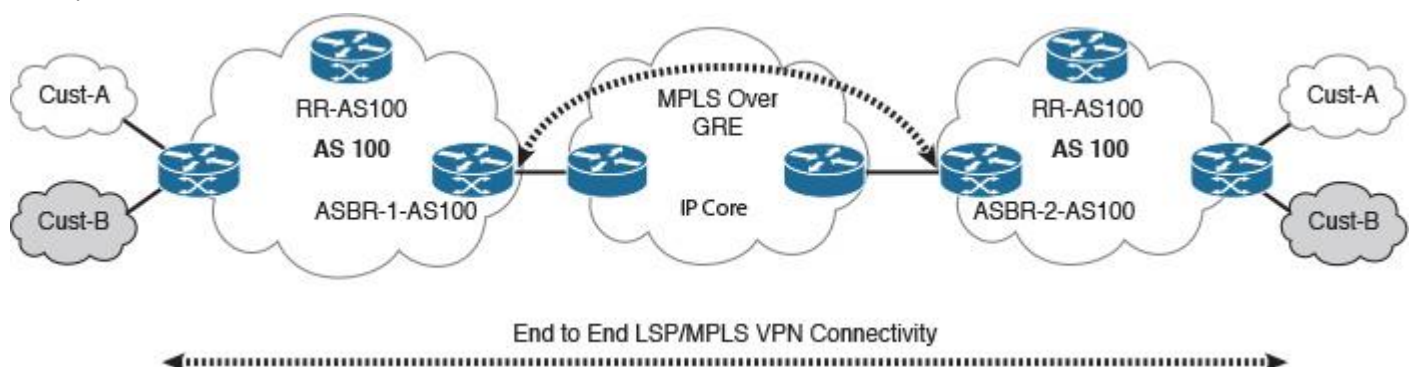


Figure 7-19 *MPLS-VPN over Native IP over Manual Overlay Tunneling*

However, in some scenarios, there might be multiple edge PEs/ASBRs that need to be interconnected over the native IP core, where the manual setup of the tunnels can add operational complexity. Therefore, instead of manually configuring tunnels, the tunnel reachability information can be sent via a BGP next-hop tunnel subsequent address family identifier (SAFI; described in draft-nalawade-kapoor-tunnel-safi-01.txt and RFC 5512). This option offers the capability of advertising which tunnel method is best to reach a given PE, such

as GRE, L2TPv3, IP in IP, or IPsec. In addition, a full-mesh overlaid topology can be created among the PE routers over native IP network, such as MPLS VPN over mGRE (0 tunnels to be deployed by the network operator), which allows network operators to provision MPLS VPN services without using LSP or LDP over the IP core. Technically, IPv4-based mGRE tunnels are used to encapsulate VPN-labeled IPv4 and IPv6 packets between dispersed PEs. Similarly, the L2TPv3 multipoint tunnel is another common mechanism that offers similar transport capability for MPLS VPN over native IP core in which only one tunnel per PE is required to be deployed, as shown in Figure 7-20.

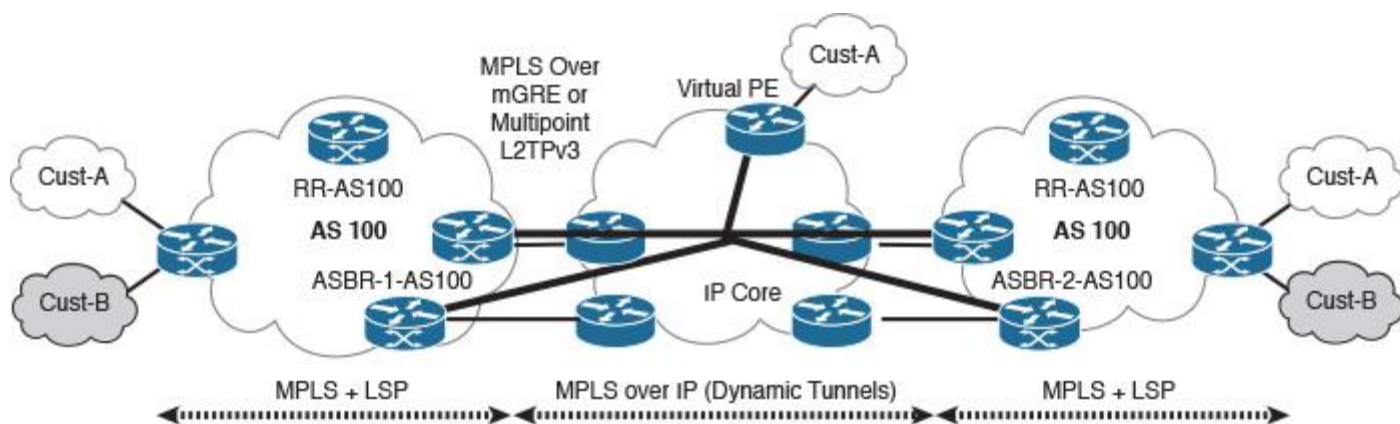


Figure 7-20 MPLS-VPN over Native IP over Dynamic Tunneling

Note

Considering staff knowledge/experience, hardware/software support, security considerations (such as encryption) and the number of nodes to be interconnected, can help to a large extent to select one option over others.

Note

IP-only backbone carriers can consider enabling BGP/MPLS IP-VPN, (specified in RFC 4364), also known as *virtual private routed network (VPRN)*, which enables IP backbone carriers to retain their native IP backbone and provide IP VPN-like services for their transit customers. With this method, each individual customer's VPN is routed to the respective PE of each VPN instance, while maintaining its own routing table space isolated, so that customers with overlapping prefixes can be transported over the same shared infrastructure. Technically, VPRN uses MPLS labels to identify paths between ingress and egress PEs, in addition the multiprotocol extensions is incorporated into BGP to facilitate carrying a route distinguisher (RD) along with each prefix when sending updates to ensure that you can use a single protocol instance to carry different routes for different VPNs even if there is overlapping addressing between VPNs.

Note

The technologies and design options discussed in this section are similar to the ones discussed in the enterprise WAN virtualization section in Chapter 4, "Enterprise Edge Architecture Design." However, the enterprise WAN virtualization was focused on how to connect the main or hub site with the remote sites. In this subsection, though, the selection of the suitable option is from the service provider point of view (transit network). In other words, the technologies and principles are all the same; however, the selection criteria might differ from a design principles point of view.

Table 7-3 summarizes the characteristics of each CSC model discussed in this chapter by highlighting the main advantages and design considerations of each.

CSC model	Advantages	Design Considerations
Non-MPLS customer over MPLS VPN carrier	Flexible and scalable B-Carrier option. C-Carrier will have more flexibility to exchange large number of prefixes B-Carrier does not need to install and process external customer prefixes. Offers transport separation (VRF/VPN) per C-Carrier across the B-Carrier network.	Requires MPLS between the C-Carrier and B-Carrier. May requires mapping between MPLS EXP and DSCP QoS marking at the C-Carrier edge to the B-Carrier. To extend IPv6 between C-Carrier Islands, you can use 6PE.
MPLS customer over MPLS VPN carrier	Same as the above (only difference is the customer is MPLS enabled).	Requires MPLS between the C-Carrier and B-Carrier. Supports seamless QoS design (MPLS EXP based). To extend IPv6 between C-Carrier Islands, you can use 6PE.
MPLS VPN customer over MPLS VPN carrier	Flexible, scalable B-Carrier option, supports expanding MPLS VPN services. B-Carrier does not need to install and process external prefixes of the C-Carrier. Offers transport separation (VRF/VPN) per C-Carrier across the B-Carrier network.	Requires MPLS between the C-Carrier and B-Carrier. Supports seamless QoS design (MPLS EXP based). To extend IPv6 between C-Carrier Islands, you can use 6VPE.

MPLS VPN customer over MPLS carrier	Simple, scalable B-Carrier option, supports expanding MPLS VPN services. With MPLS in the core, the B-Carrier does not need to install or process external customer prefixes.	Requires MPLS between the C-Carrier and B-Carrier. Supports seamless QoS design (MPLS EXP based). To extend IPv6 between C-Carrier islands, you can use 6VPE. Does not provide isolation between the different customers/tenants across the backbone.*
MPLS VPN Customer over IP-only carrier	Simple and basic native IP routing across the B-Carrier.	Has the least scalability and flexibility among other options. May requires each customer to build their own overlay to extend their routing or MPLS VPN services. May requires mapping between MPLS EXP and DSCP QoS marking at the C-Carrier edge to the B-Carrier. Does not provide isolation between the different customers/tenants across the backbone.**

C-Carrier = customer carrier, B-Carrier = backbone carrier

* Apart from networks migration and merger scenarios, it is not practically feasible to have MPLS core only without enabling L2/L3VPN and taking advantage of its benefits.

** If BGP/MPLS IP-VPN defined in RFC 4364 is deployed by the B-Carrier, in this case the (non-MPLS) IP backbone will offer a scalable solution that can support the MPLS IP L3VPN connectivity model.

Table 7-3 CSC Connectivity Options Comparison

Acquisition of an MPLS-L3VPN Service Provider Design Scenario

This design scenario focuses on the common L3VPN interconnection design options used to provide interprovider connectivity. Carrier Supporting Carrier (CSC) presented in this scenario is one of the common connectivity models offered by today's global MPLS providers. In addition, the inter-AS connectivity model is incorporated into this scenario to provide the desired connectivity solution based on the different design requirements.

Background Information

ABC Corp. is an MPLS L3VPN service provider located in the United States. ABC Corp. is currently facing several scalability and integration issues with other global MPLS VPN providers, due to the increased number of customers who require extending their L3VPN connectivity to the UK and across Europe. Consequently, ABC Corp. decided to acquire XYZ Corp., an MPLS L3VPN service provider located in the UK with several POPs distributed across Europe. This will help ABC Corp. to overcome their current issues and expand their business.

Design Requirements

- ABC Corp. requires the network merger project with XYZ Corp. to be completed as quickly as possible.
- Both ABC Corp. and XYZ Corp. must maintain their current BGP AS numbers.
- The security team within ABC Corp. dictated that this network integration should use a connectivity model that does not share internal infrastructure IP addressing of both SP networks.
- The same SLAs must be offered across the two SP networks for those customers who are located within the United States and have remote sites located in the UK or Europe that require end-to-end reliability and service consistency.
- Because of the increased number of customers who need to extend their L3VPN service, the proposed solution must be scalable enough to satisfy this increased demand.
- A secondary path is required to provide load sharing and fault tolerance.
- Cost should be considered as a tiebreaker only if there are two connectivity options that both meet the primary requirements.
- Customers who have an SLA with expedited forwarding for their delay-sensitive applications, such as VoIP, should have their traffic forwarded over the low-latency path as the primary path.

Figure 7-21 summarizes the requirements on a high level.

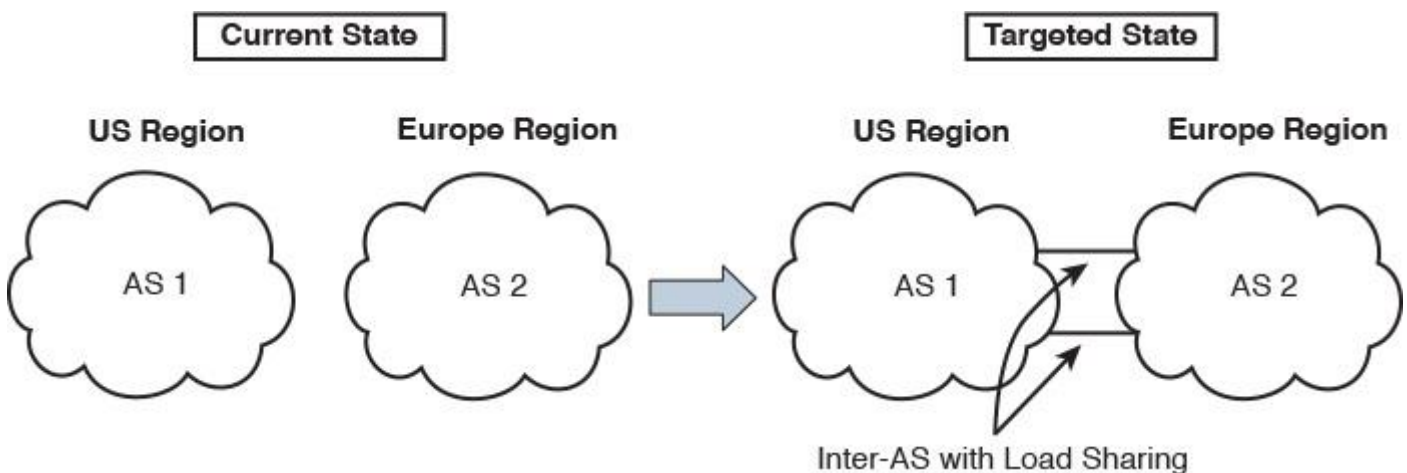


Figure 7-21 Network Merger Connectivity Requirements

Available Interconnection Options

The following are the available (offered) connectivity options for ABC Corp. for their interprovider integration project:

- **Option 1, direct point-to-point Layer 2 link provided by an international carrier Ethernet provider (L2VPN):** This high-speed (10-Gbps) link can be provisioned quickly with high service reliability and is cost-effective.
- **Option 2, carrier supporting carrier:** This high-speed (5-Gbps) option is provided by a global MPLS VPN provider and its cost is higher than Option 1. In addition, this option has good level of service reliability;

however, this link has higher latency compared to Option 1 and may require a longer time for the service and link provisioning.

■ **Option 3, carrier supporting carrier:** This option is offered by a European MPLS VPN provider that has few POPs in the United States. The offered connectivity is over high speed, 5 Gbps, but because this provider does not offer any QoS model, it has higher latency compared to Option 1 and 2. The pricing in euro is considered a bit higher for this link than other options.

■ **Option 4, overlay tunneling over the Internet:** This option offers 100 Mbps of bandwidth and is considered the cheapest and quickest in terms of provisioning and implementation. However, there might be some concerns about service reliability, security, and end-to-end latency (lack of end-to-end QoS).

Note
Both Option 1 and 2 support end-to-end six-class CoS (QoS model).

Inter-AS Connectivity Model Selection

In scenarios like this where there are multiple options with several design requirements, you should create a decision matrix, using the business-driven selection criteria depicted in Figure 7-22, to simplify the optimal design decision.

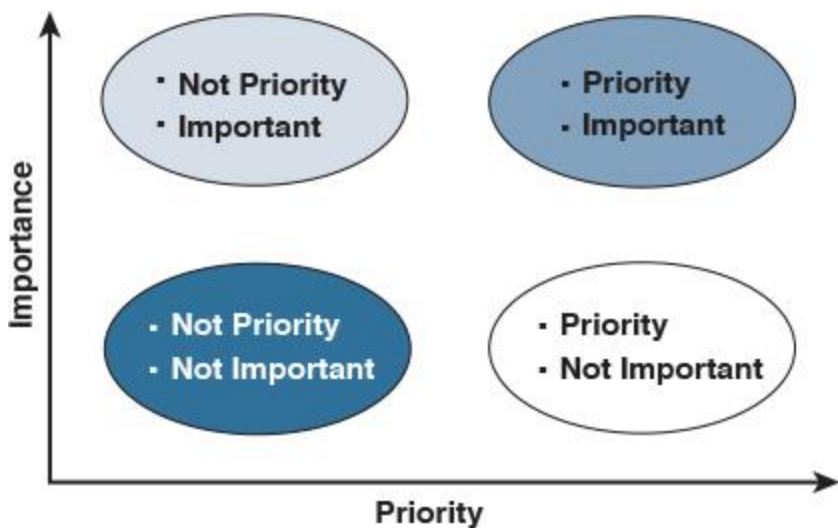


Figure 7-22 Business-Driven Selection Criteria

Table 7-3 maps each of the technical requirements to the relevant selection criteria shown in Figure 7-16 in a decision matrix style.

	Priority: Important	Priority: Not Important	Not Priority: Important	Not Priority: Not important
Reliability	X			
Scalability	X			
Security			X	
Encryption				X
Cost			X	
Service provisioning time*		X		
Consistent QoS	X			
Variety of access media				X
High capacity	X			

* Based on the design requirements, the service provisioning time is a priority (to complete the project as soon as possible), but it is not important to the business, nor will the business pay a penalty or lose customers if some delay occurs in the service provisioning.

Table 7-3 *Interprovider Connectivity Model Selection Decision Matrix*

Based on ABC Corp. requirements and the decision matrix listed in the Table 7-3, the two possible solutions that you should consider are Option 1 and 2. Option 3 cannot be considered because its lack of QoS support and the fact that it is not cheap. Options 4 has several design limitations and concerns; therefore, it will not be considered even though it is the quickest in terms of service provisioning.

Proposed Solution

Based on the selected interconnect options and the security requirements. inter-AS Option B with next-hop-self will be used in this design to establish the provider interconnect between the two SP networks over a 10-Gbps point-to-point Layer 2 Ethernet link (over L2VPN).

Note

Inter-AS Option A was not selected because scalability is a “priority and important” requirement.

In addition, the secondary link will be established over a global MPLS VPN provider using the CSC connectivity model (using inter-AS Option B as well) to establish the second inter-AS path between AS 1 and AS 2 with maximum bandwidth capacity of 5 Gbps. For the purpose of the traffic load sharing and service differentiation requirement, customers will be divided across those two paths based on the offered SLA. For example, customers with strict SLA requirements (such as need low-latency connectivity for VoIP or video traffic) will be connected over the direct L2VPN interconnect as a primary path, while other customers will be using the interconnection over the CSC provider as the primary path. Thus, if any of these links fails, the remaining path will be used to carry traffic for all customers. So, traffic load and link capacity planning for situations like this have been considered. Technically, the distribution of customer traffic across these two paths will be achieved by altering BGP attributes (for simplicity, BGP community and local preference attributes will be considered), as illustrated in Figure 7-23.

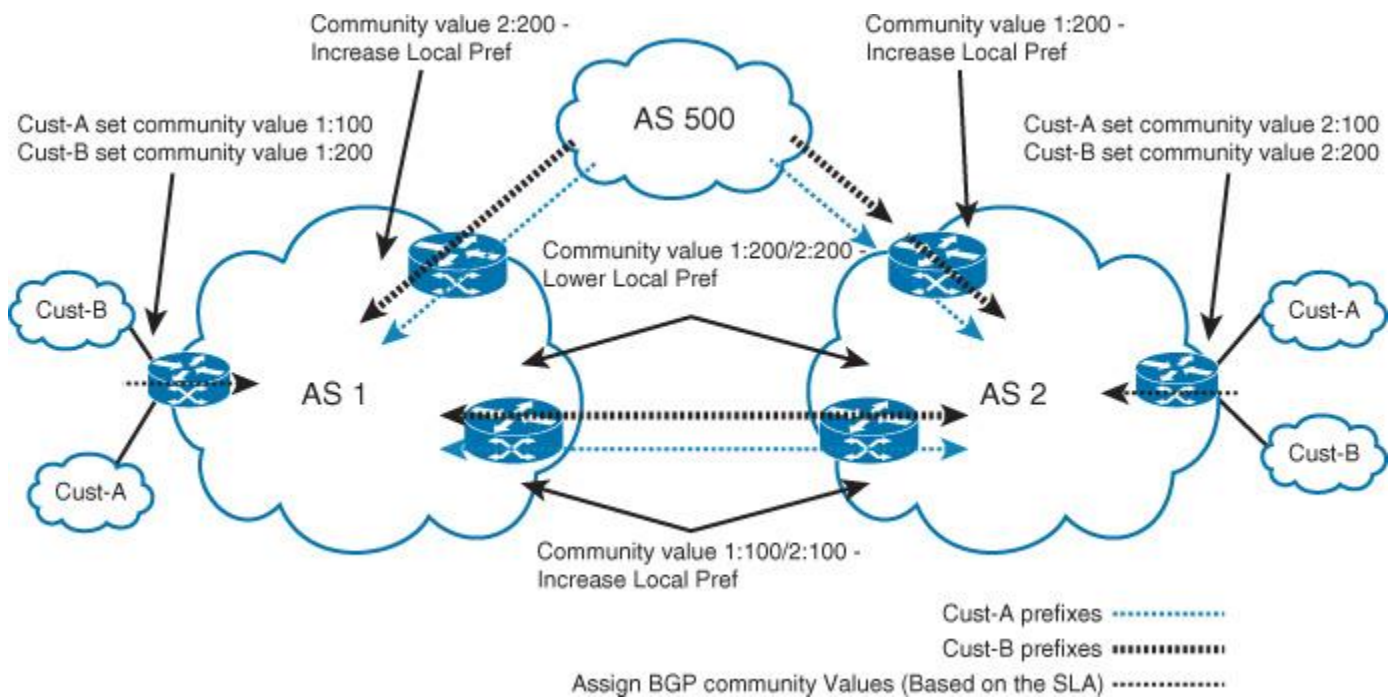


Figure 7-23 Influencing MP-BGP Path Selection over the Interprovider Links

Note

For simplicity, in this scenario customer A (Cust-A) represents customers with strict SLA, and customer B (Cust-B) represent customers that have no special requirements in their SLA such as VoIP traffic.

Figure 7-23 illustrates a logical view of the MP-BGP policies to control path selection over the interprovider links.

Furthermore, to provide a more scalable and manageable solution, BGP community values is considered in the scenario to tag traffic at each customer-facing PE with the relevant BGP community value, which can be used by the different ASBRs across both AS 1 and AS 2 to assign the routes with the desired local preference value based on its BGP community value (tag). This way, network operators do not need to maintain a large number of prefix lists at the ASBRs that contain all the customers' prefixes (because this may results in a high potential for operational complexity and errors that ultimately will result in customer dissatisfaction).

The end result of these BGP policies must provide traffic load distribution as illustrated in Figure 7-24.

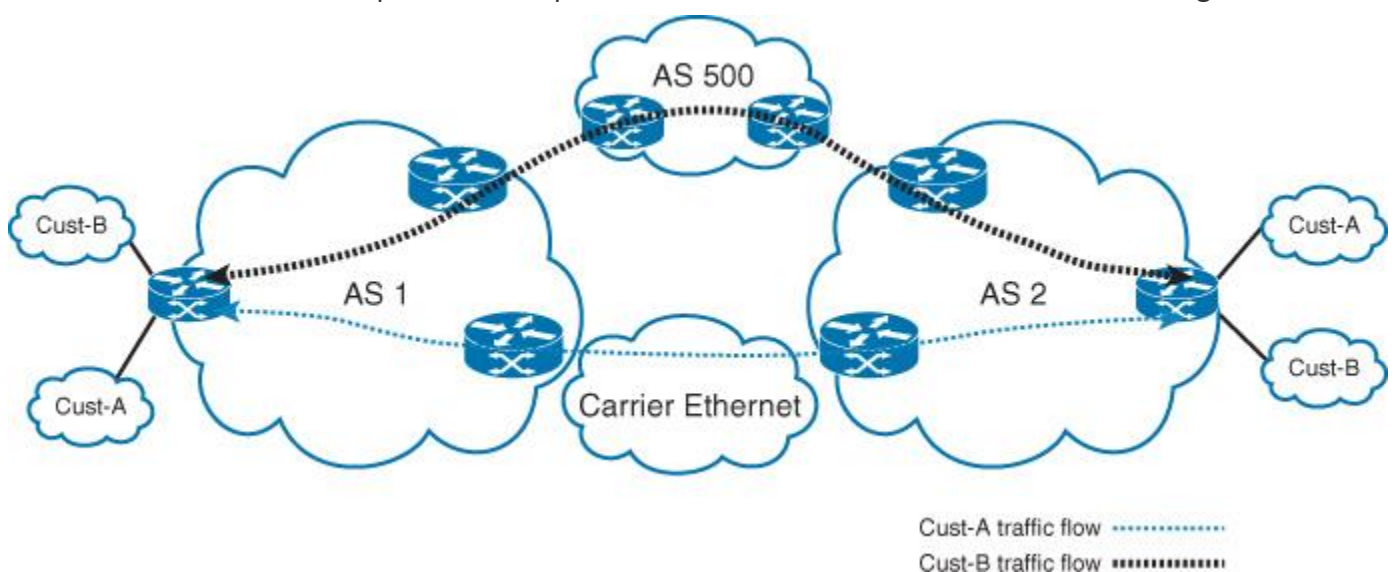


Figure 7-24 Traffic Flow Distribution Across Different Interprovider Paths

Network Merger implementation Plan

Figure 7-25 lists the required steps to move ABC Corp. from the current state to the targeted state discussed earlier.

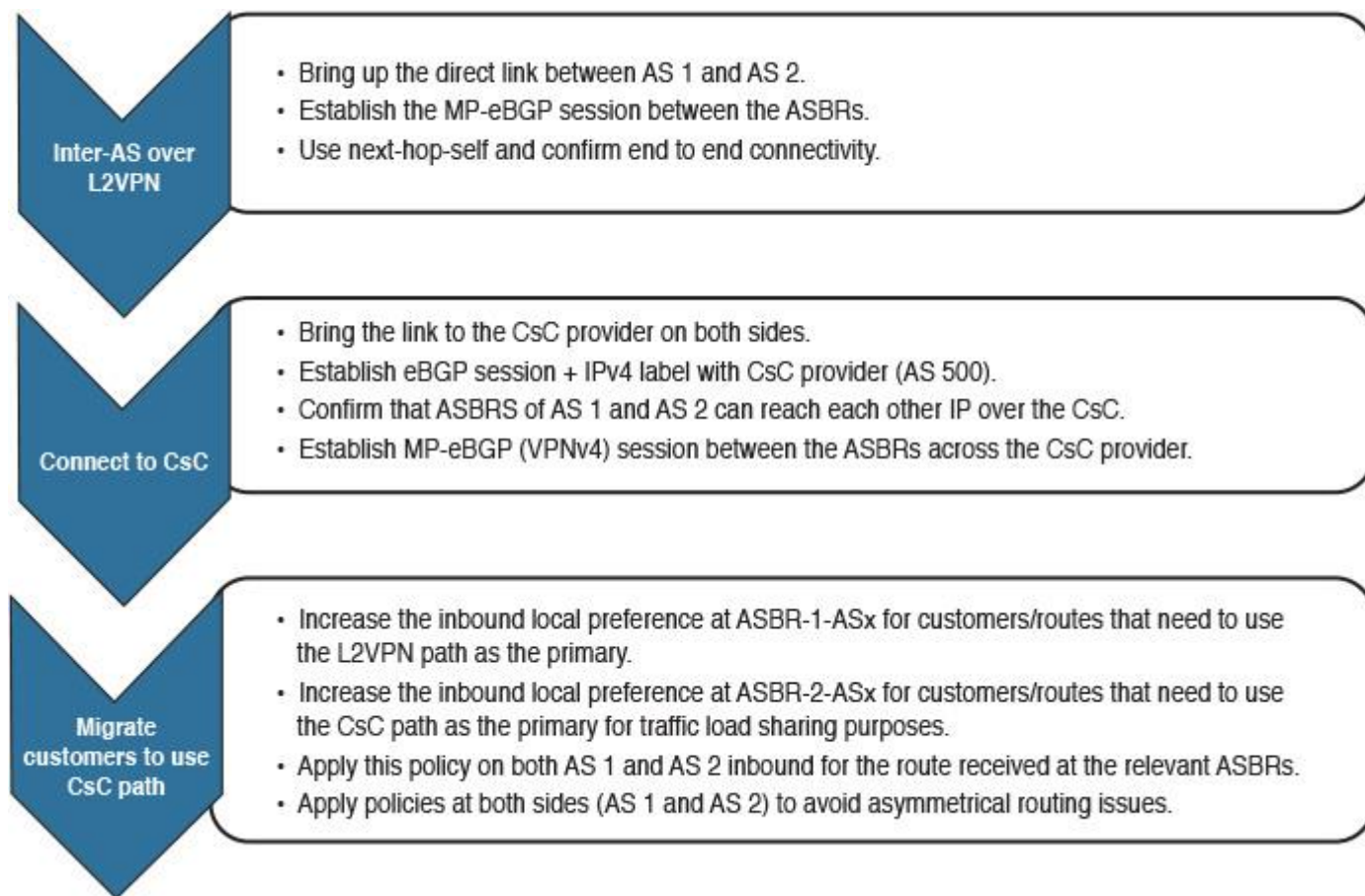


Figure 7-25 Implementation Plan to Activate the Interprovider Integrations

Summary

Interconnecting multiple autonomous systems is a common design practice used between service providers and by large global enterprise networks. As discussed in this chapter, there are various connectivity options and models. Therefore, selecting the right design connectivity model is critical, because all the following communications between the interconnected autonomous systems will flow over this interconnect. For example, interconnecting two autonomous systems that belong to the same service provider is different from connecting two autonomous systems that belong to two different service providers. Technically, both examples can use the exact same connectivity option; however, other aspects drive the design choice, such as security considerations. Besides, although inter-AS and CSC work on the same logic, each model has different characteristics and targets different design goals, as summarized in Table 7-4.

	CSC	Inter-AS
Topology model	Hub and spoke.	Point to point.
Typical SP POP connectivity	POPs of the backbone carrier (the carrier in the middle) can be interconnected over MPLS VPN, MPLS only, or Native IP.	Typically POPs of a given SP are not available in the geographic area required by their customers.
Typical customers	ISPs, MPLS VPN providers, and large-scale enterprises.	Any two autonomous systems need to interconnect, commonly with MPLS VPN services.
MPLS L3VPN support	Customer (child) provider network may or may not have MPLS enabled. Backbone carrier (parent) network can be based on MPLS VPN, MPLS only, or native IP.	Participating providers must support MPLS VPNs (either L2VPN or L3VPN).
MPLS L2VPN support	With this connectivity model, the backbone carrier that provides L2VPN transport will not participate in any label distribution or Layer 3 routing with the customer provider network (see the “L2VPN” section in Chapter 6).	L2VPN is one of the common extended services between the interconnected providers’ networks using same L3VPN inter-AS connectivity options
Route distribution	Internal routes and BGP next-hop reachability information can be shared with the backbone carrier, but external routes are not shared with the backbone carrier.	Depending on the inter-AS model used (for example, A, B, C, or D).

Table 7-4 *Inter-AS Versus CSC*

Part IV: Data Center Networks Design

Chapter 8 Data Center Network Design

Chapter 8. Data Center Networks Design

The data center network (DCN) is the primary element interconnecting the various components that all collectively comprise a data center. These components include the following computing, application, storage, and security nodes and services, where scalability, resiliency, security, and performance are the most common fundamental considerations of a successful DCN design.

This brief definition of a DCN emphasizes the important role of the data center's network infrastructure. In fact, all the communications between applications and services into and out of the data center are steered and carried by the network. As a result, any network availability or capacity issue within the underlying network infrastructure will impact the overall data center responsiveness, even if you have the best network overlay, servers, and application designs in the world. It is exactly like a building with a weak foundation; it cannot scale and be developed reliably, even if it has very promising architecture. For this reason, the data center's design and architecture must be business driven. All the business's requirements (its vision, continuity, and service priorities) should be taken into account when architecting a data center network, regardless of whether it is an enterprise data center or a service provider data center.

From a business point of view, the data center is the place where the business stores its data and applications; the data center houses the most critical assets of today's businesses. Sometimes, the data center even represents the core revenue source of the business, such as hosting data center providers (like software as a service [SaaS] and infrastructure as a service [IaaS] providers). Therefore, the design of a data center ideally should always offer a flexible and reliable design capable of accommodating the evolutions of business requirements and market trends. In addition, one unique and challenging design aspect of DCNs is that to produce a successful business-driven DCN design, you must connect with multiple players within the organization. These players usually belong to different teams, such as the server team, the storage team, the applications team, and the security team. Ideally, the input from these teams should drive the overall data center design, including the network.

This section starts by covering the traditional way of designing DCNs. It then covers the drivers and design considerations of the next generation of data centers, and how yesterday's data centers are not architected to meet today's modern data center requirements. This chapter also discusses the design options and challenges of modern data center interconnect solutions.

Note

This chapter, like other chapters in this book, covers the various architectures, protocols, and technologies used by traditional and today's data centers regardless of whether they are not recommended or whether they are commonly used and highly recommended. After all, the reality is that organizations may have considered some of these technologies or design architectures during a time when their business and application requirements differed from today's requirements. In addition, technologies are always in evolving state; this implies that what was considered a best practice or fit for any given design yesterday will probably not be the case for today's data center design. Consequently, network designers must be aware of the various design options and technologies and have a good understanding of the strengths and weaknesses of each to be able to evaluate current designs and to suggest design

optimization or a complete redesign (always taking into account, of course, the different design requirements, starting from business and application requirements all the way up to the technical traffic flow and routing).

Traditional Data Center Network Architecture

During the past decade, the classic three-tier network architecture has been the predominant network architecture for mid- and large-scale data centers, because this multitier siloed approach offered the desired level of design flexibility, availability, and performance for the various application and server requirements at that time. The overall architecture of a DCN, at a high level, is built on the same concept as the hierarchal network architecture discussed earlier in the enterprise campus section in this book, which usually consists of the core, distribution, and access layers. A common question with regard to the classic three-tier design is this: How does a three-tier DCN design differ from a typical three-tier campus LAN design? Although these two designs follow similar concepts and architecture (for example, they both employ the concept of hierarchal network architecture), they are not identical in their detailed design requirements. This is summarized in Figure 8-1.

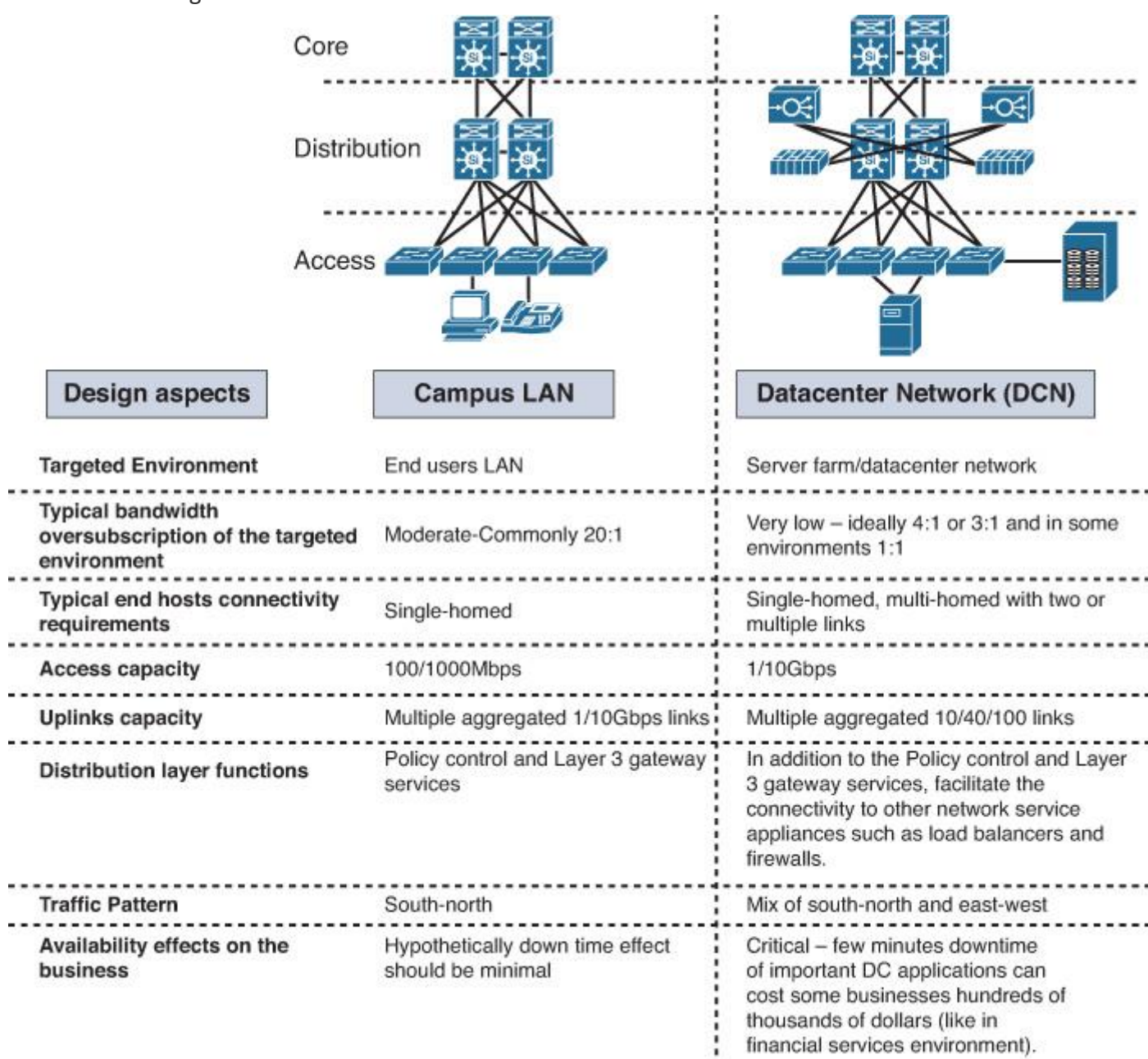


Figure 8-1 Campus LAN Versus DCN

STP-Based Data Center Network Architecture

As discussed earlier, there is not a significant difference between the overall architecture of a classic three-tier DCN and the campus design (at a high level). Therefore, all the Spanning Tree Protocol (STP) and Layer 2 considerations discussed in Chapter 2, “Enterprise Layer 2 and Layer 3 Design,” apply here. The only crucial difference is that in a DCN you need to consider the deployment model and failure scenarios for the services provisioned at the data center distribution layer. This is because, in most cases, these services (either service module or appliance based) work in active-standby or active-active modes. In these modes, Layer 2 communication is almost always required between these services across the upstream distribution layer nodes.

If the network, and Layer 2 in particular, was not designed to properly take the connected services to the distribution layer nodes into consideration, the network will most likely face a dual-active or split-brain situation during some failure scenarios. This usually occurs when two service appliances (such as firewalls), operating in active-standby mode, lose communication with each other. As a result, each device thinks that it is the only active device. (The active firewall will continue in active mode, and the standby firewall will move from standby to active mode as well.) Both firewalls handling traffic independently (operating in dual-active mode) is highly undesirable and dangerous because the risk of asymmetrical connection failure is extremely high, as shown in Figure 8-2.

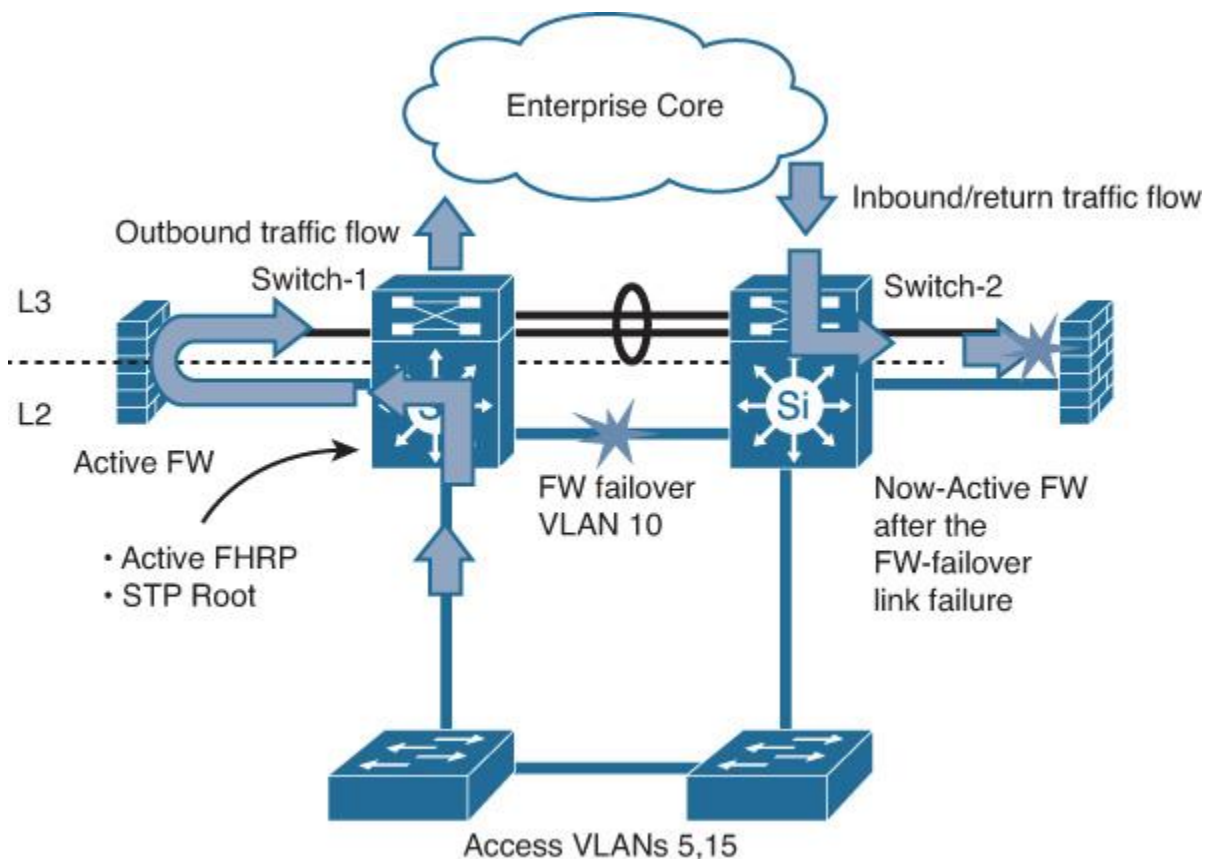


Figure 8-2 Asymmetrical Connection Failure Scenario

In Figure 8-2, a loop-free U topology is used. In case of L2 link failure between the two distribution switches that carry the data center firewall failover VLANs, both firewalls will be in active mode (split-brain). First Hop Redundancy Protocol (FHRP) active will continue to be on switch 1 because the heartbeat messages flow through the access switch, which means outbound traffic flows through distribution switch 1. If inbound traffic from the core arrives at the distribution switch 2 during this failure scenario, it attempts to flow through the now-active firewall. Traffic will be dropped because of this failure because the now-active firewall has no state information of this session. This issue can be temporarily fixed by enforcing all the inbound traffic to flow via the active FHRP switch. Nonetheless, the main reason behind this example is to highlight the different considerations when designing Layer 2 for a data center compared to a typical campus

LAN. Even though the overall topologies and features are similar to some extent, the functions and traffic flow are normally more complicated in DCNs.

As a general rule, to achieve the highest capacity in DCNs, always select the technologies and techniques that provide you with the maximum possible number of forwarding paths (taking into account that too much redundancy might not always be a good idea). Therefore, multichassis link aggregation architecture (mLAG) and Clos fabric architectures, from an L2 design point of view, are always preferred over traditional STP.

Note

The influence of introducing network service security appliances, such as firewalls and load balancers, into the data center aggregation layer is critical. It must be seriously considered, along with all the possible failure scenarios, when contemplating an STP-based access-distribution Layer 2 topology. This is applicable for both services, which are either module or appliance based.

mLAG-Based Data Center Network Architecture

As discussed earlier in this book, the mLAG architecture keeps STP in place but overcomes its shortcomings by using mLAG functions with switch clustering enabled at the aggregation layer. Compared to an STP-based design, this architecture offers a more flexible and resilient DCN with significant performance and bandwidth capacity optimization. However, scalability (scale-out) is one of the main concerns with this architecture for large-scale data centers that are based on virtualized workloads. Typically, only two aggregation switches can be clustered to serve as one logical device for the downstream switches (access switches). Consequently, access switches connecting to different clustered upstream aggregation switches have to communicate via the Layer 3 core, which may break the requirements of seamless end-to-end Layer 2 VM mobility or distributed workloads across different points of delivery (PODs), as shown in Figure 8-3.

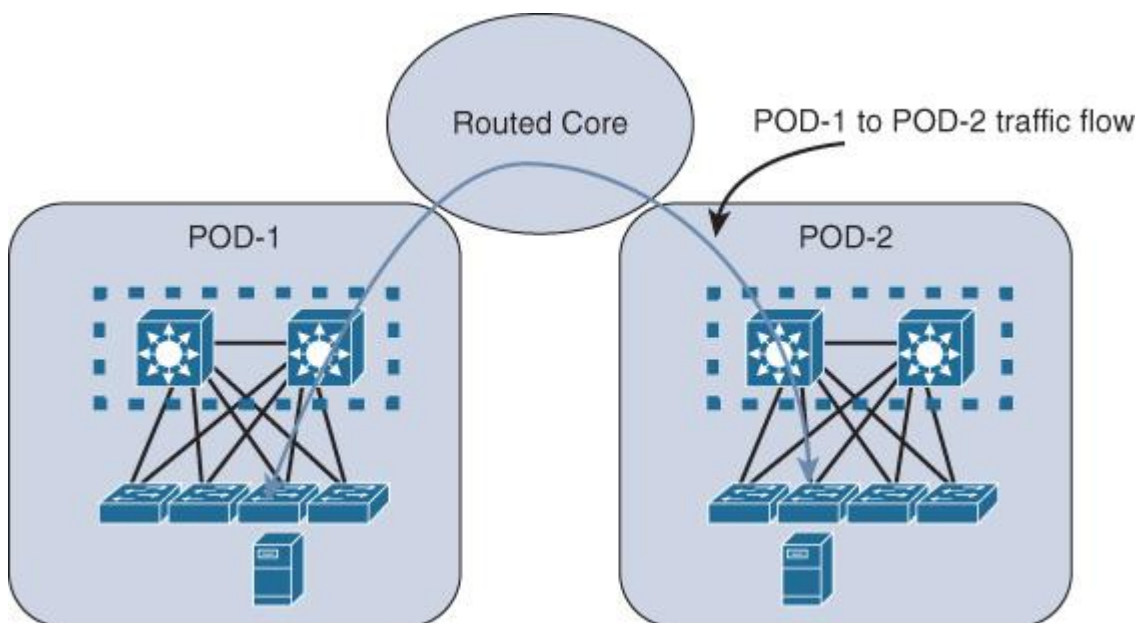


Figure 8-3 mLAG-Based DCN

Next-Generation Data Center Network Design

This section discusses the different design options and protocols used in today's modern DCNs and the drivers toward adopting these new architectures.

Data Center Virtualization and Cloud-Based Services Overview

In the world of servers, the term *virtualization* refers to the ability to run multiple operating systems and application instances on a single physical server. The abstraction layer of virtualization is usually provided by software (OS) known as a *virtual machine manager* (VMM), also commonly known as a *hypervisor*. The

VMMs/hypervisors provide the capability to control and manage multiple operating systems and emulate a dedicated environment to each server over one shared physical system using its hardware resources. As shown in Figure 8-4, the shared resources include CPU, memory, and network cards, which separate computing environments from the actual physical infrastructure. This approach has proven to reduce costs (capital expenditure [capex] and operating expenditure [opex]), and it enables IT and businesses to gain increased efficiency and flexibility from the existing computing hardware.

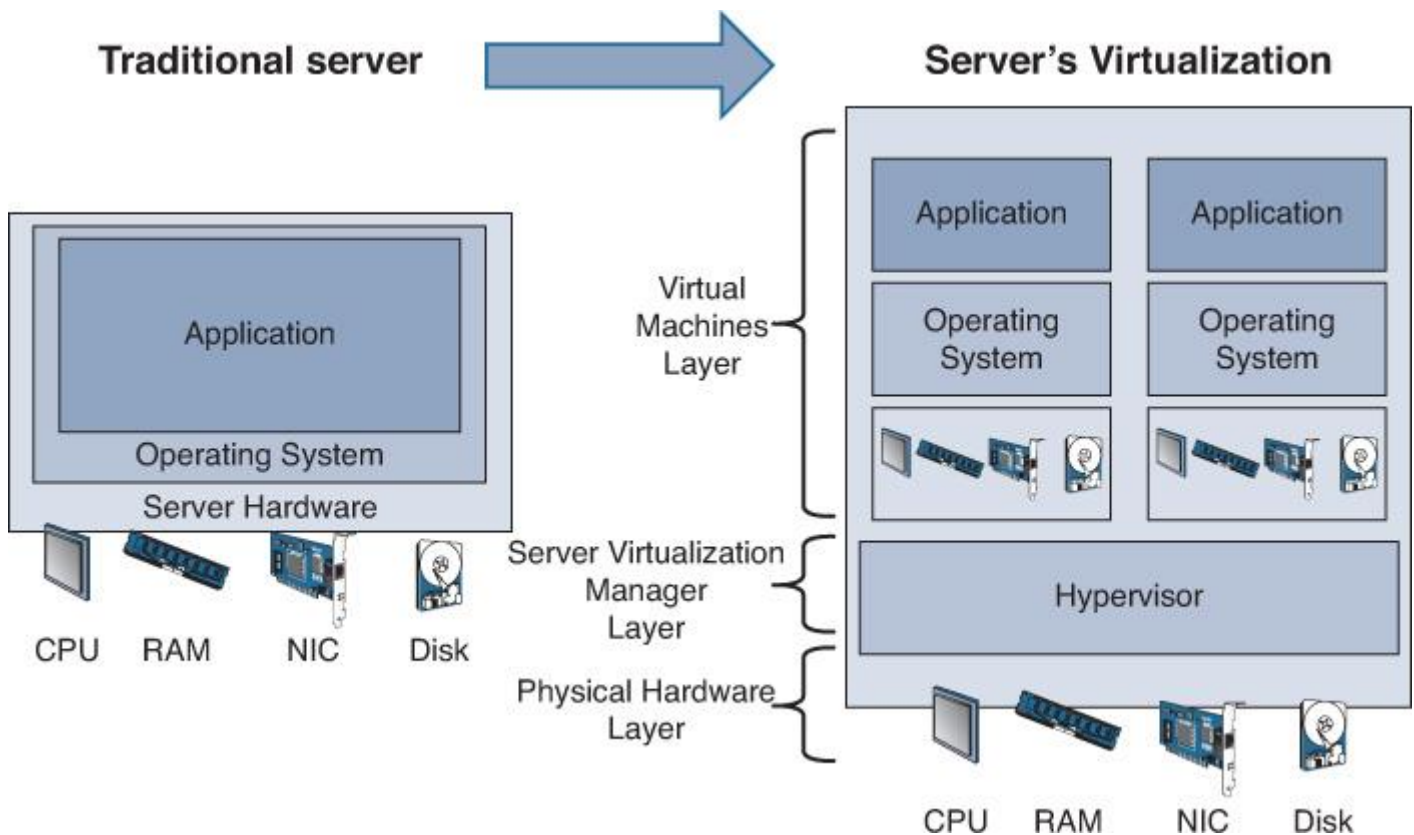


Figure 8-4 Server Virtualization

It is important for network architects and designers to understand the logical difference between the terms *change* and *transformation*. Change always takes the past as a foundation, and modifies it in a manner to make it more optimized; more scalable or more reliable. However, transformation refers to something new, such as designing a future goal or target, and then achieving it. Unlike change, transformation does not mainly rely on something that already exists, and then modifying it so as to make it more secure or reliable; instead, it's more about creating a concept that does not exist already (new)¹.

1. <http://theprimes.com/change-vs-transformation/>

Based on that, the concept of cloud or cloud computing in the world of data centers is a transformation in how data centers are designed and operated. In fact, it is a completely new era in the world of data centers. Technically, cloud computing further leverages the concept of server virtualization by adding an additional layer of virtualization and service abstraction. This offers on-demand network and service access to a shared pool of resources. These resources usually include network, storage, and computing resources, which ultimately enable multitenancy capabilities in the data center. Also *cloud services* incorporate self-service portals, automation, and orchestration to more effectively leverage these shared pools of resources.

Cloud computing also provides an X-as-a-S offering, such as IaaS and ITaaS. This is when scalable IT-related capabilities are provided “as a service,” which allows more elastic and on-demand service provisioning and access. Furthermore, cloud services can help to transform IT from being a capex model to an opex model, which will help organizations to better control costs. In sum, virtualization is fundamental to enabling cloud services in a data center. Virtualization generally takes the following forms:

- Server virtualization
- Network virtualization

- Storage virtualization
- Management virtualization and cloud orchestration

This chapter focuses on network virtualization and the new trends and architectures of next-generation data centers, which are propelling the change in DCN designs, approaches, and technologies.

That said, this does not mean that each and every single data center design needs to be transformed to a cloud-based DC model. Doing so depends entirely on the business goals, model, vision, cost (capex versus opex), and various other requirements (such as elastic business-continuity requirements), which in turn will help you to identify the current and anticipated future size of the organization and its DC network and understand mission-critical applications, the services provisioning model, traffic pattern, and so on. These collectively can help you to decide whether it is more feasible to “change” the existing DC design and architecture to a more optimized one or to “transform” it to a completely new DC model/architecture.

Drivers Toward New Fabric-Based Data Center Network Architectures

It is commonly known that the DCN sits at the core of the IT infrastructure. Therefore, it is crucial to understand how IT can deliver services and value back to the business. The new business demands and technology trends are changing the role of IT and introducing new challenges to the availability of applications; yesterday’s data centers were not designed to address these challenges, which include mobile and virtualized workloads, cloud applications, big data analytics, and dynamic policy provisioning. In this new environment, it is obvious that bandwidth demand is much higher and that the traffic pattern is unpredictable as compared to traditional data centers. Traditionally, the DCN is based on the classic three-tier hierarchy, as shown earlier in Figure 8-1.

Typically, the classic three-tier architecture introduces significant oversubscription and blocking architecture, which is unsuitable for the requirements of today’s workloads and application requirements. This is simply because the higher the traffic moving in the network hierarchy (bottom-top) of the multitiered network, the more bandwidth sharing there will be among the nodes in the lower tiers (bandwidth aggregation), as shown in Figure 8-5.

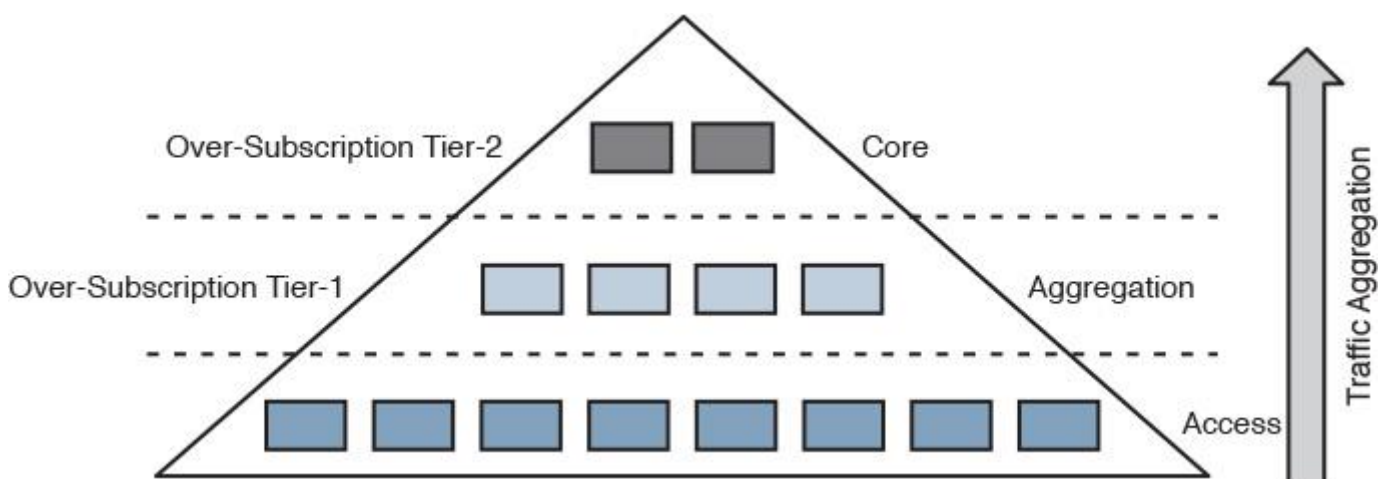


Figure 8-5 Traffic Aggregation

Note

Oversubscription and blocking in traditional architectures can be described as follows:

Oversubscription is when the ingress capacity exceeds the egress capacity. For example, for 48x 10G attached servers, the access switch needs at least 480 Gbps of port capacity toward the upstream or distribution layer to provide 1:1 oversubscription (1:1 here meaning zero or no subscription). However, if the access switch has only 2x 10G uplinks, in this case the oversubscription is 24:1. Although not every network is required to provide 1:1 performance, oversubscription needs to be minimized in modern large-scale DCNs (for example, ideally 4:1 or 3:1) [55]. *Blocking* is a situation, at a device level, caused by oversubscription, which can lead to network queuing. As a result, the overwhelmed node, such as a distribution switch handling inter-POD or rack traffic, will start blocking the new traffic flows. Ideally, during the planning phase, the network architect or designer should obtain enough information (such

as baseline conditions and application traffic patterns) before attempting to define any oversubscription standard or deciding whether a nonblocking DCN architecture is required.

Note

As a network designer, always ask for information that helps you to derive the DCN design decision based on the requirements of the systems and mission/business-critical applications hosted in the data center.

Note

The term *POD* (point of delivery) represents a standardized structured block constructed primarily of network, computing, and storage resources. In virtualized data centers, this concept serves the foundation of achieving standardized design modularity that can support high scale-out and scale-up heterogeneous workload requirements; this will ultimately help in minimizing the overall design and operational complexity.

A move from the traditional architecture to a more flexible, high-speed (nonblocking), low-latency, and agile network architecture was required to accommodate these evolutions and the transformation of modern data center requirements using fabric based infrastructure.

Fabric is defined by Gartner, Inc. as “a set of compute, storage, memory and I/O components joined through a fabric interconnect and the software to configure and manage them.”²

2. <http://www.gartner.com/it-glossary/fabric-computing>

The new switch fabric concepts and architectures came into the picture to meet today’s data centers trends and requirements, which primarily derived from the following factors:

■ **Traffic pattern:** Server virtualization and cloud-based services are a game changer in data center design concepts and architectures, in particular the traffic flow within a data center and between interconnected data centers. Traditionally, traffic flow was mainly north-south, sourced from internal or external users to the application hosted in the data center. In contrast, modern data centers are moving toward a more distributed workload and workload mobility, taking advantage of the vitalization concepts and distributed virtual machines (VMs) across one or multiple data centers between hosts, as shown in Figure 8-6. Therefore, the traffic pattern is becoming more of a combination of east-west (server-to-server) and north-south (client-server), which makes it unpredictable. As a result, the traditional three-tier DCN design is becoming more of a limiting factor for the new server and application traffic requirements (increased east-west traffic flow).

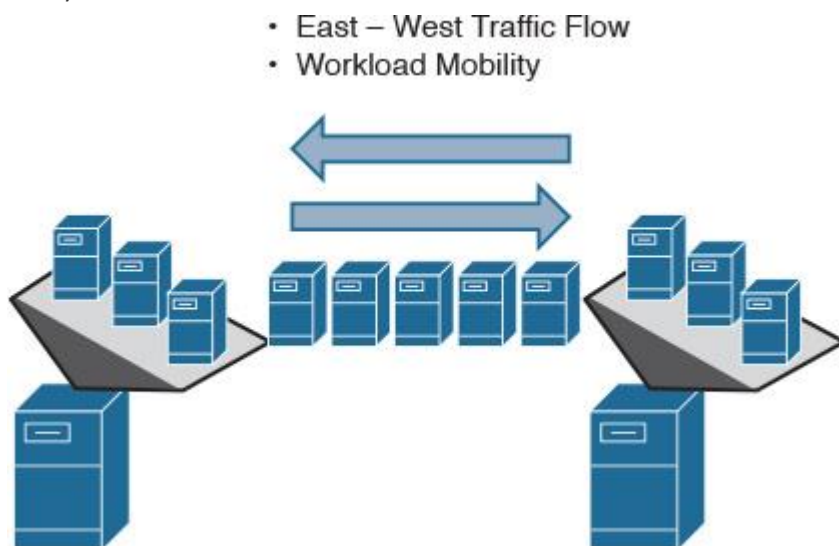


Figure 8-6 Workload Mobility

■ **Cost:** Because of the limitations of the classic multitier architecture, it will be necessary to have a larger number of switches with a larger number of ports (scale up) to handle the increase in the data center size. This will lead to extra cost (capex) and a more complicated design, which will increase the operational cost (opex) as well.

■ **Reliable scalability:** Server virtualization and VM mobility increase demand on the network infrastructure to support the ability to scale-out, considering the extremely increased number of virtual hosts compared to traditional data centers. Take, for example, an access switch that has 16 physical hosts connected to it. Traditionally, there will be 16 servers to handle; with the server virtualization concept, if these hosts have 10 VMs each, this access switch needs to handle 160 servers. Typically, these 16 physical hosts, if they used to be connected over a 1G port, now must each be connected over a 10G access port. In other words, this tremendous increase in the number of hosts (virtual) and traffic volume needs a new architecture that is built to scale without compromising bandwidth efficiency.

■ **High flexibility:** The emergence of virtualized and cloud-based data centers requires networks with architecture that supports simple and flexible automated provisioning, such as application programming interface (API) integration with cloud orchestration systems.

■ **Ultra-low-latency network:** Today, some environments require high-frequency trading, such as in financial services (where thousands of trading transactions are expected to be performed in a few seconds), which can be a big challenge when traditional data center platforms and design are used. Consequently, to meet the requirements of this type of environment, you need a network architecture that can support ultra-low-latency forwarding and a nonblocking switch fabric architecture that can forward packets in full size within microseconds. These are required to satisfy the needs of this type of environment, in which the extremely low-latency, high-performance, high-bandwidth and reliable transport form the foundation of today's high-frequency trading DCN

Modern data center architectures are moving from the classic three-tier architecture toward flatter fabric-based architectures, which are specifically built and architected for the new requirements of the next-generation data centers and to overcome all the limitations discussed earlier. Moreover, the any-to-any, nonblocking fabric architecture enables IT to align with the business needs and priorities by delivering tangible outcomes. These outcomes include a reduction in the number of full server I/O adapters (support of server virtualization and unified I/O), significant enhancement of the reliable scalability of the network, and the simplification of cabling. Furthermore, power and cooling costs are lowered by the elimination of unnecessary switching infrastructure used by the traditional data center architecture, and overall operational cost is also reduced.

Modern Data Center Network Architectures and Overlays

As discussed earlier, the traditional three-tier DCN design has many limitations when used in a virtualized or cloud-based data center with workload mobility, high-bandwidth capacity, and fast forwarding requirements, simply because at each tier in the network, the uplink bandwidth is distributed across the next-higher tier. Consequently, the more tiers traversed by traffic between servers (east-west), the greater the potential for bandwidth shortage (traffic aggregation) associated with increased latency and limited scale-out capabilities. Therefore, a new and more scalable network architecture, combined with a switch fabric architecture capable of providing any-to-any nonblocking connectivity, is required to overcome these limitations. Figure 8-7 shows the evolution of DCN architectures.

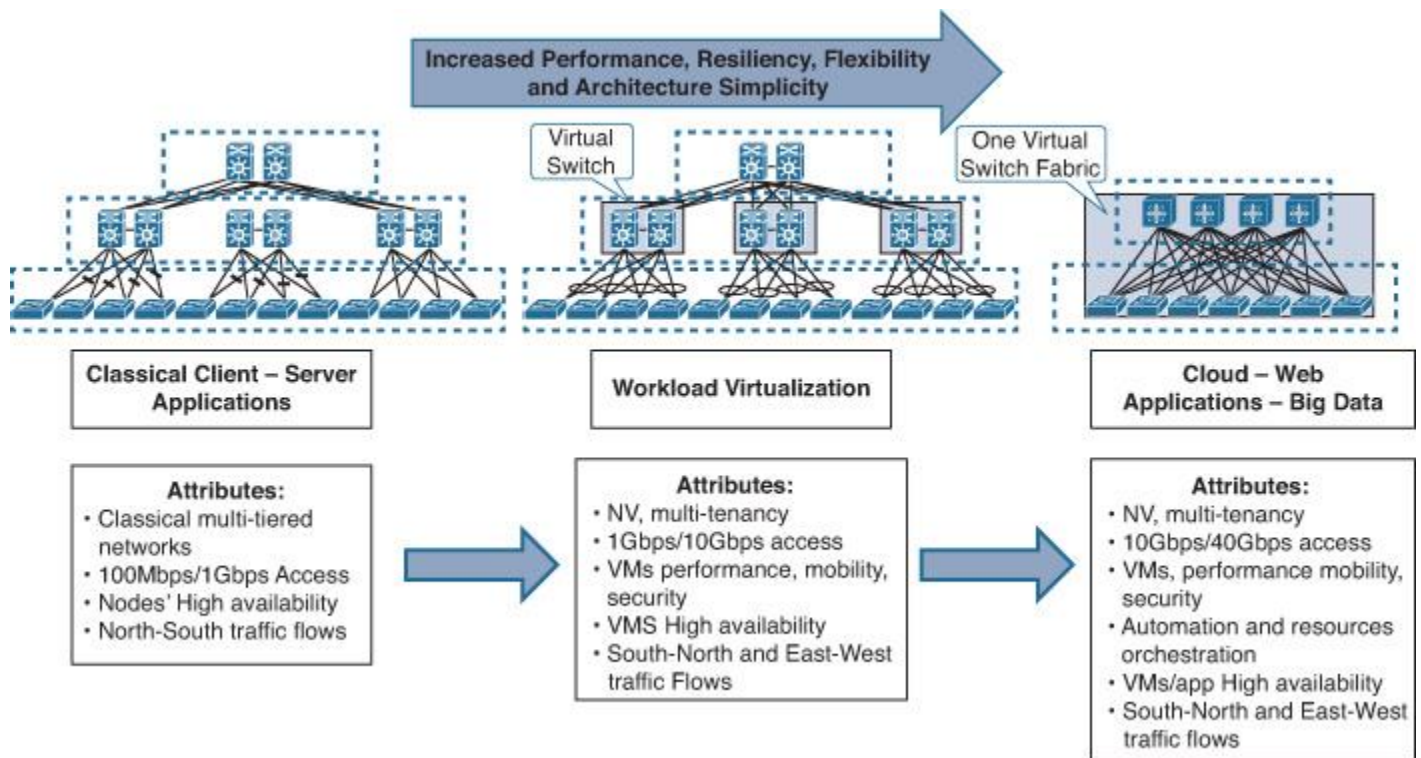


Figure 8-7 Evolution of Data Center Network Architectures

Consequently, modern data center fabrics should accelerate the deployment of applications and meet development and operations (DevOps) needs by

- Eliminating any dependencies on traditional protocols, such as STP
- Offering fast forwarding underlay for multidimensional traffic (high bisectional bandwidth)
- Providing optimized scalable forwarding (scaling forwarding tables and segments)
- Providing cloud integration and automated provisioning (such as API integration with cloud orchestrator systems)
- Providing virtualized networking to support mobility and virtual devices across the DCN
- Providing a reliable scale-out capability

At the time of this writing, there were a few modern data center fabric architectures, such as single virtual fabric and Clos architecture. This section focuses on the most common and proven architecture in today's data centers, which is the Clos architecture. Different design considerations and options are discussed, as are the drivers of each that help network designers select the right architecture to offer modern DCNs a higher degree of flexibility, reliability, scalability, and performance.

Clos Architecture

Clos architecture offers a nonblocking architecture based on multistage topology, which can maximize the overall network performance by increasing the available bandwidth to higher than what a single node can provide. The mathematical theory of this architecture was initially created by Charles Clos in 1953, hence the reason it is called Clos. In a nutshell, this architecture is based on associating each output with an input point. In other words, the number of outputs equals the number of inputs, or there is precisely one connection between nodes of one stage with those of another stage [55]. Figure 8-8 illustrates this architecture that offers zero oversubscription (1:1) where a 3-stage Clos topology is used to connect 18 host (edge ports) using 3x of 6 port switches in each stage.

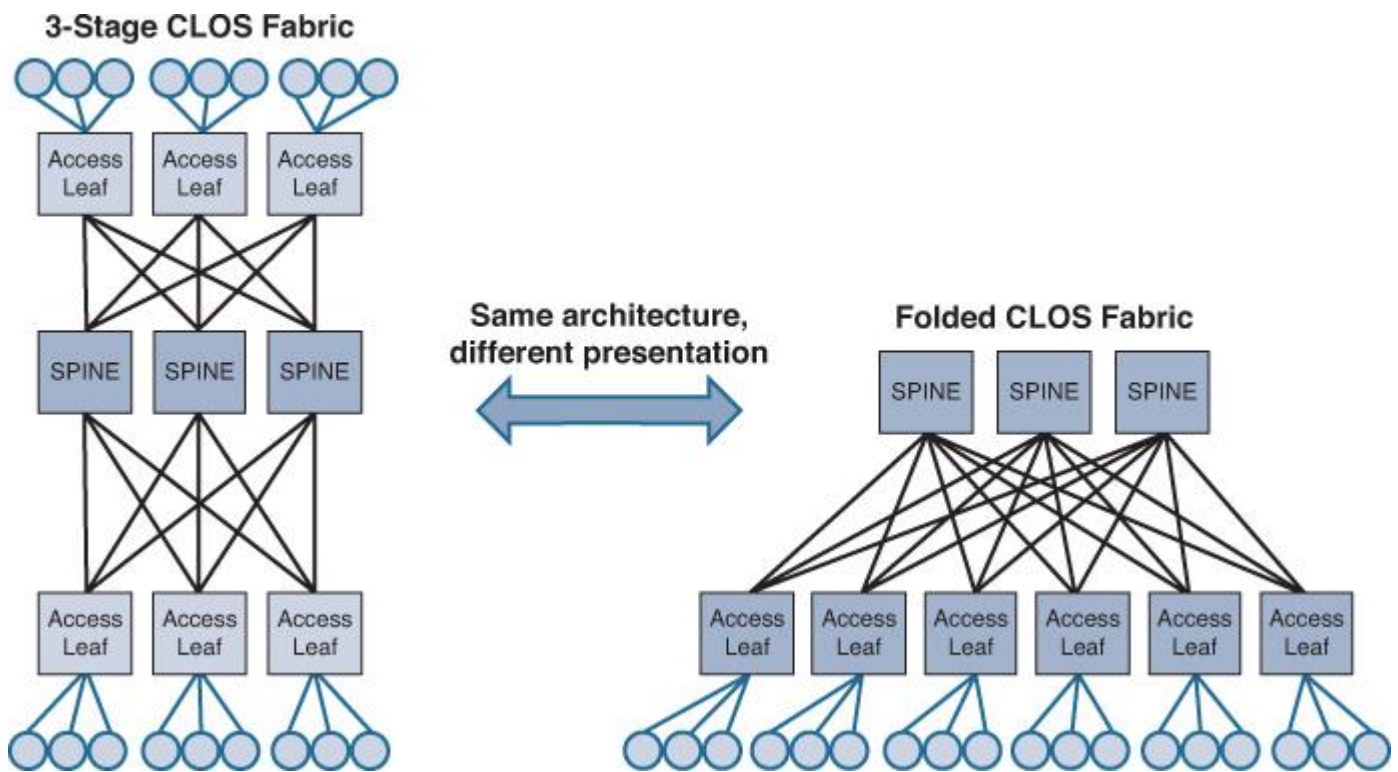


Figure 8-8 Clos Architecture

Note

Although the Clos architecture can offer scalable 1:1 (zero) oversubscription, this level of oversubscription is rarely required. Typically, 4:1 or 3:1 offers an efficient level of oversubscription with modern data centers that are built on 10/40/100G Ethernet. In fact, application and service requirements should drive the required level of oversubscription during the planning phase, taking into account the anticipated traffic load.

This architecture can be considered a two-tier hierarchy, in which the access layer switches are called leafs and the aggregation layers are called spines. With this two-tier leafs and spines architecture, each leaf switch is connected to every spine switch on the aggregation layer, which makes the DCN architecture flatten and eliminate bandwidth aggregation/oversubscription, because of bandwidth being the same at every tier. As a result, this architecture offers a high degree of network redundancy and a simplified design, and supports a large volume of bisectional traffic (east-west), as shown in Figure 8-9. Moreover, it can support multipathing, which can efficiently use network resources to support more leaf switches. Network designers can either consider scaling up, scaling out spine switches, or moving to a five-stage Clos topology for significantly larger scale DC networks.

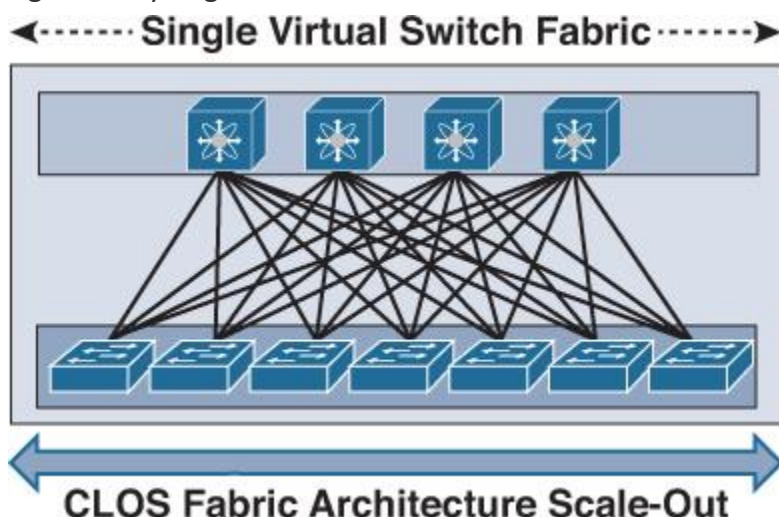


Figure 8-9 Clos Architecture Scale-Out

Unlike traditional data center architecture where Layer 3 gateway services and other network and security services are provided by the aggregation layer, in Clos (spines and leafs) architecture, the role of the spine is only to forward traffic between the leafs in the same manner that P routers work in a Multiprotocol Label Switching (MPLS) network. Furthermore, DC services and Layer 2 and Layer 3 demarcation points are moved in the architecture down to the leaf (also known as the *border leaf*), taking advantage of the high capacity and performance of the bisectional bandwidth of this architecture, as shown in Figure 8-10. As a result, additional spine switches can be introduced to the network to increase its scalability and reliability without any changes required to network and security services. In other words, unlike the traditional design, Clos is not tied to a pair of aggregation switches only. Therefore, Clos offers the optimal architecture for today's multitenant virtualized and cloud-based data centers.

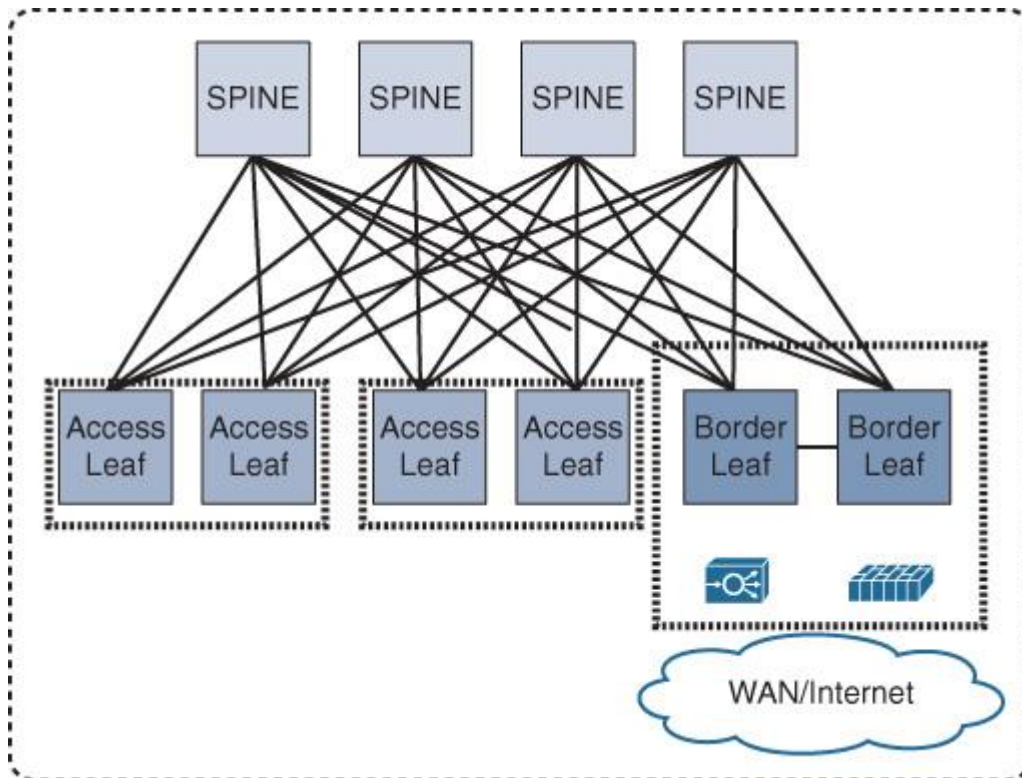


Figure 8-10 Clos: Border Leafs

Note

The definition of a *tenant* varies based on the targeted business and the data center environment. For example, in an enterprise environment, the tenants can be the different departments or business units (like in ITaaS environment), whereas in a cloud-hosting service provider, the most typical tenants are the external customers.

Clos Transport Protocols

To provide any-to-any nonblocking seamless connectivity and eliminate the reliance on STP, Clos fabric architectures rely on transport protocols that offer the ability to use all the available links between any two leaf switches over the intermediate spine nodes (equal-cost multipath routing [ECMP]).

Overlay Transport

In fact, network overlays are the representations of virtual networks of interconnected nodes that share an underlying physical network, allowing deployment of applications that require specific network topologies without the need to modify the underlying network. Ultimately, this can offer to the Clos fabric design the optimal level of flexibility, scalability, and performance required by today's modern data centers.

At the time of this writing, the following are the primary protocols considered by the majority of modern data centers to offer scalable and flexible ECMP forwarding over the Clos fabric architecture:

- MAC-in-MAC

- MAC-in-IP
- MPLS based

MAC-in-MAC

As mentioned earlier in this chapter, a primary goal of the Clos architecture is to eliminate any reliance on Layer 2 STP and optimize traffic distribution in multiple directions. To achieve this and facilitate the MAC-in-MAC encapsulation to work over all available paths, Intermediate System-to-Intermediate System (IS-IS) is used instead of traditional STP, because it is actually a Layer 2 routing protocol that does not rely on IP for carry frames. Therefore, IS-IS routing is used to distribute link-state information and to calculate the shortest paths through the Clos network (usually between the multiple leaf stages via the spine nodes) to form the underlay of the Clos fabric network.

MAC-in-MAC, however, can be achieved using different protocols, such as IETF TRILL (Transparent Interconnection of Lots of Links), IEEE 802.1aq SPB (shortest path bridging), or Cisco FabricPath. Furthermore, TRILL/FabricPath brings the stability and performance of Layer 3 routing to Layer 2 switched networks, which offer a highly agile and reliable Layer 2 fabric that supports high scale-out DCN design requirements,³ as shown in Figure 8-11.

3. “Scale Data centers with Cisco FabricPath,” <http://www.cisco.com>

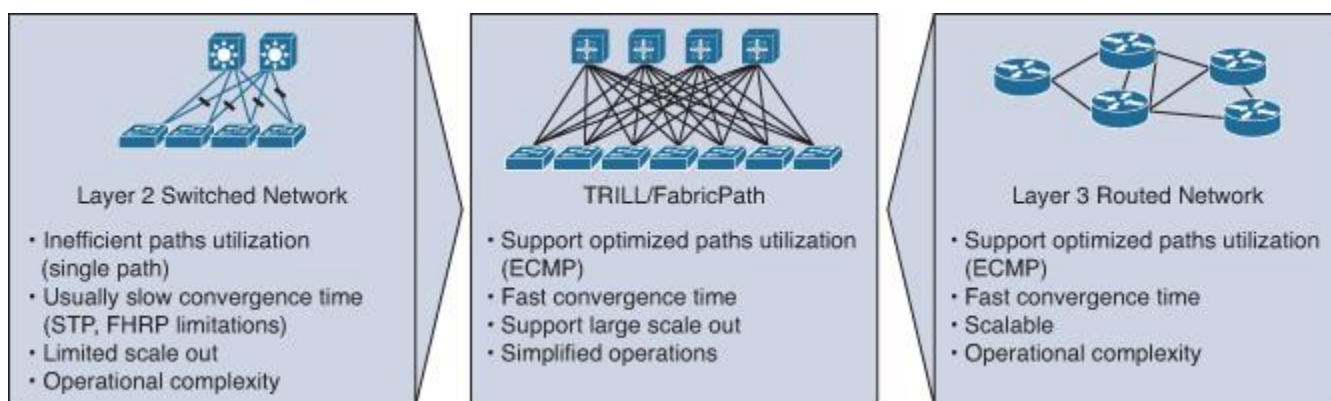


Figure 8-11 The Evolution of the TRILL/FabricPath Concept

Although TRILL and FabricPath are similar conceptually, FabricPath offers a more flexible and scalable design than TRILL when applied to the DCN. For example, the conversational MAC address learning of FabricPath offers a higher degree of MAC learning and forwarding efficiency to large-scale networks with a large number of MACs (such as a virtualized DC environment). In addition, FabricPath allows network designers and data center operators to optimize DCN efficiency and manageability by considering the FabricPath multitopology design, as shown in Figure 8-12.

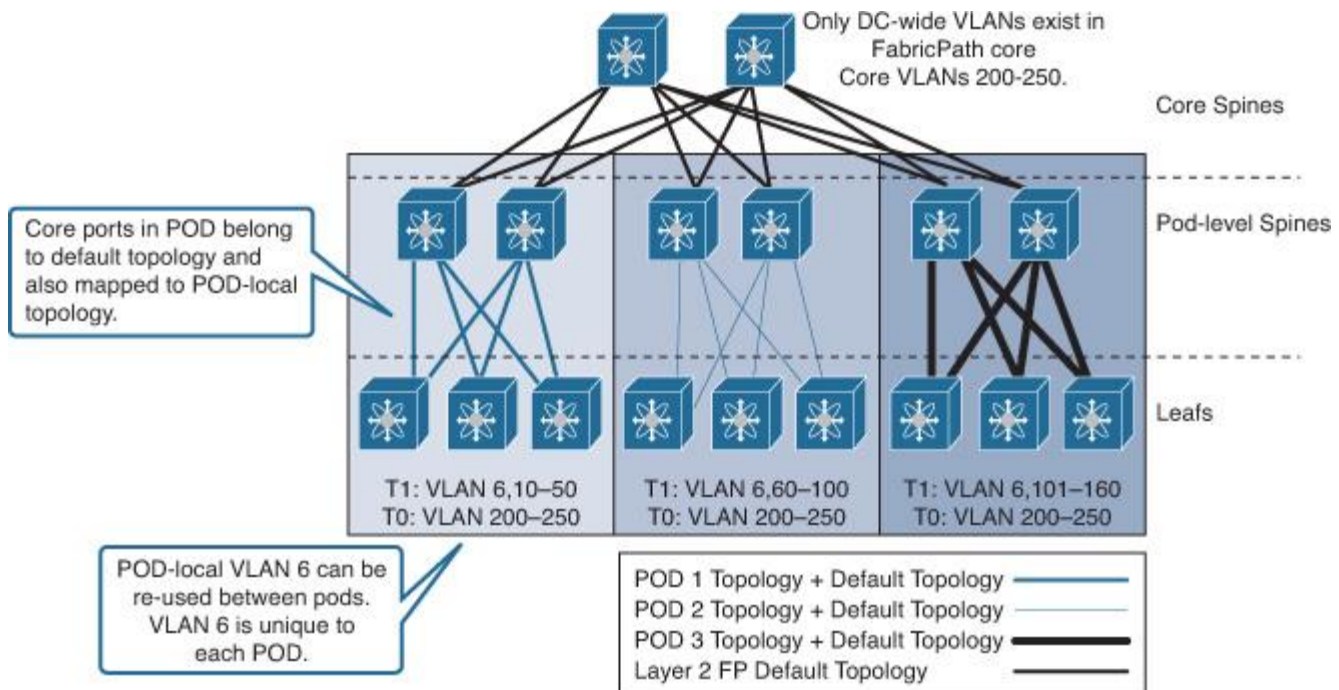


Figure 8-12 FabricPath Multitopology

As shown in Figure 8-12, the core is not required to hold all the VLANs from different downstream PODs if they are not used for inter-POD communication (local to the POD only). It can also be feasible in scenarios such as migration designs, VLAN localization, VLAN reuse, traffic engineering, and security. Network designers can take advantage of this multitopology architecture to optimize not only unicast but multdestination as well. This optimization is achieved by isolating and containing BUM (broadcast, unknown unicast, and multicast) traffic of the POD's local VLANs with the POD boundary without flooding this traffic unnecessarily to the core and other PODs, because each FabricPath topology can have its own multdestination tree (MDT), as shown in Figure 8-13.

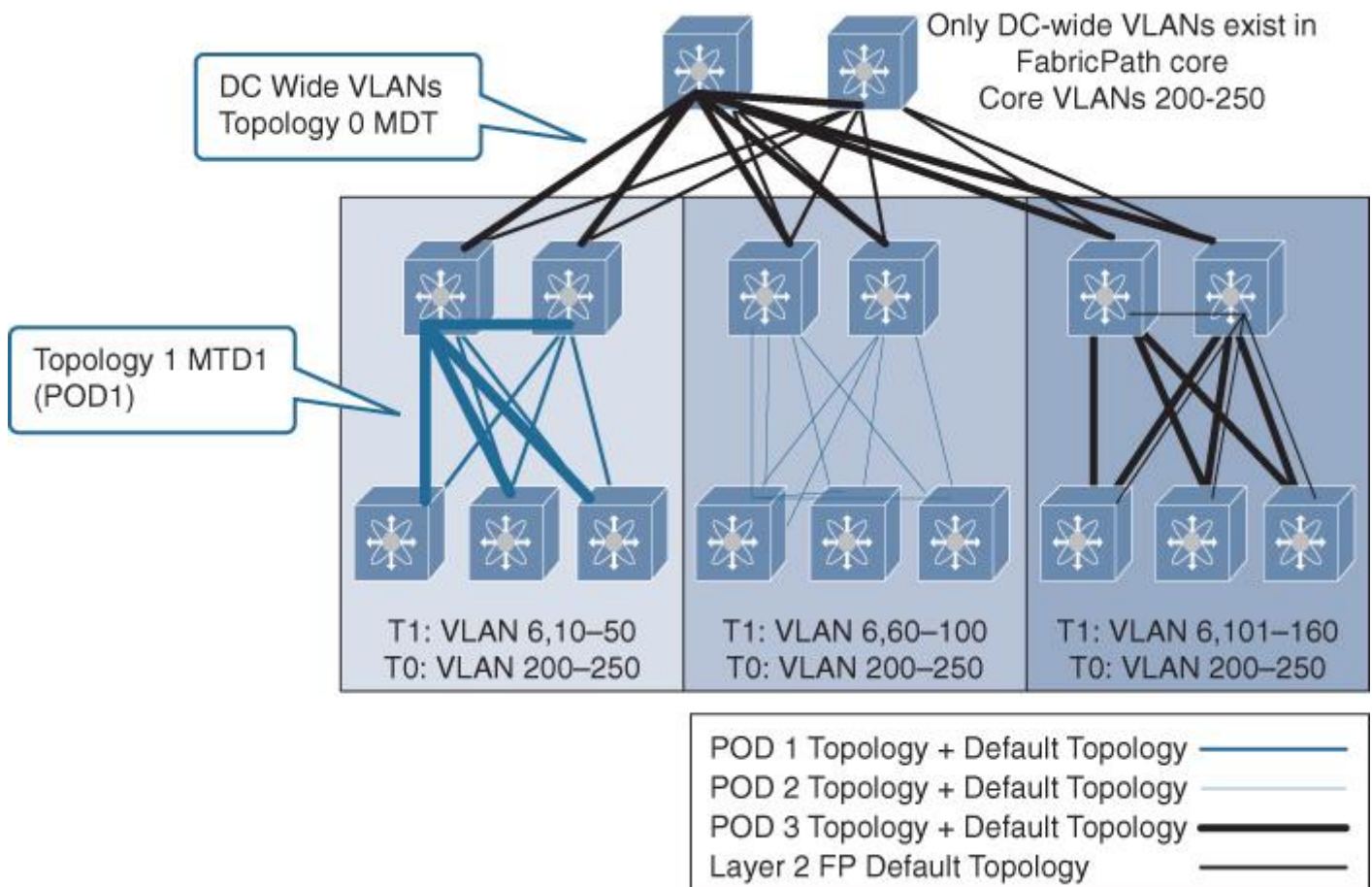


Figure 8-13 FabricPath MDT Tree

Furthermore, FabricPath enhances FHRP gateway services by offering the ability to provide four active HSRP peers in active-active mode (also known as *anycast HSRP*), whereas TRILL forwards traffic based on the traditional active-standby mode, as shown in Figure 8-14. Traffic forwarding based on the traditional active-standby mode usually offers lower forwarding efficiency and higher convergence time in the event of an FHRP gateway failure.

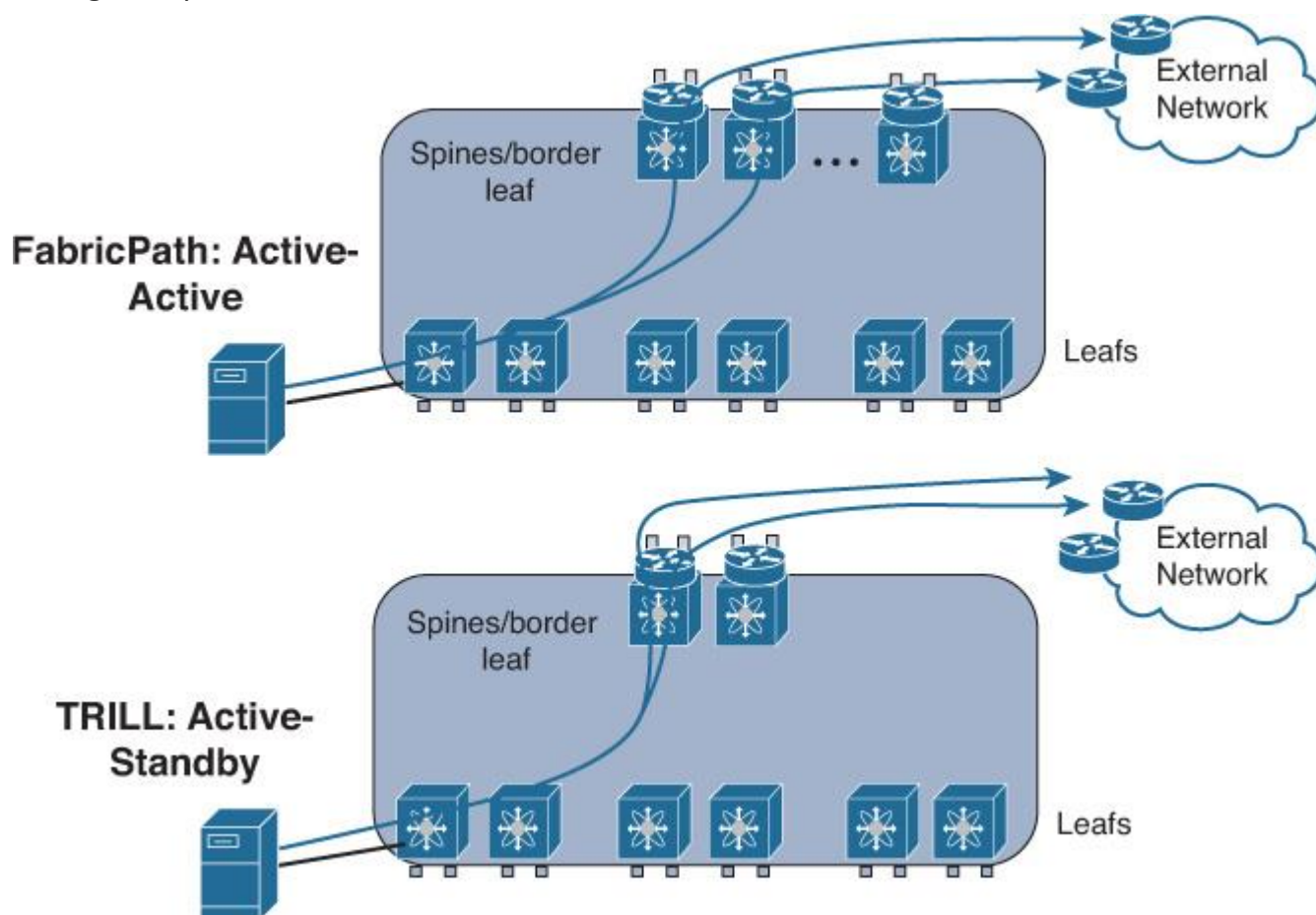


Figure 8-14 FHRP: TRILL Versus FabricPath

Note

With TRILL/FabricPath design, the Layer 3 gateway services can be deployed either at the spine nodes or can be moved to the border leaf nodes. Therefore, in Figure 8-14 the spine nodes are labeled as spines/border leaf.

Table 8-1 highlights the primary differences between FP, TRILL, and SPB.

	TRILL	FabricPath	SPB
Standard	IEEE	Cisco propriety	IETF
Frame routing: ECMP	Yes	Yes	Yes
Frame routing: TTL	Yes	Yes	No
Active-active FHRP	No	Yes, anycast HSRP	Yes
End-Host reachability and distribution	Flood and learn	Flood and learn with conversational MAC learning	Flood and learn with Provider Backbone Bridging (PBB) (802.1ah) (Shortest-Path Bridging-MAC Address [SPBM])
Service instances	~4K	~16M (by extending 12-bit VLAN ID to a 24-bit segment ID)	~16M with MAC in MAC
C-MACs learned only at the edge	Yes	Yes	Yes (SPBM)
Applicability	Enterprise & SP grade DC	Enterprise & SP grade DC	Enterprise DC and service provider (SP)-grade network

Table 8-1 *FP Versus TRILL Versus SPB*

Note

The Shortest Path Bridging VLAN ID (SPBV) version of SPB is Q-in-Q rather than MAC-in-MAC.

MAC-in-IP

With MAC-in-IP, it is obvious that the transport will be IP based from the network underlay point of view. Typically (not always), designing a Clos with MAC-in-IP overlay will rely on IS-IS to form the underlay of the Clos fabric network (ECMP). However, unlike MAC-in-MAC, the concept of MAC-in-IP overlay transport is based on the formation of virtual Layer 2 topologies on top of a physical Layer 3 network (IP encapsulation) combined with the concept of virtual network identifier, which is primarily used to forward frames to the required virtualized network context. Furthermore, MAC-in-IP overlay provides network based overlay, and it is also considered to be a host-based overlay solution offering multitenancy at scale with more simplified and automated workload provisioning capabilities. Figure 8-15 illustrates this concept with VxLAN applied as the MAC-in-IP overlay and the VxLAN tunnel endpoints (VTEPs) created at different levels.

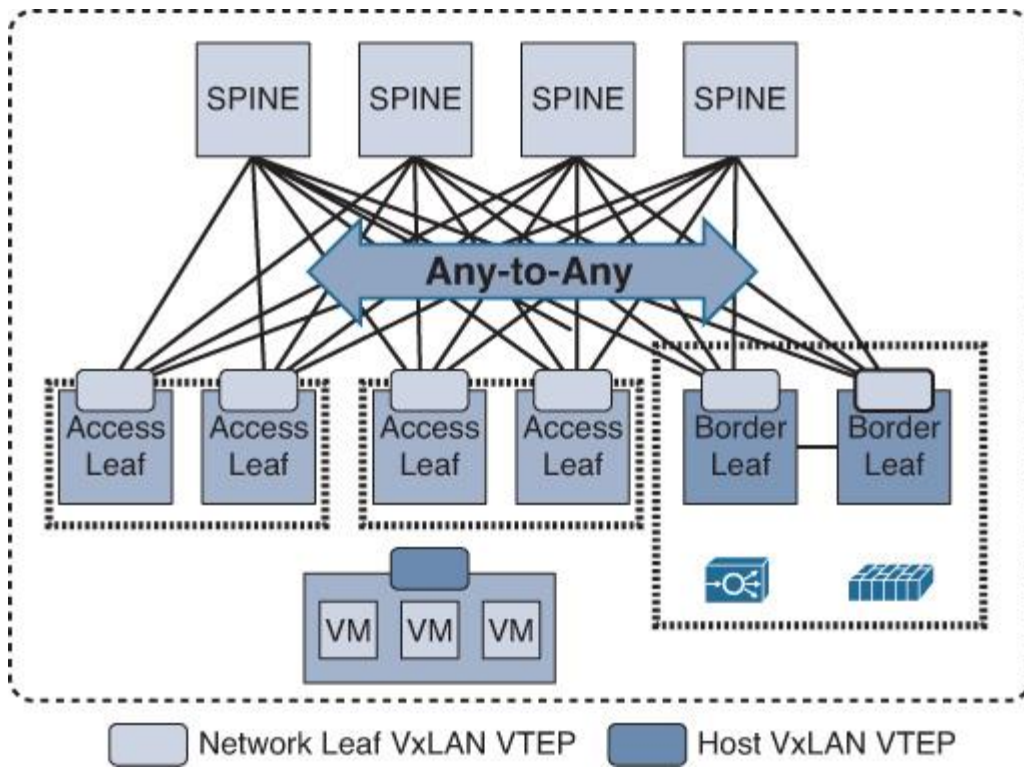


Figure 8-15 MAC-in-IP Network Overlay

At the time of this writing, the following are the primary protocols used for this purpose:

- Network Virtualization Using Generic Routing Encapsulation (NVGRE)
- Virtual Extensible LAN (VxLAN)
- Stateless Transport Tunneling (STT)

Table 8-2 compares the characteristics of these protocols.

	NVGRE	VxLAN	STT
Standards	IETF draft, sridharan-virtualization-nvgre.	IETF RFC 7348.	IETF draft, davie-stt
Encapsulation	Generic routing encapsulation (GRE)-based encapsulation.	UDP-based encapsulation.	TCP-based encapsulation
Widely supported/considered by vendors	Moderate: Microsoft, Arista, Emulex, Huawei, and HP.	Highly: Cisco, VMware, Arista, Brocade, Citrix, Red Hat, and Broadcom.	Least: VMware and Broadcom
Overlay identification	24-bit virtual subnet identifier (VSID).	24-bit virtual network ID (VNI).	64-bit context ID
Layer 2 forwarding of BUM traffic	Each VSID is mapped to a multicast group; multiple VSIDs can share the same multicast group.	Each VNI is mapped to a multicast group; multiple VNIs can share the same multicast group.	Either ingress replication or encapsulate IP multicast as destination IP
Overlay ECMP	32 bits of VSID plus the flow ID for ECMP purposes.	UDP based (derived from a hash of the inner headers).	TCP based (derived from a hash of the inner headers)
Underlay ECMP hashing	Hashing based on GRE header is not common (not support by most if not all switches).	UDP 5-tuple-based hashing.	TCP 5-tuple-based hashing

Table 8-2 MAC-in-IP Protocol Comparison

Although host-based overlay may offer capabilities that optimize the overall architecture and its flexibility with regard to workload mobility and automated provisioning, a robust and reliable underlay transport still forms the foundation of this overlay. If there is any inefficiency within the underlay design or performance, it will be reflected directly on the overlay efficiency as well. Therefore, a successful overlay DCN design must be architected over a reliable networked infrastructure that supports the desired level of resiliency and performance.

Note

The Cisco Application Centric Infrastructure (ACI) provides optimized traffic forwarding over a Clos architecture associated with flexible network-based VxLAN forwarding where each leaf performs encapsulation-normalization,

which offers the freedom for DC operators and server administrators to consider any type of host-level tagging, such as 802.1Q, network virtualization using GRE (NVGRE), VxLAN, or untagged.

Note

Today's modern and cloud-based data centers are moving rapidly toward the concept of a centralized control plane that offers an abstracted centralized view of the network and can manage traffic and deploy services centrally (manually and automatically provision services) to meet today's data centers needs, which are mainly built around simplicity, high-speed forwarding, and fast services provisioning with automation without having to touch each individual network node in the forwarding plane across the DCN. This concept is known as *software-defined networking* (SDN). Although SDN is beyond the scope of this book, it is highly recommended that network designers (regardless preparing for the CCDE or not) have a good understanding of this concept and how it can simplify and optimize virtualized and cloud-based data center designs.

MPLS Based

Although other protocols discussed earlier in this chapter provide end-to-end path separation optimally by using different mechanisms, with MPLS the design can be feasible for scenarios requiring integration with other MPLS domains (for instance, data centers that belong to MPLS L3VPN providers that offer hosting services to their customers as well). Therefore, enabling MPLS L3VPN across the DCN can facilitate end-to-end MPLS L3VPN integration and offer seamless access with a scalable architecture.

That said, using MPLS across the leaf and spine architecture is becoming less common in today's DCNs as other protocols and overlay technologies such as VxLAN offer more simplified and flexible provisioning and integration with cloud orchestration (host based and network based) at a high scale. Therefore, data center operators can consider any hybrid approach, where you can use one of the protocols discussed earlier across the DC Clos fabric, such as TRILL or VxLAN, while MPLS L3VPN can be extended from the Clos (border leaves) at the DC edge, as shown in Figure 8-16. In addition, the flexibility and efficiency of this design offers a simplified migration approach for data center operators to move from a pure MPLS-based DC to FabricPath or VxLAN based while retaining MPLS VPN enabled at the data center Clos fabric border leaves. In addition, this approach offers a seamless integration with other MPLS domains such as MPLS L3VPN, as shown in Figure 8-16.

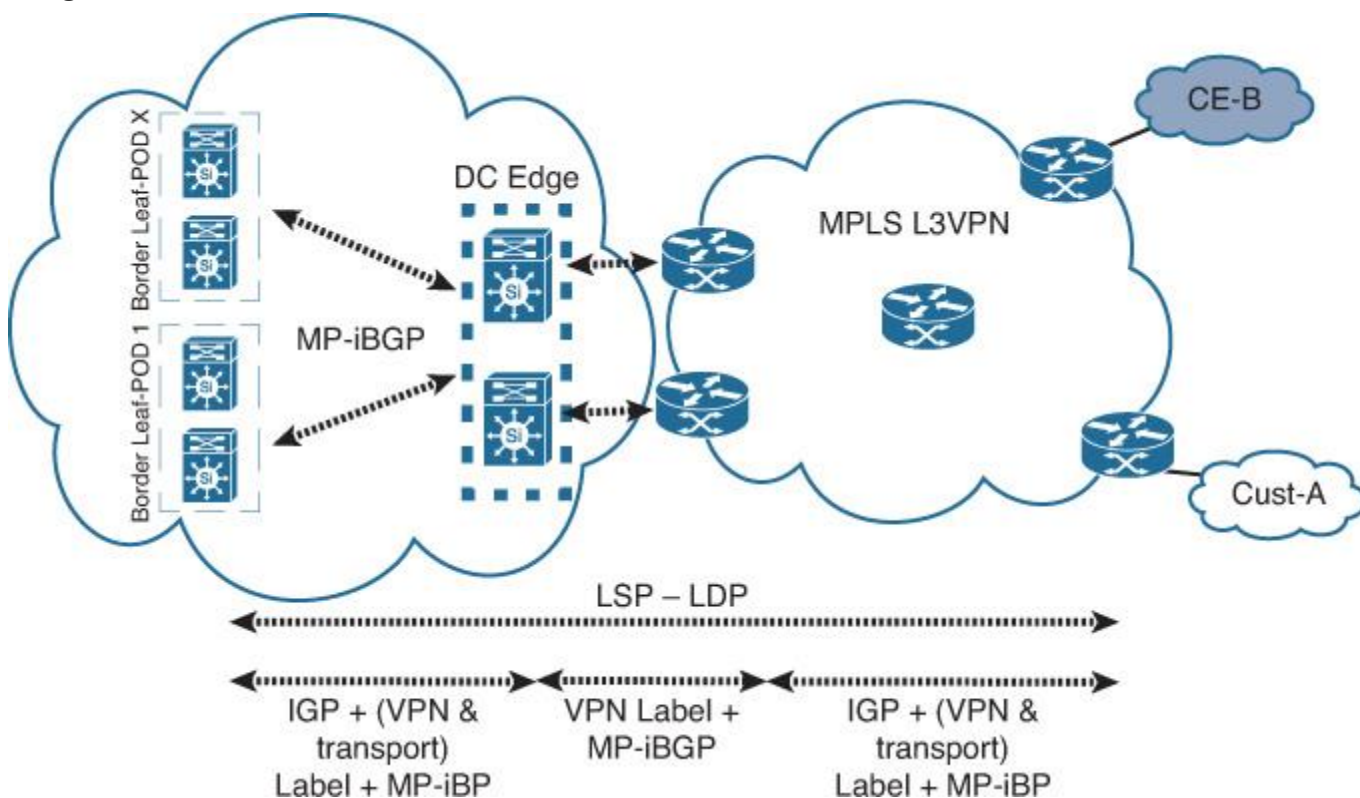


Figure 8-16 MPLS at the Border Leafs of the DCN

Note

If the leaf switches do not support MPLS, virtual routing and forwarding (VRF), or Multiprotocol Border Gateway Protocol (MP-BGP), MPLS can be enabled at the DC edge nodes only to interconnect with other MPLS L3VPN domains. This approach requires extending the relevant DC VLANs to the DC edge nodes that will operate like an SP provider edge (PE) nodes, which in turn will increase its operational complexity.

Note

MPLS over Clos is still a valid and doable design option; however it depends on the requirements and situation whether to consider it; see Table 8-3 for more detail.

Note

In this design, Layer 2 communications between different PODs can be facilitated by considering inter-POD L2VPN.

Note

In general, overlays can lead to increased operational complexity, because normally these overlays will require network operators to investigate the mapping of the virtual topology on top of the physical topology.

Layer 3-Based Transport (Nonoverlay)

This design model is based on the concept of using Layer 3 routing only to handle the packet forwarding across the Clos fabric network by extending Layer 3 routing across the leaf and spine nodes, as shown in Figure 8-17. Therefore, this model is not actually overlaid. Instead, it is based on an end-to-end routed network from leaf to leaf (top-of-rack [ToR] switches) over the spines. This design model offers an increased level of network scalability and stability by limiting Layer 2 flooding domains to the minimum (usually up to the ToR/leaf switches).

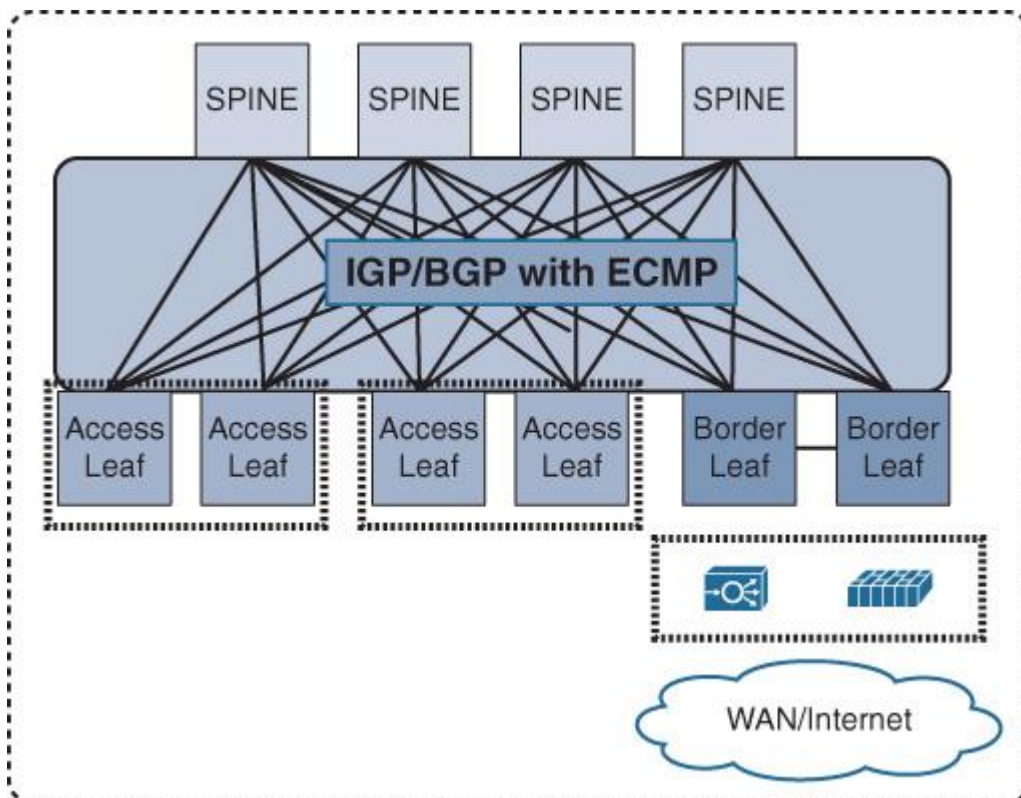


Figure 8-17 Layer 3-Based Clos

Furthermore, this design model facilitates to a high degree Clos architecture scale-out in large-scale DCNs. In addition, using one unified control plane protocol can significantly simplify overall network manageability and troubleshooting. Therefore, this routed Clos fabric architecture can support large-scale data centers that host tens of thousands of systems (physical or virtual). However, this design approach is more suitable for large-scale data centers where the highest priority is a stable and scalable network that can host extremely large number of servers (instead of using large Layer 2 adjacencies across the network) [61]. However, the

degree of the network scalability and stability with this architecture is still dependent on the characteristic of the used Layer 3 control plane protocol and its design.

Typically, you can use either an interior gateway protocol (IGP) (such as link-state routing protocols) or BGP as the control plane protocol to provide the control plane routing and forwarding. However, link state requires some fine-tuning to the protocol to optimize its performance and reduce the amount of link-state flooding and processing overhead over such a topology (Clos). In addition, the shortest path first (SPF) algorithm limitations on Clos topology can add some design and operational complexities, because each leaf might appear as a transit path to reach a spine or another leaf switch.

In contrast, using BGP as the control plane protocol within the data center over the Clos fabric architecture, in particular external BGP (eBGP), can offer a more controlled, simplified, and scalable routed Clos fabric design as compared to IGP. However, this approach may introduce some limitations around network convergence time. As highlighted earlier, this design model is more suited for environments that do not require extending Layer 2 across different leaf nodes or PODs; therefore, if VM mobility is required in this architecture, you can use a MAC-in-IP overlay, which means another protocol to manage, and this translates into higher design and operational complexity in this case.

Table 8-3 compares the different transport options that you can use with the Clos DCN architecture covered in this section.

	MAC-in-MAC	Layer 3 Based	MAC-in-IP	MPLS
Fault domain	Large	Minimal	Moderate, depending on the tunnel (VTEP) termination point	Minimal with L3VPN, larger with L2VPN
Scalability	Large	Extremely Large when used with eBGP as the control plane	Very large when used with BGP as a control plane	Large
Encapsulation	Frame encapsulation	N/A	IP encapsulation	MPLS label
Support extended Layer 2 within a DC and multiple DCs	Yes	Requires an overly such as VxLAN or multitier Overlay Transport Virtualization (OTV)	Yes	Yes (L2VPN)

Underlay control-plane (static, IGP, BGP)	IS-IS	IGP/BGP	Any unicast routing	Any unicast routing
ECMP	Yes	Yes	Yes	Yes
Support host-based overlay	No	No (requires other protocols)	Yes	No (unless the end host can do MPLS encapsulation)
Requires staff routing knowledge	Not required	Advanced	Minimal	Advanced
Operational complexity	Low	Generally high, depending on the size and routing design	Moderate	Moderate to high, depending on the size and design (for example, L3VPN with L2VPN)

Table 8-3 DCN Transport Protocol Options Comparison

Note

Dynamic fabric automation is a variation of the Clos architecture that offers automation and API northbound integration with the cloud orchestrator. For instance, with Cisco Dynamic Fabric Automation (DFA), when a VM is provisioned, the DFA leaf node autodetects the VM, and pulls and applies the policies associated with the VM on the connected port. The fabric tracks a VM and moves these port profiles within the fabric automatically [38]. The Cisco DFA fabric can apply this automation workflow to a similar physical server.

Comparison of Data Center Network Architectures

Table 8-4 compares the primary characteristics of each of the most common design models of DCNs available today and discussed earlier in this chapter.

	Three-Tier Classic STP Based	Three-Tier mLAG Based	Clos Fabric Architected
Scalability	Limited. Usually each access node is limited to only 2x upstream nodes.	Limited. Each access node is limited to only 2x upstream nodes.	Can scale out up to the limit of the HW/SW of the underlay and overlay protocols, such as up to 32 upstream nodes (spines) (web scale).
Operational complexity	High (manage STP, FHRP LAG configuration, and so on)	Moderate. The larger the network, the more mLAG port channels to be created and managed.	Simple. No LAG or mLAG is required.*
Seamless Layer 2 communication	With this design model in large networks, access switches in different PODs can end up communicating via the L3 core.	With this design model in large networks, access switches in different PODs can end up communicating via the L3 core.	The nature of two-tier switch fabric Clos architecture supports large scale-out and end-to-end L2 communications between the leafs without L3 core in the path. (seamless end-to-end Layer 2)

Performance and bandwidth efficiency	This design model offers the least because of STP limitations and the limited upstream switches and bandwidth aggregation toward the core.	This design model offers enhanced network performance and bandwidth capacity compared to STP based, but still has limited upstream switches (2x) and bandwidth aggregation toward the core.	This design model (two-tier leaf and spine) offers the optimal design model with regard to performance and increased bandwidth capacity (leaf to leaf).
Availability	In the event of aggregation switch failure, there will be usually 50 percent loss of the available bandwidth for each of the affected access switches.	In the event of aggregation switch failure, there will be usually 50 percent loss of the available bandwidth for each of the affected access switches.	If the design is based on 4x upstream switches (spines), the failure of a spine switch reduces the available bandwidth at the leaf switch by only 25 percent.
Convergence	Depends on the STP design, but in general this design model is lowest in terms of convergence time.	Fast. Failure of one mLAG member link or upstream switch will technically have minimal to zero impact.	Fast. Failure of leaf uplink or a spine switch will technically have minimal to zero impact
Layer 2 - VM mobility and distributed workload	Supported with limited scale (when access to access switch communicate via a Layer 3 node/core switch).	Supported with limited scale (when access to access switch communicate via a Layer 3 node/core switch).	Seamless support end to end at a web scale.
Suitability	Small to medium DCs. Limited bandwidth requirements. Limited hosts connectivity. Flexibility requirements. Nonvirtualized and very limited scale of virtualized workload.	Small to medium DCs. More scalable bandwidth requirements. Flexible hosts connectivity. Nonvirtualized and limited scale of virtualized workload.	Large to very large scale (web scale). Highly scalable bandwidth. Flexible hosts connectivity. Nonvirtualized and virtualized workload.

* If pure Layer 3 routing is used as the Clos control plane, there will be added operational complexity in configuring and managing a large-scale routed network.

Table 8-4 DCN Architectures Comparison

Note

Network designers might sometimes need to integrate one of the modern DC designs discussed here, such as TRILL or VxLAN, with a DC environment that is based on a traditional design (STP based), which is common in migration scenarios. Therefore, in this type of scenario, one of the primary design principles that must be considered is that flooding domains (failure domains) have to be minimalized and contained. For instance, the classic Layer 2 VLANs of an existing DC LAN can terminate at the VxLAN gateway node that performs bridging and routing between the different DCN domains.

Data Center Interconnect

The concept of distributing business applications and services across geographically dispersed data centers has existed for years; it is not a new concept or a design approach. Primarily, it is driven by the need to maintain business resilience and provide reliable disaster recovery (DR) capabilities for the data center IT infrastructure and resources. The emergence of cloud computing has further covered the need for extremely robust network resilience strategies, addressing security, availability, and virtual machine (VM) mobility while maintaining the flexibility and agility of a cloud model. Cloud computing promises to offer services that can have dynamic resource allocation and are accessible from anywhere (over the Internet) in a completely virtualized on-demand environment. Accordingly, the availability of the cloud data center and its elements (network, computing, and storage) are critical to supporting business continuity.

The nature of a cloud data center built on a fully virtualized environment introduces some challenges to maintaining business continuity, in comparison to traditional data centers. For example, the physical components of the cloud data center (computing, network, and storage) may not all be located within the same physical location. Instead, they may be distributed across more than one physical location and interconnected over a transport mechanism that maintains an end-to-end path isolation. To achieve the desired level of data center business continuity, there must be a reliable DR plan and procedures in place.

Businesses and stakeholders are realizing the importance of business continuity. For instance, the loss of connectivity to a data center in a financial services business, for a few minutes, may cost the business hundreds of thousands of dollars. This makes considerations of a DR plan a common business requirement to achieve the desired level of business continuity during and after failure situations. Typically, disasters can be classified based on the degree and geographic coverage of the impact (local, metro, regional, or global) and can result from

- **Natural catastrophe:** Such as fire, flood, or earthquake

- **System outage:** Such as power outage at the Main Distribution Frame (MDF) that provides the external connection to the DC

- **Human error:** Such as misconfiguration, which may lead to bringing one or all interconnected data centers into a blackout situation

Based on the business's targeted level of continuity and data protection, you can use a different remote DR site model along with the appropriate DR solution. A remote DR site refers to the site to where some or all of the vital business operations and applications can be moved in the event of any outage (whether a planned or unplanned outage). Typically, a DR site is one that is dedicated to the data center's failover purposes, such as a secondary data center. Nonetheless, sometimes it is expensive for businesses to maintain a dedicated DR site. Therefore, these businesses consider other sites, such as a regional data center, with its primary function being to serve regional remote sites as the primary DC. At the same time, it operates as a DR data center for the centralized or primary data center.

In addition, some public cloud providers offer DR as service (DR-as-a-S). This service is for customers who do not need to build or dedicate a data center. These customers can benefit from this cost-effective approach, because the data center operator does not need to build and maintain cooling, power, cabling, and internal and external network connectivity. Moreover, data center operators that already have their own private cloud can consider what is known as a *hybrid cloud*. A hybrid cloud is where the DR site can be hosted by

one of the public cloud providers as well, for the same purpose, while maintaining connectivity between the two cloud DCs using an intercloud solution.

Besides the remote DR site consideration, the DR solution should be selected based on how long the organization can wait for services to be restarted (in particular, the extent to which the business can afford to lose data after the failover happens). Should the business restart in a degraded mode, or must all services be fully available immediately after the switchover? Financial services normally require a recovery time objective (RTO) of less than one hour with a recovery point objective (RPO) equal to zero without a degraded mode [39]. This is a fairly widespread practice so that no transactions are lost. The data center interconnect (DCI) plays a primary role in reaching and delivering these requirements. Table 8-5 compares the typical DR models in terms of the levels of RTO and RPO each model can support; this relationship also shown in Figure 8-18.

DR Model	RTO	RPO	Operational Mode
Cold standby	High	High	Active-standby
Warm standby	Moderate	Moderate	Active-standby/active-active
Hot standby	Low	Low	Active-active

Table 8-5 DR Models

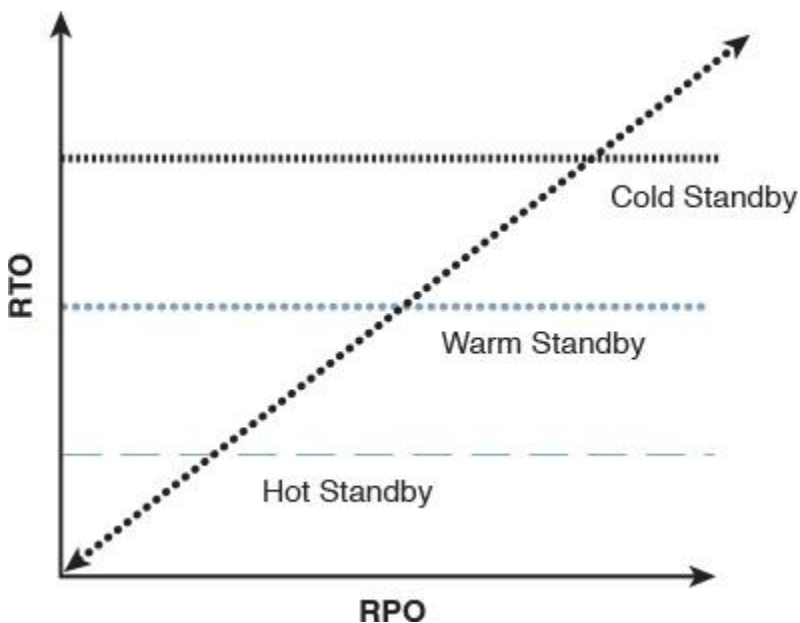


Figure 8-18 DR Models and RTO-RPO Relationship

Note

Business continuity (BC) is not analogous to DR. Typically, BC should answer the questions of “How can you keep the business running at the recovery site?” and “How can you eventually transfer business functions back to the primary site?” DR answers the question, “How can you recover/move services on the recovery site during and after a disaster or data center failure?” In other words, a DR plan is one component to achieving BC.

Note

Even though the cold standby has high RPO and RTO, it might not be suitable for a business that has systems capable of operating offline and storing data locally then replicating it later to the data center. In other words, network design

decisions always depend on the business goals and needs and on the characteristics of the systems and applications used.

Note

Although the focus of this section is on modern data centers based on virtualization and cloud services with VM mobility, all the design options and considerations discussed in this section apply to traditional data centers. In fact, with traditional data centers where little or no virtualization has taken place, the job is much easier because there is no VM mobility. Instead, the main focus is on selecting which DR approach to use, and how L3 routing is designed to address inbound and outbound traffic flow when the interconnected data centers are designed to work in either an active-standby or active-active manner.

Note

BC aims to continue business activities (business as usual) after a disaster or failure. From the network point of view, this goal is to be achieved across two or more data centers where a redundant external network path (communication with the external users WAN, MAN, or Internet) is in place. The subsequent sections assume the external connection to the WAN or Internet is already in place (locally per DC) and focus on the design consideration of the DCI part and how to optimize traffic flows to the different DCs to meet BC goals.

DCI Building Blocks

The following elements are the primary functional components that comprise the DCI building block architecture to provide transport elasticity, flexibility, transparency, and resiliency between two or more sites:

■ DCI Connectivity

■ **Extended LAN:** Provides transparent transport for application and operating system mobility that requires Layer 2 adjacency, such as geoclustering.

■ **Layer 3 based:** Provides the typical routed connectivity between the interconnected data centers and provides a reliable transport for workload mobility for the systems that support it over Layer 3.

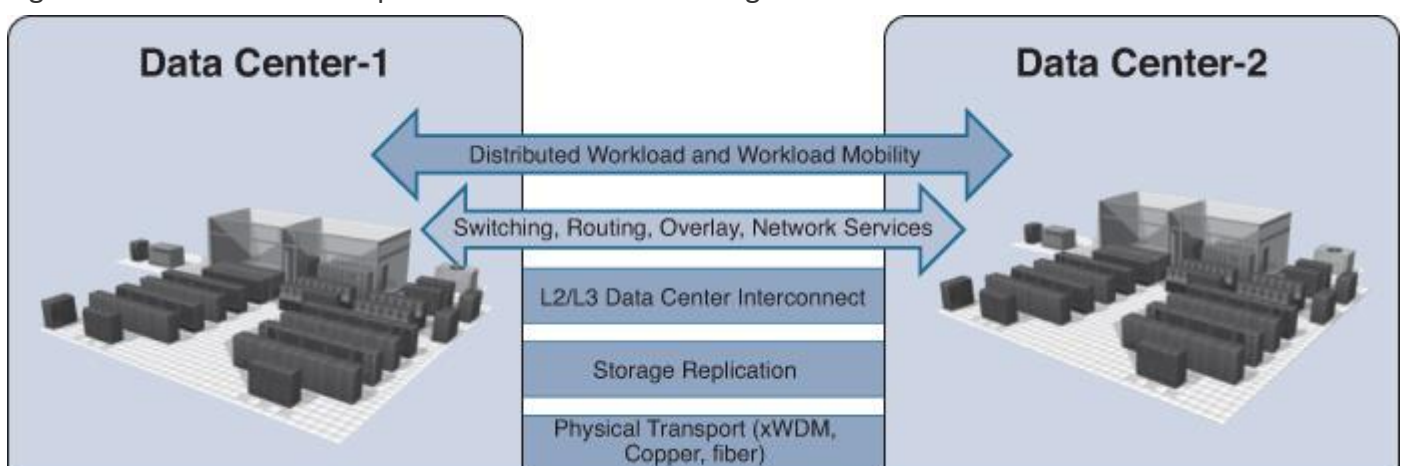
■ **Hybrid:** This connectivity option is usually based on having a mixture of both Layer 3 and Layer 2 DCI, where the only LAN extension across the DCI is provisioned used for certain VLANs that require Layer 2 extension, while all other subnets will communicate over the DCI using the Layer 3 based DCI model.

■ **Intra and inter-DC routing:** Provides remote site and end-user access to the DC resources, such as applications and cloud services, in addition to facilitating the communications between the segmented networks across the interconnected data centers. (Usually each DC should have its own WAN or Internet connectivity to external networks.)

■ **Path optimization:** Enhances (ideally localizes) ingress and egress (south-north) traffic flows and server-to-server (east-west) workflows.

■ **SAN extension:** Provides data access for applications (either locally or remotely) along with the desired data replication type (such as synchronous and asynchronous replication).

Figure 8-19 shows the conceptual view of the DCI building blocks.



The design options, considerations, and drivers of each of the DCI functional components in the preceding list are covered in the subsequent sections.

DCI Connectivity Options

Workload mobility and distributed workload of the virtualized applications among data centers across dispersed geographic locations offer businesses and IT teams flexible efficiency to meet various operational requirements, such as data capacity expansion, workload allocation flexibility, seamless application and system migrations, disaster avoidance, and so on. The design of the DCI solution forms the foundation of today's interconnected data centers. Failure to meet the desired requirements or to understand how the different services and applications behave before, during, and after any failure event can take the entirety of the data centers out of service, as would be the case in a dual-active scenario when the applications are not designed to perform in this manner (active-active).

Simple VM mobility from one DC to another can introduce a blackout situation to the data centers or the DCI. Similarly, a lack of good understanding about how the various systems and applications are integrated and communicate with each other can also introduce a blackout situation. Therefore, the design of a DCI solution must address the DR and BC goals, in addition to workload functions (mobile or distrusted) between geographically diverse data centers, to ensure nonstop access to business-critical applications and information and to prevent complicating the design and operations of the data center.

From the DCI design point of view, the main drivers toward considering a Layer 2 or a routed DCI solution are the requirements supporting the elasticity of applications and storage access. Commonly, stretched Layer 2 (LAN) DCI is required to enable the transparent movement of VMs and the transparent access of stored data. However, this is not always the case. Therefore, the decision whether to go with a routed or with a Layer 2-based DCI solution is primarily derived from the requirements of the applications and systems at the higher layers to meet the overall BC goals. For example, it is important that network architects distinguish between workload mobility and distributed workload design requirements. Typically, with workload mobility, the moved VMs from one DC to another should maintain their IP information. This can be achieved by either extending the Layer 2 LAN segment over the DCI or by considering an IP mobility solution over a routed DCI, such as Locator/ID Separation Protocol (LISP).

The distributed workload, however, refers to the concept of geoclustering (the clustering of servers over DCI), which might not always be deployed as virtualized systems. Typical geoclustering systems require a nonroutable connectivity between the distributed systems for the heartbeat sessions, whereas the front-end virtual IP (VIP) of the cluster can follow the same concept of IP mobility or extend the LAN DCI to fail over after any failure event. In other words, besides understanding the business goals in terms of DR and BC, identifying systems and applications architectures to achieve these goals is a key point to achieve a successful DCI design [39, 40]. This section covers the various DCI connectivity options and highlights the strength and weaknesses of each.

Note

Although many modern data center architectures offering seamless workload mobility over long-distance DCI tend to support a robust DR level, mobility itself is not the DR solution. The DR plan can take advantage of VM mobility to optimize the overall desired BC goal, but there are many components other than VM mobility (for example, data replication, external connection failover, and geoclustering).

Note

This section does not intend to recommend one design or connectivity option over another. Nevertheless, it will analyze potential design and connectivity options and highlight the strengths and weaknesses of each. The best choice is always the one that meets business goals, DR/BC requirements, and the service and application requirements, which

collectivity drive the design choice. Therefore, a good understanding of each design option and its components is essential.

Routed DCI

This design model is based on Layer 3 routed interconnection between the different data centers. In fact, this design model follows the same design concepts discussed in Chapter 4, “Enterprise Edge Architecture Design,” in the “Enterprise Multihomed Internet Design Considerations” section.

Although extended Layer 2 (LAN) as a DCI solution is more common, this does not mean that it is the optimal or more reliable DCI design. In general, the routed DCI design model offers a simpler and more predictable design compared to the extended LAN model. However, as covered earlier, the design choice must always be derived from the upper-layer requirements (applications, workload, and virtualized and nonvirtualized) that align with the business DR and BC goals. Therefore, if the Layer 2 extension is technically not required to achieve the desired goals, the routed DCI can be considered in this case. The routed DCI can help to avoid several design and operational complexities (as discussed later in this chapter, along with the possible solutions). Furthermore, with the routed DCI model, fault domains will be contained within each site (fault-isolation principle). This offers a more stable and reliable DCI solution, as shown in Figure 8-20.

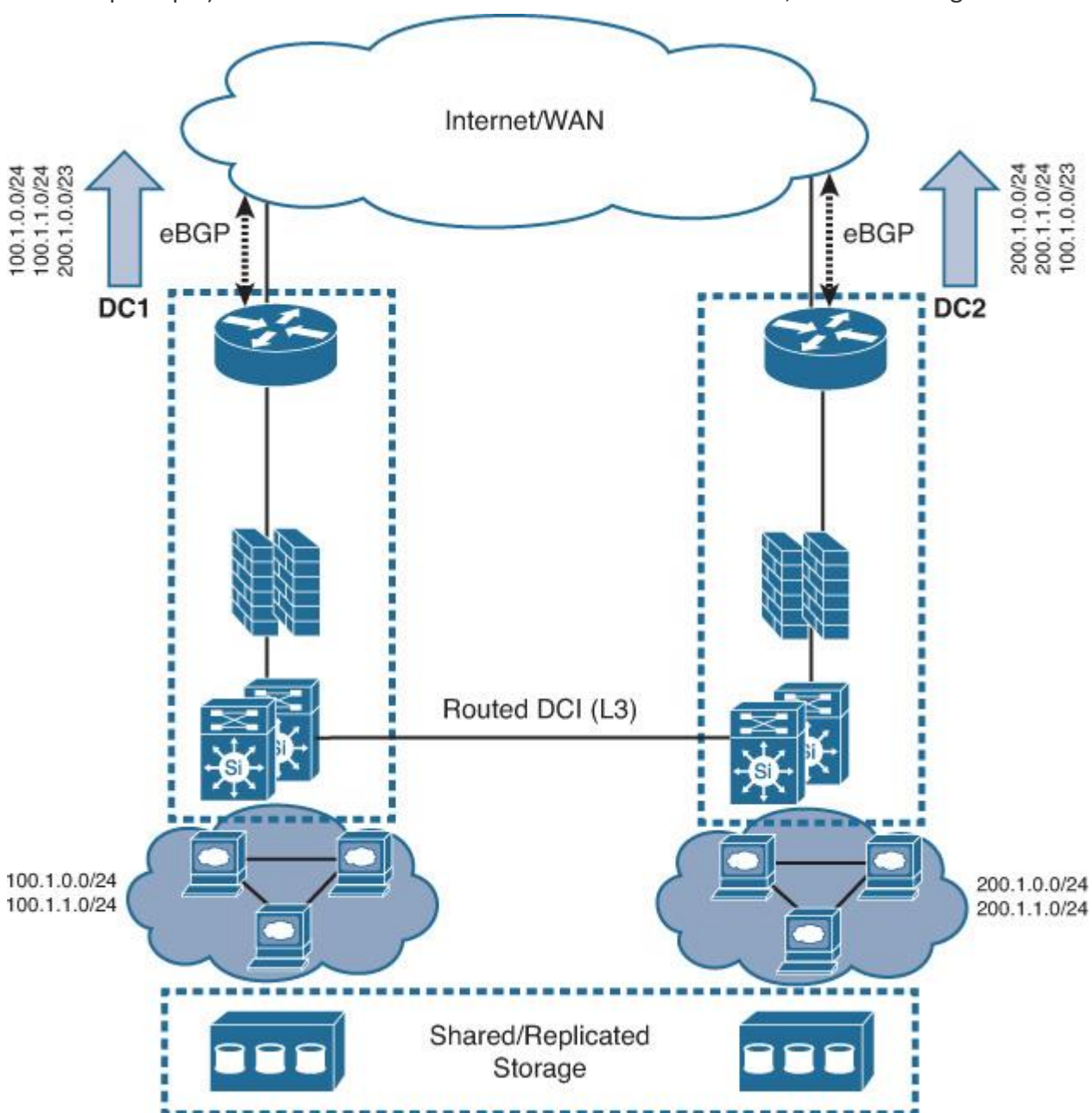


Figure 8-20 Routed DCI

The DCI solution shown in Figure 8-20 reflects typical interconnected data centers over a routed DCI, where each site has its own IP range (either full range or a divided range). At the same time, each site advertises the other site's IP range (such as summary of the both site's IP range if its summarizable) for failover purposes. However, in some scenarios, the functional requirements of the interconnected DCs require each DC to provide connectivity to its local resources only in alignment with the architectures of the organization's applications.

From the networking design point of view, this means that the DCI must not be used for failover of traffic coming from the external networks (WAN/Internet) and that the load distribution for external connections should be controlled via higher layers, such as global load balancers (also known as *global site selector* [GSS]). Consequently, in this design approach, each DC location will advertise only its own local IP range. This ensures that each DC will provide connectivity only to the local resources and will not provide connectivity to the other DC during some failure scenarios. Data replication between the DCs is handled by the storage design and hypothetically should ensure that both data centers can have valid and updated replicas of the data that aligns with the BC goals by achieving the desired level of RTO and RPO. This example demonstrates how the entire typical routed DCI design can change based on business, functional, or application requirements. Another example is that some data center operators advertise the same IP ranges from multiple data centers to offer an "anycast" style of load distribution; traffic will usually always prefer the closest data center in scenarios where multiple data centers exist across different geographic areas and advertise the exact same IP subnets.

As with any design solution, network designers must understand the different aspects of the routed DCI design to be able to identify when and why to position it as a reliable DCI solution for a given DCN design (ideally based on the top-down approach). Table 8-6 summarizes the main strengths and design concerns of the routed DCI.

Strengths	Design Concerns
Offers more stable DCI solution	Workload mobility with IP mobility can add design and operational complexity.
Provides a scalable solution that can support multiple interconnected DCNs	Not all geoclustering supports routed DCI.
Offers more deterministic control of ingress and egress traffic flows to avoid and minimize asymmetrical routing issues	May lead to DCI congestion in the event of one DC external link failure (WAN/Internet).
Supports VM mobility with an IP mobility solution	Active-active can be a challenging design to achieve if it is not aligned well with other services and components at higher layers such as global site load balancers.
Supports active-active and active-standby interconnected DCs design model	

Table 8-6 Routed DCI Strengths and Design Concerns

Layer 3 DCI Connectivity Options

The provisioning of a routed DCI connectivity between data centers can be achieved with different types of layer 3 transport mechanisms, including the following:

- MPLS L3VPN WAN
- Tunneling over the Internet